OM6239 2.4G Wireless SOC

V0.1

Liuc

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1 Introduction

The OM6239 is a member of the low-cost, high-performance family of intelligent 2.4 GHz RF transceivers with embedded microcontrollers. The OM6239 is optimized to provide a single chip solution for Ultra Low Power (ULP) wireless applications. The combination of processing power, memory, low power oscillators, real-time counter and a range of power saving modes provides an ideal platform for implementation of RF protocols. Benefits of using OM6239 include tighter protocol timing, security, lower power consumption and improved co-existence performance.

1.1 Pin assignment

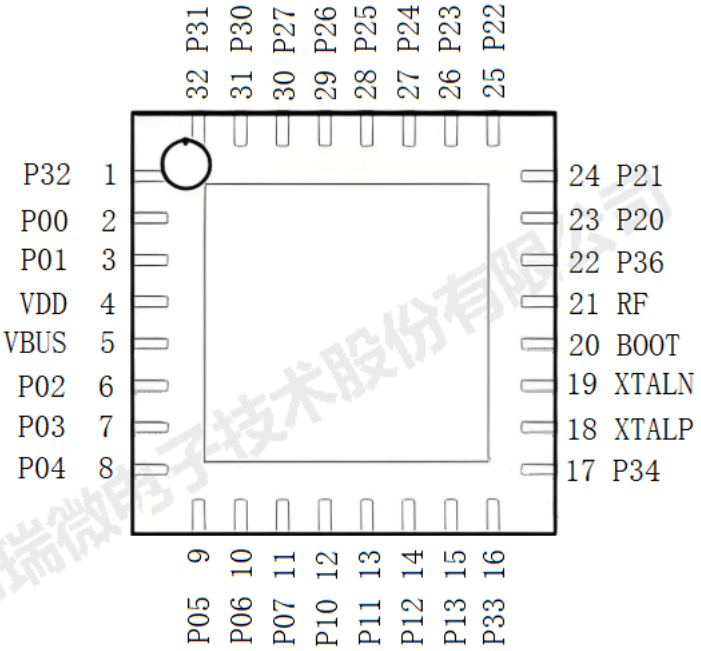


Figure1.1 OM6239 pin assignment

1.2 Pin function

Table1.1 OM6239 pin function

Pin No.	Pin name	Description
1	P32	GPIO
2	P00	GPIO
3	P01	GPIO
4	VDD	3.3 LDO output, need 2.2uF~10uF capacitor
5	VBUS	5V power supply, need capacitor at least 4.7uF
6	P02	GPIO

.

7	P03	GPIO
8	P04	GPIO
9	P05	GPIO
10	P06	GPIO
11	P07	GPIO
12	P10	GPIO
13	P11	GPIO
14	P12	GPIO
15	P13	GPIO
16	P33	GPIO
17	P34	GPIO
18	XTALP	16M Crystal +
19	XTALN	16M Cystal -
20	BOOT	BOOT pin, 0 for flash, 1 for sram
21	RF	RF pin
22	P36	GPIO
23	P20	GPIO
24	P21	GPIO
25	P22	GPIO
26	P23	GPIO
27	P24	GPIO
28	P25	GPIO
29	P26	GPIO
30	P27	GPIO
31	P30	GPIO
32	P31	GPIO

1.3 Reference design

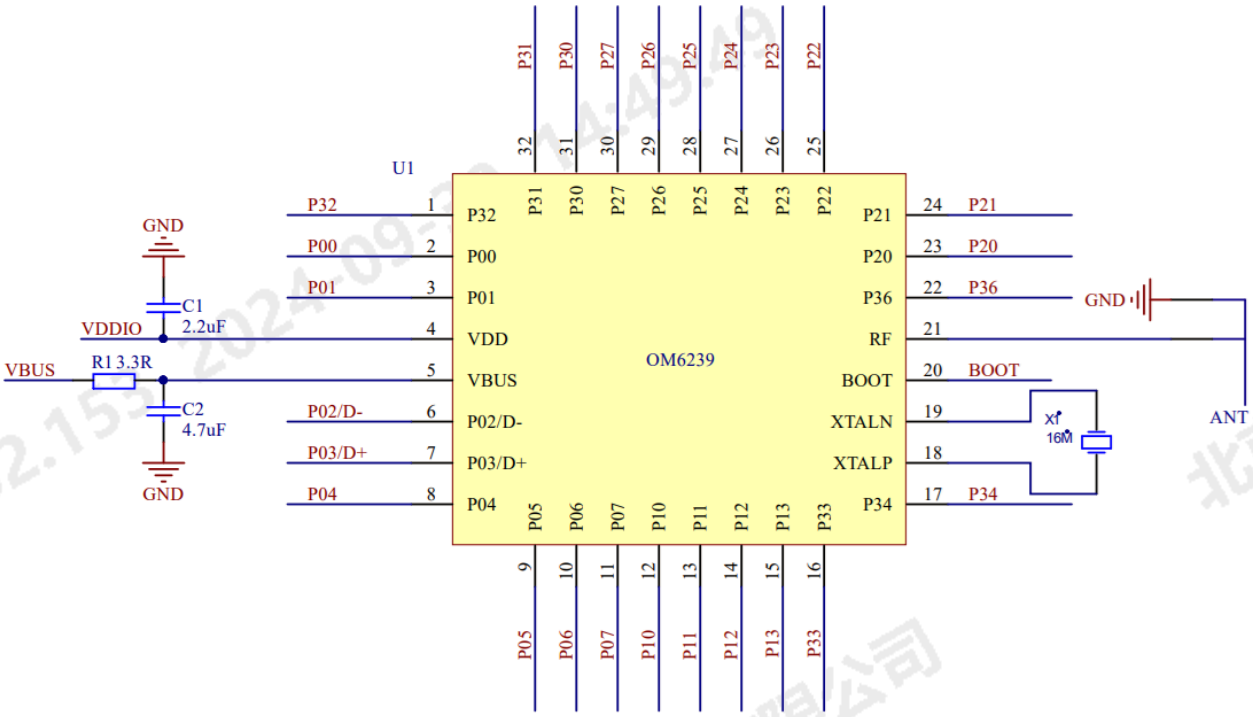
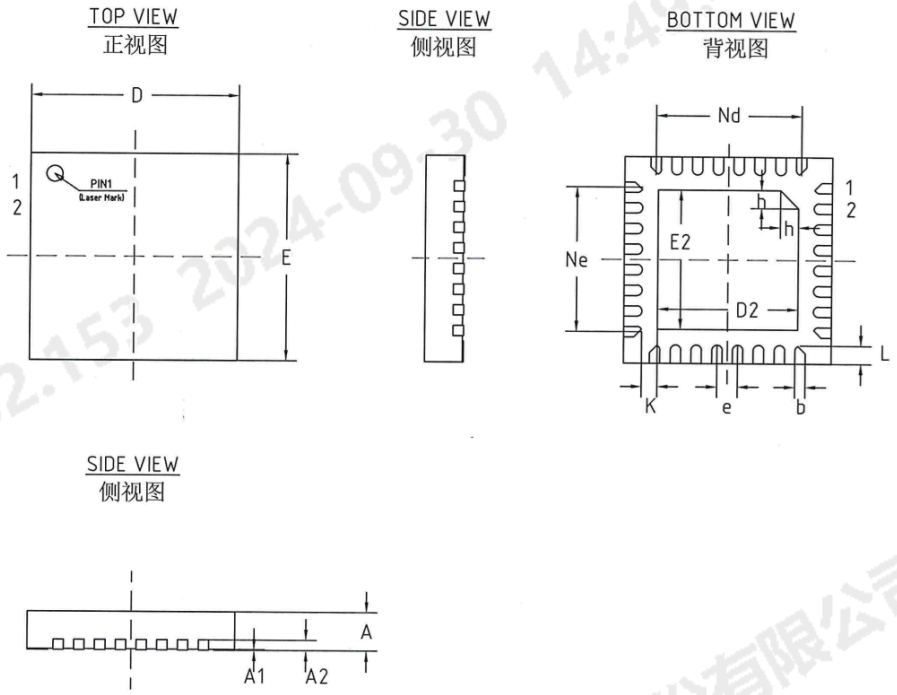


Figure1.2 OM6239 reference design

1.4 Package information

OM6239 is a QFN32 4mm X 4mm package, the information is as below.



机械尺寸/mm			
字符 SYMBOL	最小值 MIN	典型值 NOMINAL	最大值 MAX
A	0.70	0.75	0.80
A1	-	0.02	0.05
A2	0.203 REF		
b	0.15	0.20	0.25
D	3.90	4.00	4.10
D2	2.65	2.70	2.75
E	3.90	4.00	4.10
E2	2.65	2.70	2.75
e	0.40 BSC		
K	0.25	0.30	0.35
L	0.30	0.35	0.40
h	0.30	0.35	0.40
Ne	2.80 BSC		
Nd	2.80 BSC		

Figure1.3 OM6239 QFN32 4mm X 4mm package

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2 Product overview

2.1 Features

Features of the OM6239 include:

- Fast 32-bit microcontroller:
- Memory:
  - Internal RAM: 12Kbyte
  - Cache RAM: 1Kbyte
- A number of on-chip hardware resources are available through programmable multi-purpose input/output pins:
  - 26 GPIO
  - Full duplex serial port-UART
  - USB 2.0 full speed Device
  - 8 PWMs
  - External interrupts
- High performance 2.4 GHz RF transceiver
  - True single chip GFSK/FSK transceiver
  - Enhanced Shock-Burst™ link layer support in HW:
    - Packet assembly/disassembly
    - Address and CRC computation
    - Auto ACK and retransmit
  - On the air data rate 1 Mbps or 2 Mbps
  - 125 RF channels operation, with 79 (2.402 GHz –2.480 GHz) channels within 2.400–2.4835 GHz
  - Short switching time enable frequency hopping
- System reset and power supply monitoring:
  - On-chip power-on and brown-out reset
  - Watchdog timer reset
  - Software reset and power domain reset
  - Power-fail comparator with programmable threshold and interrupt to MCU
- On-chip timers:
  - One 32-bit timer/counter operating at the system clock
  - One 32-bit timer/counter operating at the low frequency clock (32kHz)
- On-chip oscillators:
  - 16 MHz crystal oscillator XOSC16M
  - 48MHz PLL
  - 32kHz clock divided by 16MHz crystal
- Power management function:
  - VBUS 4.5V~5.5V, VDD/VDDRF 1.9V~3.6V
  - Low power design supporting fully static stop/standby/retention/deepsleep
  - Programmable MCU clock frequency
  - On chip voltage regulators supporting low power mode

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- Watchdog and wakeup functionality running in low power mode
- Power fail detection

2.2 Blocks

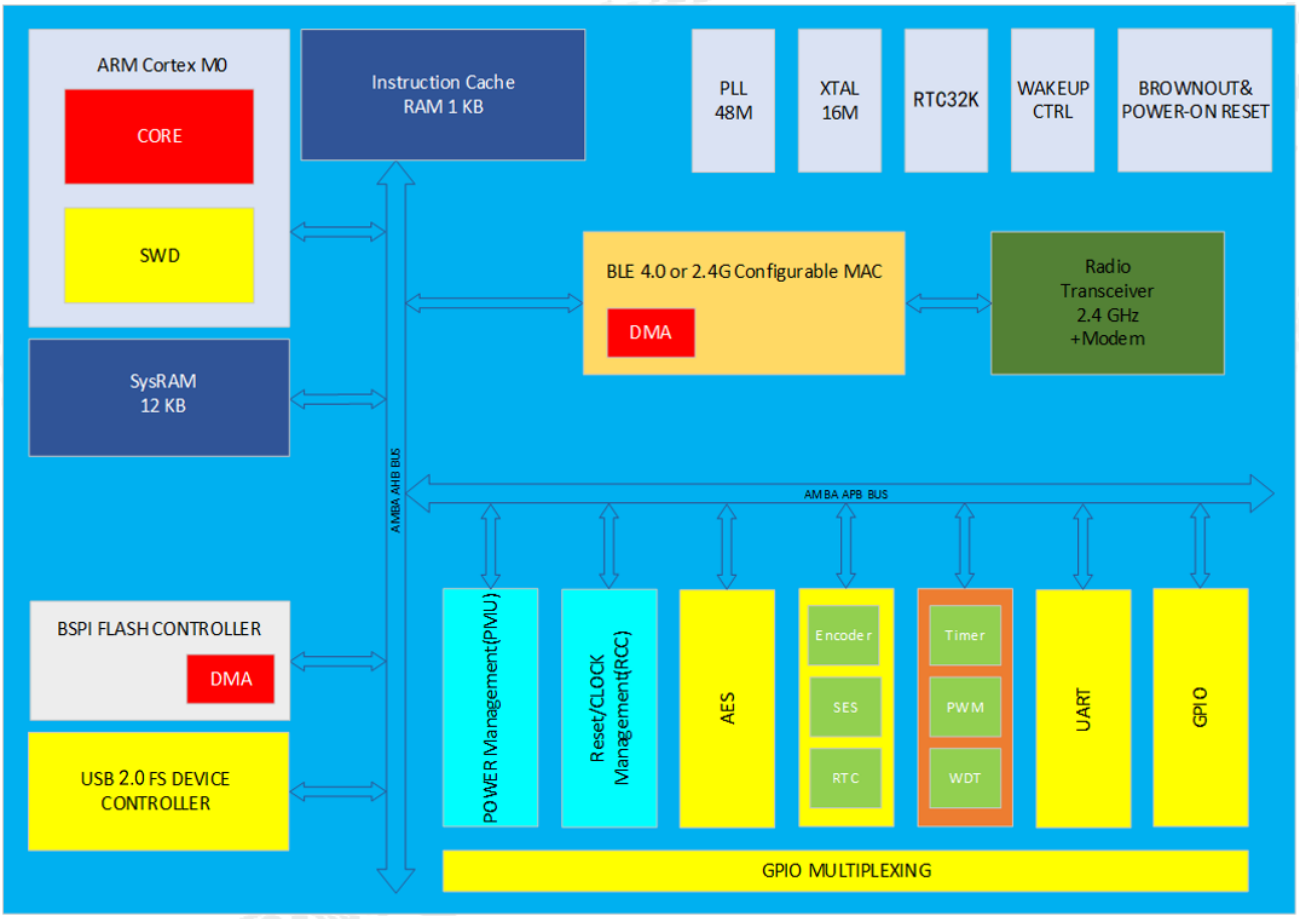


Figure2.1 OM6239 block diagram

2.3 Memory Map

0x4100 0000 - 0x41FF FFFF	USB	AHB
0x0080 0000 - 0x008F FFFF or 0x5200 0000 - 0x53FF FFFF	Flash Controller	
0x4200 0000 - 0x42FF FFFF	RF	
0x4300 0000 - 0x43FF FFFF	GPIO	
0x4000 0000 - 0x4000 00FF	PMU	APB
0x4000 0100 - 0x4000 01FF	AES	
0x4000 0200 - 0x4000 02FF	RTC	
0x4000 0300 - 0x4000 03FF	UART	
0x4000 0400 - 0x4000 04FF	CACHE	

0x4000 0500 - 0x4000 05FF	TIMER	
0x4000 0600 - 0x4000 06FF	RCC	
0x4000 0700 - 0x4000 07FF	WDT	
0x4000 0800 - 0x4000 08FF	PWM	
0x4000 0900 - 0x4000 09FF	POF	
0x4000 0A00 - 0x4000 0AFF	LEDC	
0x4000 0B00 – 0x4000 0BFF	ENCODER	
0x4000 0C00 – 0x4000 0CFF	SES	
0x2000 0000 - 0x2000 2FFF or 0x0020 0000 - 0x0020 2FFF 0x2000 4000 – 0x2000 6FFF or 0x0020 0000 – 0x0020 6FFF … 0x2000 c000 – 0x2000 EFFF or 0x0020 C000 – 0x0020 EFFF	Embedded SRAM (12K)	AHB
0x2000 3000 - 0x2000 FFFF or 0x0020 3000 - 0x0020 FFFF 0x2000 7000 -0x2000 7FFF or 0x0020 7000- 0x0020 7FFF … 0x2000 F000- 0x2000 FFFF or 0x0020 F000 – 0x0020 FFFF	Cache RAM(1K)	

2.4 Embedded SRAM

There is up to 12Kbytes of static SRAM. It can be accessed as bytes, half-words (16 bits) or full words (32 bits). The SRAM start address is 0x2000 0000 or 0x0020 0000.

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3 Electrical specification

Operating conditions							
No.	Symbol	Description	Note	Min	Typ	Max	Unit
1	VDD	3.3V Supply voltage		1.9		3.6	V
2	VBUS	5V Supply voltage		4.5		5.5	V
3	TEMP	Operating temperature		-40		85	℃

MCU parameters							
No.	Symbol	Description	Note	Min	Typ	Max	Unit
1	VIH	Input high voltage		0.7*VDD		VDD	V
2	VIL	Input low voltage		0		0.3*VDD	V
3	VOH	Output high voltage		VDD-0.3		VDD	V
4	VOL	Output low voltage		0		0.3	V
5	CLK_{MCU}	MCU clock				48	MHz
6	Ideepsleep	Deep sleep current	Support GPIO wake up		15		uA
7	Iretention	Register retention current	Retention 12KB sram		30		uA
8	Istb	Standby current	Crystal on, rtc32k on		60		uA
9	Istop1	Stop1 current			80		uA
10	Iactive	Active current	Run 16MHz		1.2		mA

RF parameters							
No.	Symbol	Description	Note	Min	Typ	Max	Unit
1	Freq	Operating frequency		2400		2525	MHz
2	TX_{POWER}	TX output power	VDD=3V	-40		8	dBm
3	$TX_{CURRENT}$	TX current	VDD=3V, TX_{POWER}=0dBm		15		mA
4	Pin_max	Max RX power				-10	dBm
5	RXsens_1M	RX sensitivity @ 1Mbps	10% error packet, h_idx=0.5		-90		dBm
6	RXsens_2M	RX sensitivity @ 2Mbps	10% error packet, h_idx=0.5		-87		dBm
7	$RX_{CURRENT}$	RX current			17		mA
8	Freq_step	PLL frequency step			1		MHz
9	Df_1M	Frequency deviation @ 1Mbps			250		KHz
10	Df_2M	Frequency deviation @ 2Mbps			320	500	KHz
11	Freq_xtal	Crystal frequency			16		MHz

12	CL_xtal	Crystal CL			12		pF
13	Df_xtal	Crystal tolerance	For BLE Df_xtal is +/- 30ppm	-60		60	ppm

4 RF transceiver

The OM6239 uses the same 2.4 GHz GFSK RF transceiver with embedded protocol engine. The RF transceiver is designed for operation in the world-wide ISM frequency band at 2.400–2.4835 GHz and is very well suited for ultra-low power wireless applications.

The radio transceiver, the baseband processor, and the qualified Bluetooth® low energy stack is compliant with the Bluetooth® Low Energy 4.2 standard.

The RF transceiver module is configured and operated through the RF transceiver map. This register map is accessed by the MCU through AHB and is available in all power modes of the RF transceiver module. The embedded protocol engine enables data packet communication and supports various modes from manual operation to advanced autonomous protocol operation. Data FIFOs in the RF transceiver module ensure a smooth data flow between the RF transceiver module and the OM6239 MCU.

The rest of this chapter is written in the context of the RF transceiver module as the core and the rest of the OM6239 as external circuitry to this module.

4.1 Features

Features of the RF transceiver include:

- General
  - Worldwide 2.4 GHz ISM band operation
  - Common antenna interface in transmit and receive
  - GFSK/FSK modulation
  - 1 Mbps and 2 Mbps on air data rate
- Transmitter
  - Programmable output power: -40~8dBm
  - 15mA at 0dBm output power
- Receiver
  - Integrated channel filters
  - 17mA at 2 Mbps
  - -87dBm sensitivity at 2 Mbps
  - -90dBm sensitivity at 1 Mbps
- RF Synthesizer
  - Fully integrated synthesizer
  - 1 MHz frequency programming resolution
  - Accepts low cost ±60ppm 16 MHz crystal
  - 1 MHz non-overlapping channel spacing at 1 Mbps
  - 2 MHz non-overlapping channel spacing at 2 Mbps
- Protocol Engine
  - 1 to 32 bytes dynamic payload length
  - Automatic packet handling (assembly/disassembly)
  - Automatic packet transaction handling (auto ACK, auto retransmit)
- 3 data pipe Multi-Slave for 3:1star networks

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4.2 Block

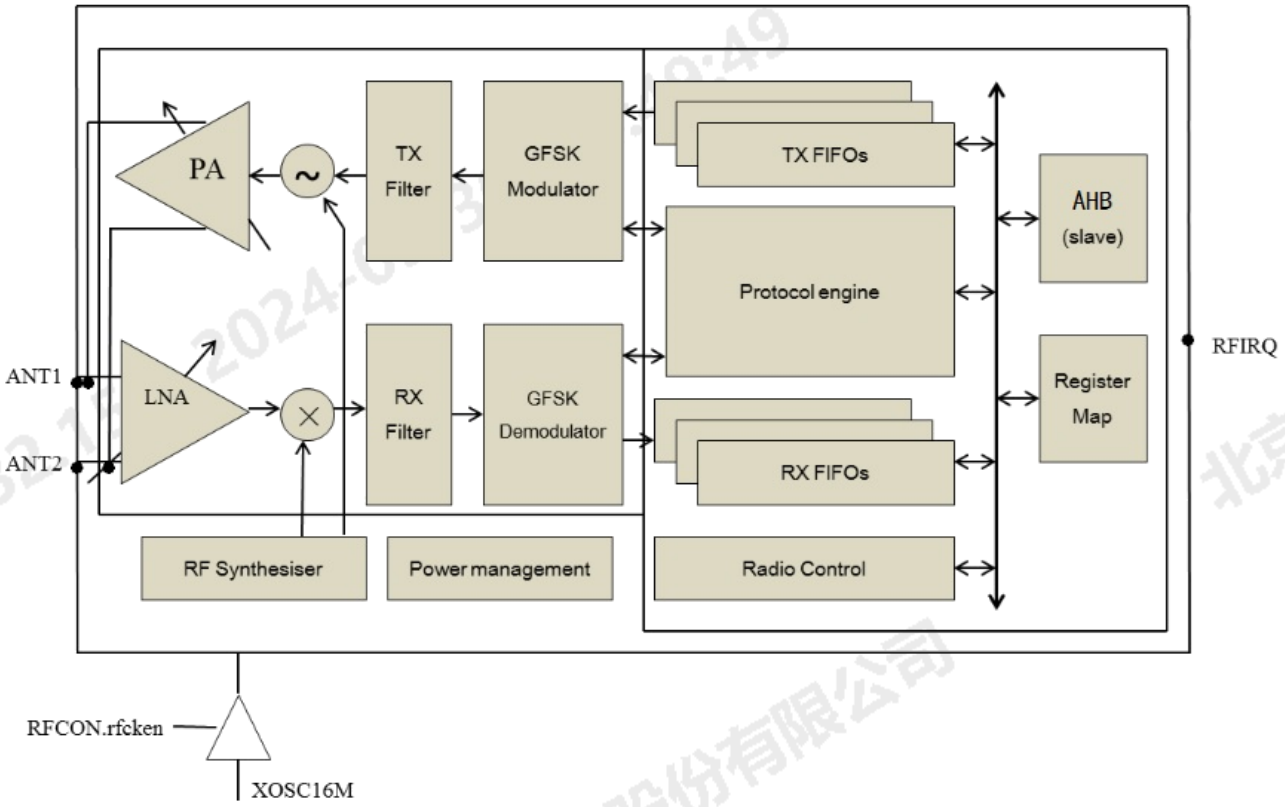


Fig.3.1 RF transceiver block diagram

4.3 Functional description

This section describes the different operating modes of the RF transceiver and the parameters used to control it.

The RF transceiver module has a built-in state machine that controls the transitions between the different operating modes. The state machine is controlled by register RFCE and RF transceiver register CONFIG.

4.4 Operational Modes

The OM6239 has a built-in state machine that controls the transitions between the chip's operating modes. The state machine takes input from user defined register values and internal signals.

The state diagrams in Figure 3.2 shows the operating modes and how they function. There are three types of distinct states highlighted in the state diagram:

- Recommended operating mode: is a recommended state used during normal operation.
- Possible operating mode: is a possible operating state, but is not used during normal operation.
- Transition state: is a time limited state used during start –up of the oscillator and settling of the PLL.

When the VDD reaches 1.8V or higher OM6239 enters the Power on reset state where it remains in reset until entering the Power-Down mode.

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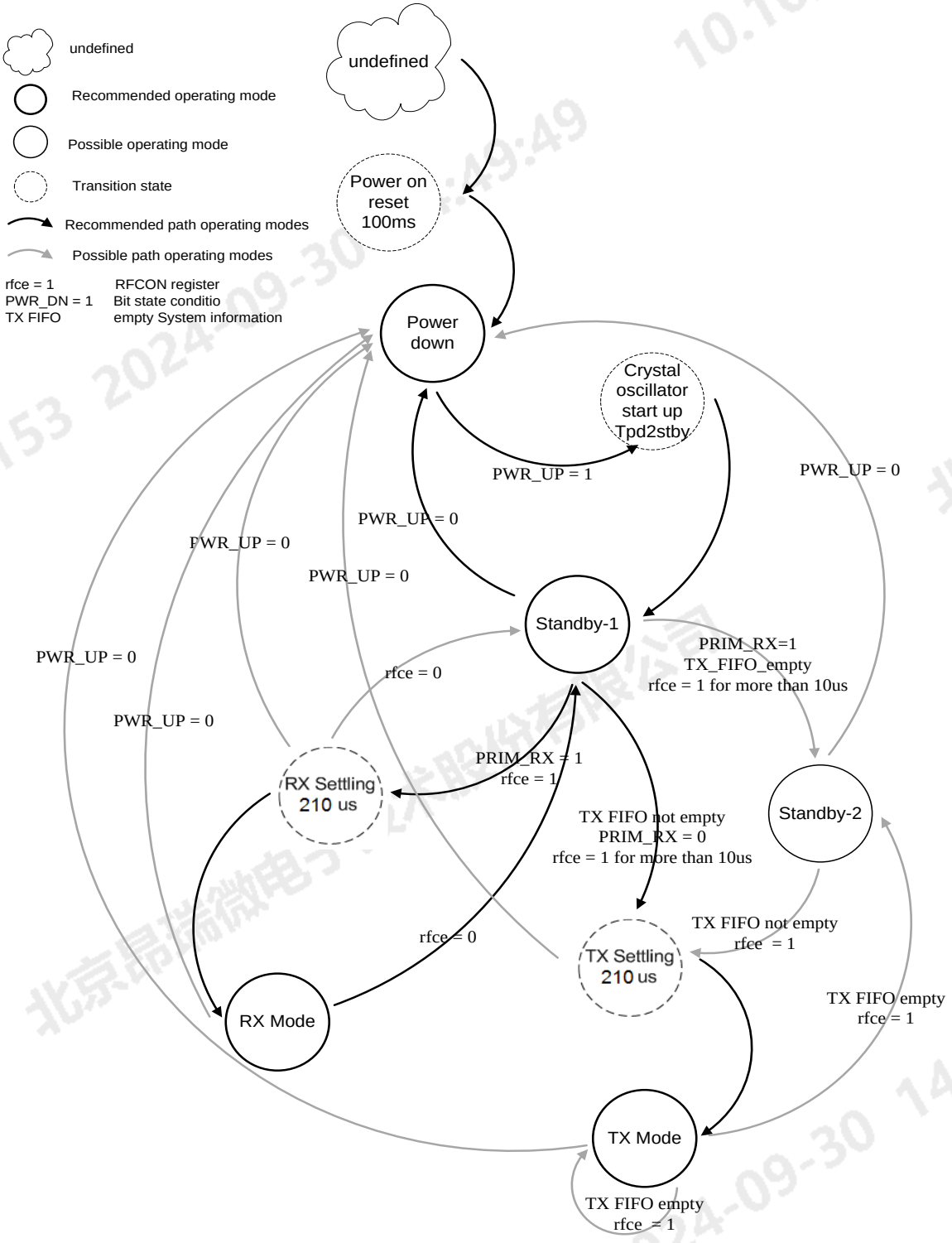


Fig. 3.2 Radio control state diagram

4.4.1.1 Power-Down Mode

In power down mode, the OM6228is disabled using minimal current consumption. All register values available are maintained and the SPI is kept active, enabling change of configuration and the uploading/down-loading of data registers. For starting up times see Table3.2. Power down mode is entered when the chip is in deep sleep or register retention mode (Details are descripted in chapter 7).

4.4.1.2 Standby Modes

4.4.1.3 Standby-I mode

When the chip is power up or wake up from deep sleep or register retention mode, the device enters standby-I mode. Standby-I mode is used to minimize average current consumption while maintaining short start up times. In this mode only part of the crystal oscillator is active. Change to active modes only happens if CE is set high and when CE is set low, the OM6239 returns to standby-I mode from both the TX and RX modes.

4.4.1.4 Standby-II mode

In standby-II mode extra clock buffers are active and more current is used compared to standby-I mode. The OM6239 enters standby-II mode if CE is held high on a PTX device with an empty TX FIFO. If a new packet is uploaded to the TX FIFO, the PLL immediately starts and the packet is transmitted after the normal PLL settling delay (210μs). Register values are maintained and the SPI can be activated during both standby modes.

Notice: From Standby-I mode to standby-II mode CE more than 20us

4.4.1.5 RX mode

The RX mode is an active mode where the OM6239 radio is used as a receiver. To enter this mode, the chip must have PRIM_RX bit and the CE pin set high.

In RX mode the receiver demodulates the signals from the RF channel, constantly presenting the demodulated data to the baseband protocol engine. The baseband protocol engine constantly searches for a valid packet. If a valid packet is found (by a matching address and a valid CRC) the payload of the packet is presented in a vacant slot in the RX FIFOs. If the RX FIFOs are full, the received packet is discarded.

The chip remains in RX mode until the MCU configures it to standby-I mode or power down mode. However, if the automatic protocol features in the baseband protocol engine are enabled, the chip can enter other modes in order to execute the protocol.

In RX mode a Received Power Detector (RPD) signal is available. The RPD register is a signed number which corresponds to different level of received power in dBm. The highest bit of RPD is sign bit. The range of the received power is -100 ~ +10dBm. The RPD has about +/-5dBm deviation from the real level.

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4.4.1.6 TX mode

The TX mode is an active mode for transmitting packets. To enter this mode, the chip must have the PWR_UP bit set high, PRIM_RX bit set low, a payload in the TX FIFO and a high pulse on the CE for more than 20µs.

The OM6239 stays in TX mode until it finishes transmitting a packet. If CE = 0, the chip returns to standby-I mode. If CE = 1, the status of the TX FIFO determines the next action. If the TX FIFO is not empty the OM6239 remains in TX mode and transmits the next packet. If the TX FIFO is empty the chip goes into standby-II mode. The OM6239 transmitter PLL operates in open loop when in TX mode. It is important never to keep the OM6239 in TX mode for more than 4ms at a time. If the protocol features are enabled, the chip is never in TX mode longer than 4ms.

4.4.1.7 Operational modes configuration

The following (table 1) describes how to configure the operational modes:

Mode	PWR_UP register	PRIM_RX register	CE	FIFO state
RX mode	1	1	1	-
TX mode	1	0	1	Data TX FIFO. Will empty all level in TX FIFO[a]
TX mode	1	0	Minimum 20us high pulse	Data TX FIFO. Will empty one level in TX FIFO[b]
Standby-2	1	0	1	TX FIFO empty
Standby-1	1	-	0	No ongoing packet transmission
Power Down	0	-	-	-

Table 3.1 RF main modes

a. If CE is held high all TX FIFO is emptied and all necessary ACK and possible retransmits are carried out. The transmission continues as long as the TX FIFO is refilled. If the TX FIFO is empty when the CE is still high, the chip enters standby-II mode. In this mode the transmission of a packet is started as soon as the CSN is set high after an upload (UL) of a packet to TX FIFO.

b. This operating mode pulses the CE high for at least 20µs. This allows one packet to be transmitted. This is the normal operating mode. After the packet is transmitted, the chip enters standby-1mode.

4.4.1.8 Timing Information

The timing information in this section relates to the transitions between modes and the timing for the CE pin. The transition from TX mode to RX mode or vice versa is the same as the transition from the standby modes to TX mode or RX mode (max. 210µs), as described in Table 3.2.

Name	The chip	Notes	Max.	Min.	Comments
$T_{pd2stby}$	Power down→Standby mode		3ms		Denpend on crystal
T_{stby2a}	Standby mode→TX/RX mode		210us		

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T_{hce}	Minimum CE high			20us	
$T_{pece2csn}$	Delay from CE positive edge to CSN low			4us	

Tab. 3.2 operational timing of the OM6239 chip

4.4.1.9 Air data rate

The air data rate is the modulated signaling rate the chip uses when transmitting and receiving data. It can be 1Mbps or 2Mbps. Using lower air data rate gives better receiver sensitivity than higher air data rate. But, high air data rate gives lower average current consumption and reduced probability of on-air collisions. The air data rate is set by the RF_DR bit in the RF_SETUP register. A transmitter and a receiver must be programmed with the same air data rate to communicate with each other.

4.4.1.10 RF channel frequency

The RF channel frequency determines the center of the channel used by the chip. The channel occupies a bandwidth of less than 1MHz at 1Mbps and a bandwidth of less than 2MHz at 2Mbps. The chip can operate on frequencies from 2.400GHz to 2.525GHz. The programming resolution of the RF channel frequency setting is 1MHz.

At 2Mbps the channel occupies a bandwidth wider than the resolution of the RF channel frequency setting. To ensure non-overlapping channels in 2Mbps mode, the channel spacing must be 2MHz or more. At 1Mbps the channel bandwidth is the same as or lower than the resolution of the RF frequency.

The RF channel frequency is set by the RF_CH register according to the following formula:

$$F0= 2400 + RF_CH\ [MHz]$$

You must program a transmitter and a receiver with the same RF channel frequency to communicate with each other.

4.4.1.11 Received Power Detector measurements

Received Power Detector (RSSI), located in register 09, is a signed number which corresponds to different level of received power in dBm. The highest bit of RSSI is sign bit. The range of the received power is -100 ~ +10dBm.

The RSSI can be read out at any time while the chip is in received mode. This offers a snapshot of the current received power level in the channel. The status of RSSI is correct when RX mode is enabled and after a wait time of T_{stby2a} +T_{delay_AGC}= 210us + 20us. The RX gain varies over temperature which means that the RSSI value also varies over temperature.

4.4.1.12 PA control

The PA (Power Amplifier) control is used to set the output power from the chip power amplifier. In TX mode PA control has four programmable steps, see Table 3.3.

The PA control is set by the RF_PWR bits in the RF_SETUP register.

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SPI RF-SETUP (RF_PWR)	RF output power	DC current consumption
1111	8dBm	37mA
0110	4dBm	21.5mA
0011	0dBm	15mA
0010	-4dBm	12.5mA

Conditions: VDD = 3.0V, VSS = 0V, TA = 27ºC

Tab.3.3 RF output power setting for the OM6239

4.4.1.13 RX/TX control

The RX/TX control is set by PRIM_RX bit in the CONFIG register and sets the OM6239 chip in transmit/receive mode.

4.5 Protocol Engine

Protocol engine is a packet based data link layer that features automatic packet assembly and timing, automatic acknowledgement and retransmissions of packets. Protocol engine enables the implementation of ultralow power and high performance communication. The Protocol engine features enable significant improvements of power efficiency for bi-directional and unidirectional systems, without adding complexity on the host controller side.

4.6 Features

The main features of Protocol engine are:

- 1 to 32 bytes dynamic payload length
- Automatic packet handling
- Automatic packet transaction handling
  - Auto Acknowledgement with payload
  - Auto retransmit
- 3 data pipe for 1:3 star networks

4.7 Protocol engine overview

Protocol engine uses self-defined protocol for automatic packet handling and timing. During transmit, Protocol engine assembles the packet and clocks the bits in the data packet for transmission. During receive, Protocol engine constantly searches for a valid address in the demodulated signal. When Protocol engine finds a valid address, it processes the rest of the packet and validates it by CRC. If the packet is valid the payload is moved into a vacant slot in the RX FIFOs. All high speed bit handling and timing is controlled by protocol engine.

Protocol engine features automatic packet transaction handling for the easy implementation of a reliable bi-directional data link. A protocol engine packet transaction is a packet exchange between two transceivers, with one transceiver acting as the Primary Receiver (PRX) and the other transceiver acting as the Primary Transmitter (PTX). A protocol engine packet transaction is always initiated by a

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packet transmission from the PTX, the transaction is complete when the PTX has received an acknowledgment packet (ACK packet) from the PRX. The PRX can attach user data to the ACK packet enabling a bi-directional data link.

The automatic packet transaction handling works as follows:

1. You begin the transaction by transmitting a data packet from the PTX to the PRX. Protocol engine automatically sets the PTX in receive mode to wait for the ACK packet.
2. If the packet is received by the PRX, Protocol engine automatically assembles and transmits an acknowledgment packet (ACK packet) to the PTX before returning to receive mode.
3. If the PTX does not receive the ACK packet immediately, Protocol engine automatically retransmits the original data packet after a programmable delay and sets the PTX in receive mode to wait for the ACK packet.

In Protocol engine it is possible to configure parameters such as the maximum number of retransmits and the delay from one transmission to the next retransmission. All automatic handling is done without the involvement of the MCU.

4.8 Protocol engine packet format

The format of the Protocol engine packet is described in this section. The Protocol engine packet contains a preamble field, address field, packet control field, payload field and a CRC field. Figure3.3 shows the packet format with MSB to the left.

Preamble 1 byte	Address 4-5 byte	2byte guard	Packet control field 9 bit	Payload 0-32 bytes	CRC 1-2 bytes

Figure3.3 A Protocol engine packet with payload (0-32 bytes)

4.9 Preamble

The preamble is a bit sequence used to synchronize the receivers demodulator to the incoming bit stream. The preamble is one byte long and is either 01010101 or 10101010. If the first bit in the address is 1 the preamble is automatically set to 10101010 and if the first bit is 0 the preamble is automatically set to 01010101. This is done to ensure there are enough transitions in the preamble to stabilize the receiver.

4.10 Address

This is the address for the receiver. An address ensures that the packet is detected and received by the correct receiver, preventing accidental cross talk between multiple OM6228 systems. You can configure the address field width in the AW register to be 5 bytes or 4 bytes address.

4.11 Guard

Figure 3.3 shows the format of the 2bytesguard packet has better synchronous characteristics.

4.12 Packet Control Field (PCF)

Figure 3.4 shows the format of the 9 bit packet control field, MSB to the left.

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Payload length 6bit	PID 2bit	NO_ACK 1bit

Figure 3.4 Packet control field (PCF)

The packet control field contains a 6 bit payload length field, a 2 bit PID (Packet Identity) field and a 1 bit NO_ACK flag.

Payload length

This 6 bit field specifies the length of the payload in bytes. The length of the payload can be from 0 to 32 bytes.

Coding: 000000 = 0 byte (only used in empty ACK packets, The 0 length packet also need to be read out use R_RX_PAYLOAD with no data following) 100000 = 32 byte, 100001 = Don't care

This field is only used if the Dynamic Payload Length function is enabled.

PID (Packet identification)

The 2 bit PID field is used to detect if the received packet is new or retransmitted. PID prevents the PRX operation from presenting the same payload more than once to the MCU. The PID field is incremented at the TX side for each new packet received through the SPI. The PID and CRC field are used by the PRX operation to determine if a packet is retransmitted or new. When several data packets are lost on the link, the PID fields may become equal to the last received PID. If a packet has the same PID as the previous packet, the RF transceiver compares the CRC sums from both packets. If the CRC sums are also equal, the last received packet is considered a copy of the previously received packet and discarded.

No Acknowledgment flag (NO_ACK)

The Selective Auto Acknowledgement feature controls the NO_ACK flag.

This flag is only used when the auto acknowledgement feature is used. Setting the flag high, tells the receiver that the packet is not to be auto acknowledged.

On the PTX you can set the NO_ACK flag bit in the Packet Control Field with this command:
W_TX_PAYLOAD_NOACK

However, the function must first be enabled in the FEATURE register by setting the EN_DYN_ACK bit. When you use this option, the PTX goes directly to standby-I mode after transmitting the packet. The PRX does not transmit an ACK packet when it receives the packet.

4.13 Payload

The payload is the user defined content of the packet. It can be 0 to 32 bytes wide and is transmitted on-air when it is uploaded to the device.

Protocol engine provides two alternatives for handling payload lengths; static and dynamic.

The default is static payload length. With static payload length all packets between a transmitter and a receiver have the same length.

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Static payload length is set by the RX_PW_Px registers on the receiver side. The payload length on the transmitter side is set by the number of bytes clocked into the TX_FIFO and must equal the value in the RX_PW_Px register on the receiver side.

Dynamic Payload Length (DPL) is an alternative to static payload length. DPL enables the transmitter to send packets with variable payload length to the receiver. This means that for a system with different payload lengths it is not necessary to scale the packet length to the longest payload.

With the DPL feature the OM6239 can decode the payload length of the received packet automatically instead of using the RX_PW_Px registers. The MCU can read the length of the received payload by using the R_RX_PL_WID command.

> ***Note:*** Always check if the packet width reported is 32 bytes or shorter when using the R_RX_PL_WID command. If its width is longer than 32 bytes then the packet contains errors and must be discarded. Discard the packet by using the Flush_RX command.

In order to enable DPL the EN_DPL bit in the FEATURE register must be enabled. In RX mode the DYNPD register must be set. A PTX that transmits to a PRX with DPL enabled must have the DPL_P0 bit in DYNPD set.

4.14 CRC (Cyclic Redundancy Check)

The CRC is the error detection mechanism in the packet. It may either be 1 or 2 bytes and is calculated over the address, Packet Control Field and Payload.

The polynomial for 1 byte CRC is $X^8 + X^2 + X + 1$. Initial value is 0Xff.

The polynomial for 2 byte CRC is $X^{16} + X^{12} + X^5 + 1$. Initial value is 0Xffff.

The number of bytes in the CRC is set by the CRCO bit in the CONFIG register. No packet is accepted by protocol engine if the CRC fails.

4.15 Automatic packet assembly

The automatic packet assembly assembles the preamble, address, packet control field, payload and CRC to make a complete packet before it is transmitted.

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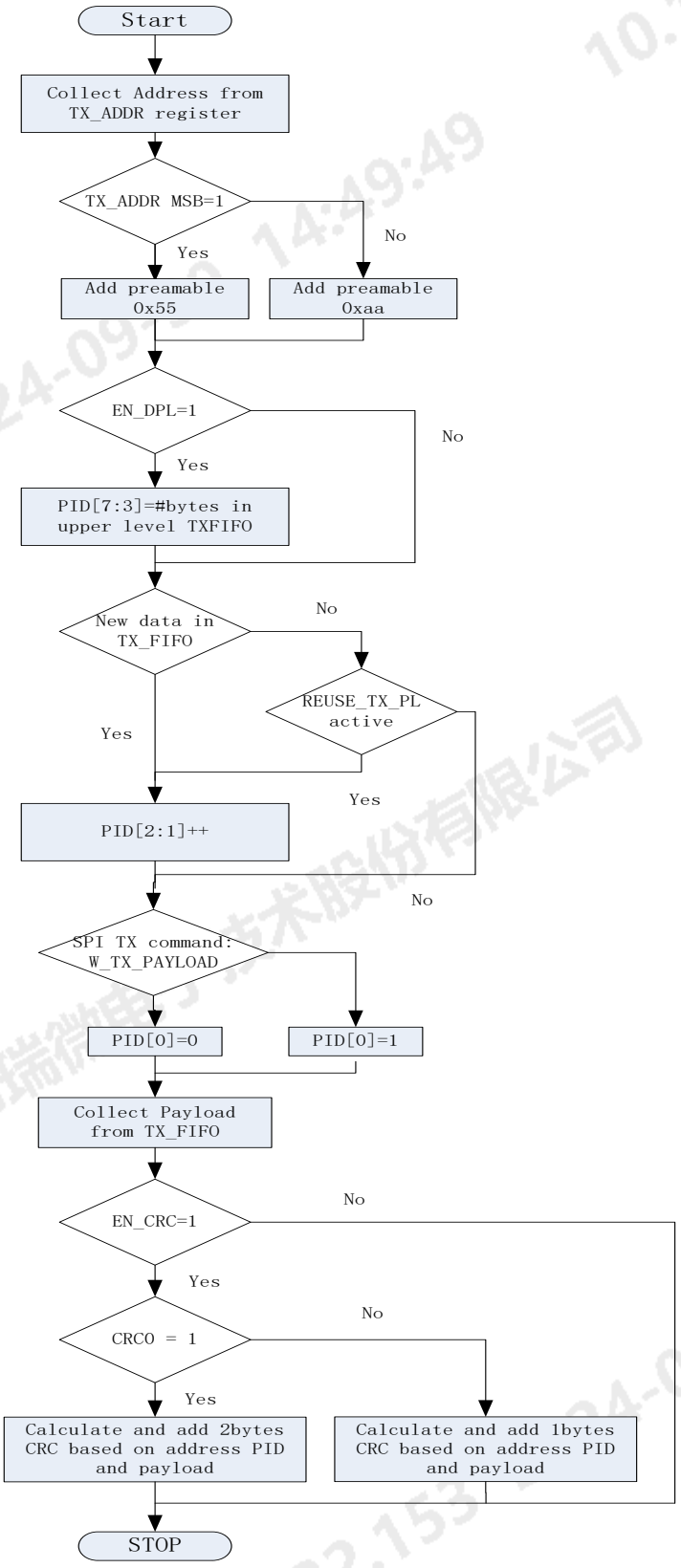


Figure 3.5 Automatic packet assembly

4.16 Automatic packet disassembly

After the packet is validated, Protocol engine disassembles the packet and loads the payload into the RX FIFO, and asserts the RX_DR IRQ.

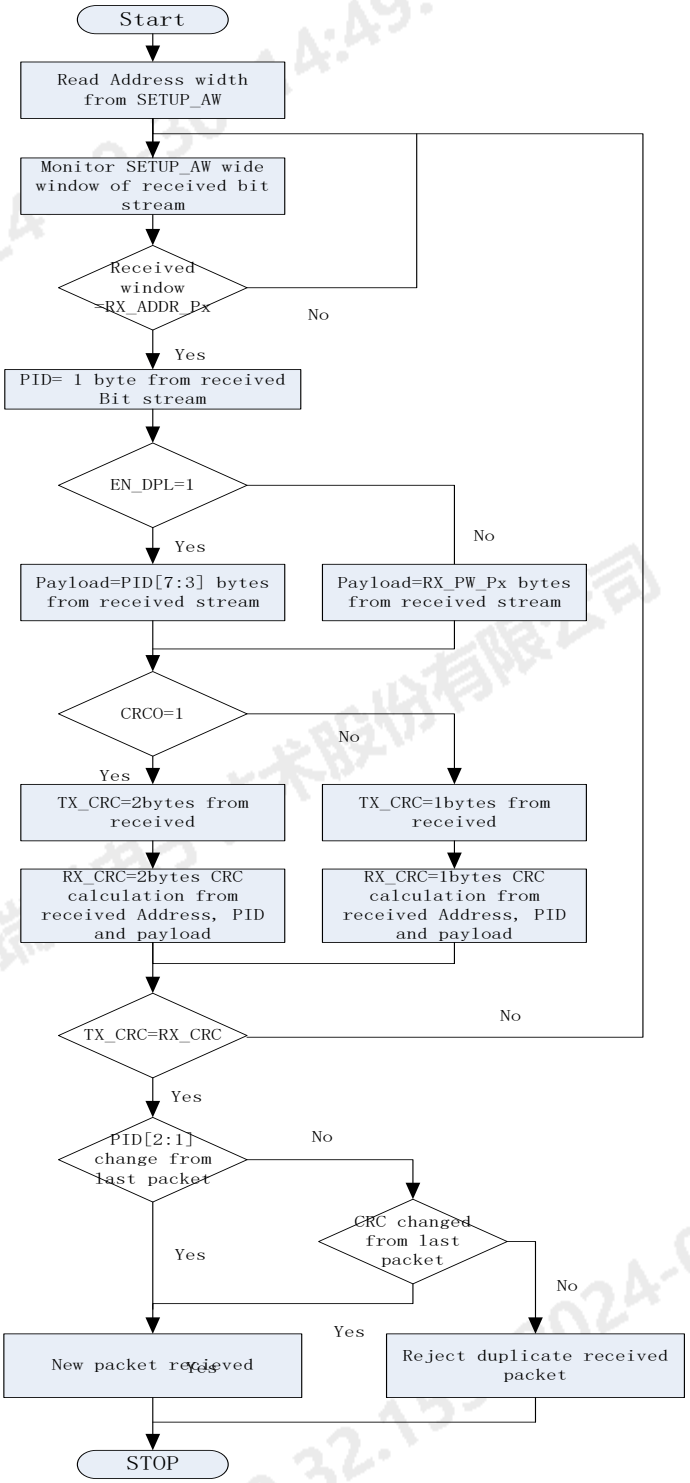


Figure 3.6Automatic packet disassembly

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4.17 Automatic packet transaction handling

Protocol engine features two functions for automatic packet transaction handling; auto acknowledgement and auto re-transmit.

4.18 Auto Acknowledgement

Auto Acknowledgment is a function that automatically transmits an ACK packet to the PTX after it has received and validated a packet. The Auto Acknowledgement function reduces the load of the system MCU and reduces average current consumption. The Auto Acknowledgement feature is enabled by setting the EN_AA register.

Note: If the received packet has the NO_ACK flag set, auto acknowledgement is not executed.

An ACK packet can contain an optional payload from PRX to PTX. In order to use this feature, the Dynamic Payload Length (DPL) feature must be enabled. The MCU on the PRX side has to upload the payload by clocking it into the TX FIFO by using the W_ACK_PAYLOAD command. The payload is pending in the TX FIFO (PRX) until a new packet is received from the PTX. The RF transceiver can have three ACK packet payloads pending in the TX FIFO (PRX) at the same time.

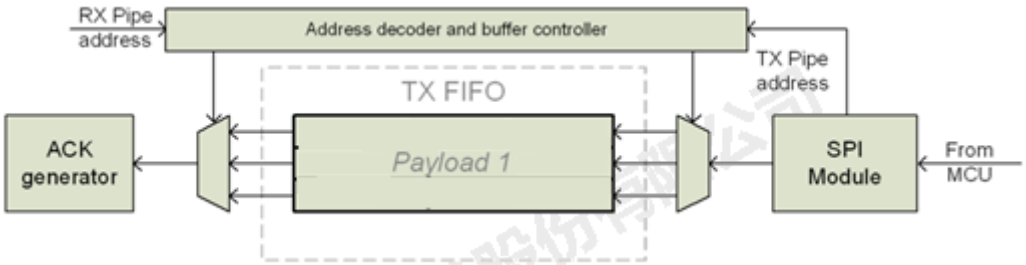


Fig. 3.7TX FIFO (PRX) with pending payloads

Figure 3.7shows how the TX FIFO (PRX) is operated when handling pending ACK packet payloads. From the MCU the payload is clocked in with the W_ACK_PAYLOAD command. The address decoder and buffer controller ensure that the payload is stored in a vacant slot in the TX FIFO (PRX). When a packet is received, the address decoder and buffer controller are notified with the PTX address. This ensures that the right payload is presented to the ACK generator.

If the TX FIFO (PRX) contains more than one payload to a PTX, payloads are handled using the first in–first out principle. The TX FIFO (PRX) is blocked if all pending payloads are addressed to a PTX where the link is lost. In this case, the MCU can flush the TX FIFO (PRX) by using the FLUSH_TX command.

In order to enable Auto Acknowledgement with payload the EN_ACK_PAY bit in the FEATURE register must be set.

4.19 Auto Retransmission (ART)

The auto retransmission is a function that retransmits a packet if an ACK packet is not received. It is used in an Auto Acknowledgement system on the PTX. When a packet is not acknowledged, you can set the number of times it is allowed to retransmit by setting the ARC bits in the SETUP_RETR register. PTX enters RX mode and waits a time period for an ACK packet each time a packet is transmitted. The amount of time the PTX is in RX mode is based on the following conditions:

- Auto Retransmit Delay (ARD) has elapsed.

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- No address match within 256μs.
- After received packet (CRC correct or not) if address match within 256μs.

The RF transceiver asserts the TX_DS IRQ when the ACK packet is received.

The RF transceiver enters standby-I mode if there is no more un-transmitted data in the TX FIFO and the CE pin is low. If the ACK packet is not received, the RF transceiver goes back to TX mode after a delay defined by ARD and retransmits the data. This continues until acknowledgment is received, or the maximum number of retransmits is reached.

Two packet loss counters are incremented each time a packet is lost, ARC_CNT and PLOS_CNT in the OBSERVE_TX register. The ARC_CNT counts the number of retransmissions for the current transaction. You reset ARC_CNT by initiating a new transaction. The PLOS_CNT counts the total number of retransmissions since the last channel change. You reset PLOS_CNT by writing to the RF_CH register. It is possible to use the information in the OBSERVE_TX register to make an overall assessment of the channel quality.

The ARD defines the time from the end of a transmitted packet to when a retransmit starts on the PTX. ARD is set in SETUP_RETR register in steps of 256μs. A retransmit is made if no ACK packet is received by the PTX.

There is a restriction on the length of ARD when using ACK packets with payload. The ARD time must never be shorter than the sum of the startup time and the time on-air for the ACK packet.

- For 2 Mbps data rate and 5-byte address; 15 byte is maximum ACK packet payload length for ARD=256μs (reset value).
- For 1 Mbps data rate and 5-byte address; 5 byte is maximum ACK packet payload length for ARD=256μs (reset value).

ARD=512μs is long enough for any ACK payload length in 1 or 2 Mbps mode.

- For 500kbps data rate and 5-byte address the following values apply:

ARD	ACK packet size (in byte)
1536us	All ACK payload sizes
1280us	<=24
1024us	<=16
768us	<=8
512us	Empty ACK with no payload

Table 3.4 Maximum ACK payload length for different retransmit delays

As an alternative to Auto Retransmit it is possible to manually set the RF transceiver to retransmit a packet a number of times. This is done by the REUSE_TX_PL command. The MCU must initiate each transmission of the packet with a pulse on the CE pin when this command is used.

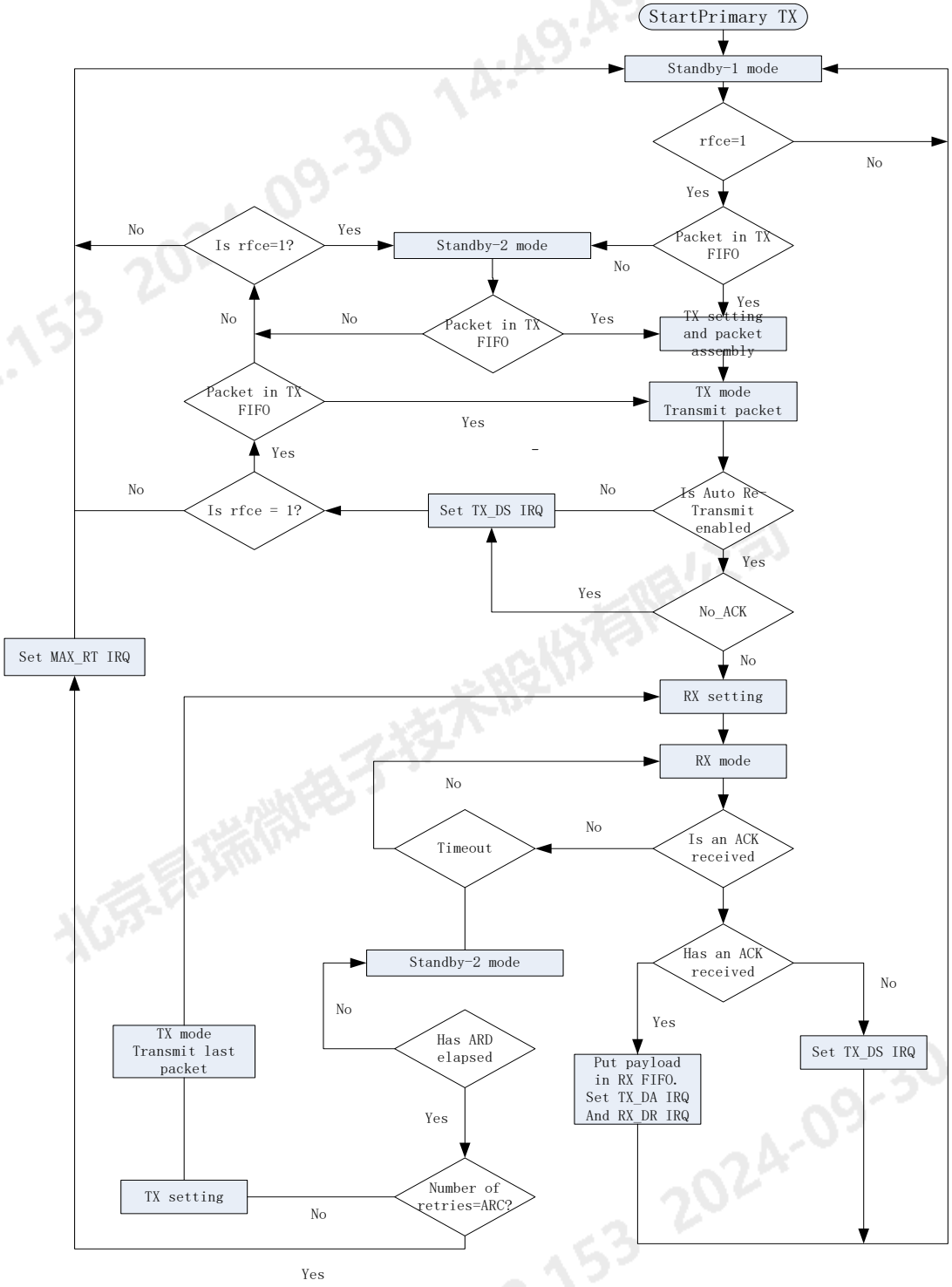
4.20 Protocol engine flowcharts

This section contains flowcharts outlining PTX and PRX operation in Protocol engine.

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4.21 PTX operation

The flowchart in Figure 3.8 outlines how a RF transceiver configured as a PTX behaves after entering standby-I mode.



Note:Protocol engine operation is outlined with a dashed square.

Figure 3.8 PTX operations in Protocol engine

Activate PTX mode by setting the rfce in the RFCON register high. If there is a packet present in the TX FIFO the RF transceiver enters TX mode and transmits the packet. If Auto Retransmit is enabled, the state machine checks if the NO_ACK flag is set. If it is not set,

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the RF transceiver enters RX mode to receive an ACK packet. If the received ACK packet is empty, only the TX_DS IRQ is asserted. If the ACK packet contains a payload, both TX_DS IRQ and RX_DR IRQ are asserted simultaneously before the RF transceiver returns to standby-I mode.

If the ACK packet is not received before timeout occurs, the RF transceiver returns to standby-II mode. It stays in standby-II mode until the ARD has elapsed. If the number of retransmits has not reached the ARC, the RF transceiver enters TX mode and transmits the last packet once more.

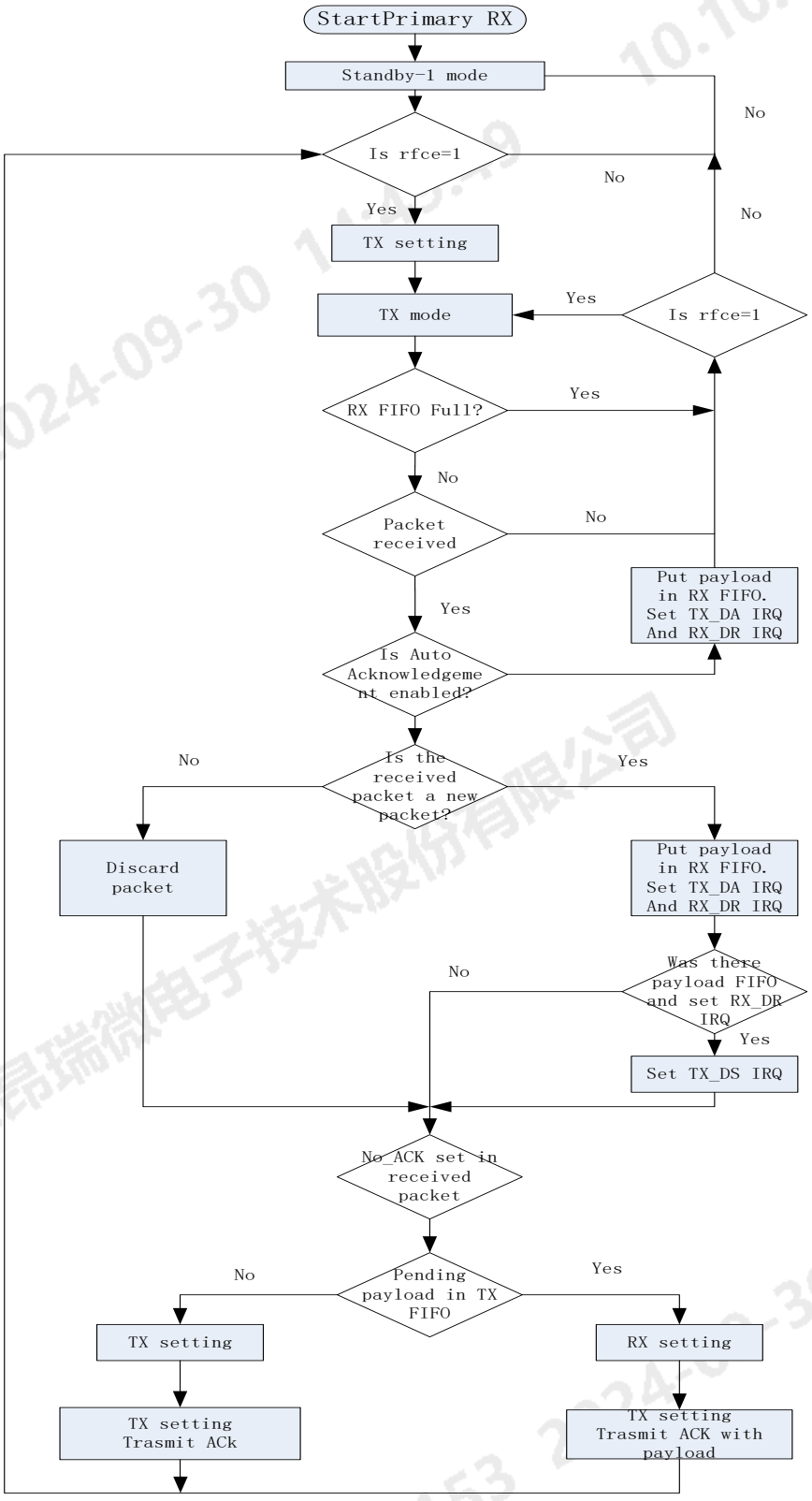
While executing the Auto Retransmit feature, the number of retransmits can reach the maximum number defined in ARC. If this happens, the RF transceiver asserts the MAX_RT IRQ and returns to standby-I mode.

If the rfce bit in the RFCON register is high and the TX FIFO is empty, the RF transceiver enters Standby-II mode.

4.22 PRX operation

The flow chart in Figure 3.9 outlines how a RF transceiver configured as a PRX behaves after entering standby-I mode.

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Note: Protocol engine operation is outlined with a dashed square.

Figure3.9 PRX operations in Protocol engine

Activate PRX mode by setting the rfce bit in the RFCON register high. The RF transceiver enters RX mode and starts searching for packets. If a packet is received and Auto Acknowledgement is enabled, the RF transceiver decides if the packet is new or a copy of a

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previously received packet. If the packet is new payload is made available in the RX FIFO and the RX_DR IRQ is asserted. If the last received packet from the transmitter is acknowledged with an ACK packet with payload, the TX_DS IRQ indicates that the PTX received the ACK packet with payload. If the No_ACK flag is not set in the received packet, the PRX enters TX mode. If there is a pending payload in the TX FIFO it is attached to the ACK packet. After the ACK packet is transmitted, the RF transceiver returns to RX mode.

A copy of a previously received packet might be received if the ACK packet is lost. In this case, the PRX discards the received packet and transmits an ACK packet before it returns to RX mode.

4.23 MultiSlave

MultiSlave is a feature used in RX mode that contains a set of three parallel data pipes with unique addresses. A data pipe is a logical channel in the physical RF channel. Each data pipe has its own physical address (data pipe address) decoding in the RF transceiver.

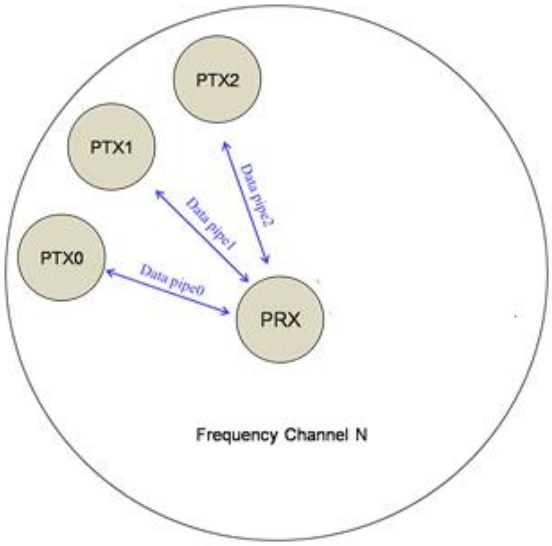


Fig.3.10 PRX using Multi-Slave

The RF transceiver configured as PRX (primary receiver) can receive data addressed to three different data pipes in one frequency channel as shown in Figure 3.10. Each data pipe has its own unique address and can be configured for individual behavior.

Up to six RF transceivers configured as PTX can communicate with one RF transceiver configured as PRX. All data pipe addresses are searched for simultaneously. Only one data pipe can receive a packet at a time. All data pipes can perform Protocol engine functionality. The following settings are common to all data pipes:

- CRC enabled/disabled (CRC always enabled when Protocol engine is enabled)
- CRC encoding scheme
- RX address width
- Frequency channel
- Air data rate
- LNA gain

The data pipes are enabled with the bits in the EN_RXADDR register. By default only data pipe 0 and 1 are enabled. Each data pipe address is configured in the RX_ADDR_PX registers.

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Note: Always ensure that none of the data pipes have the same address.

Each pipe can have up to a 5 byte configurable address. Data pipes 0-2 share the four most significant address bytes. The LSByte must be unique for all six pipes. Figure 3.11 is an example of how data pipes 0-2 are addressed. Only pipe0 can have up to a 5 byte configurable address, other's pipes have 1bytes configurable address.

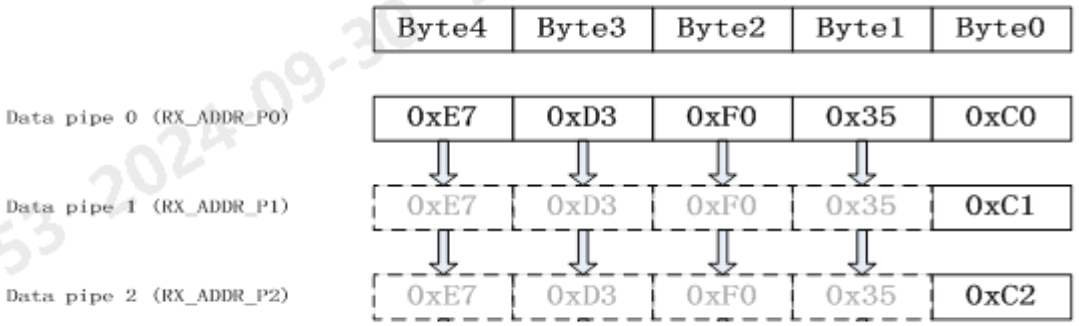


Figure 3.11 Addressing data pipes 0-2

The PRX, using Multi-Slave and Protocol engine, receives packets from more than one PTX. To ensure that the ACK packet from the PRX is transmitted to the correct PTX, the PRX takes the data pipe address where it received the packet and uses it as the TX address when transmitting the ACK packet. Figure 3.12 is an example of an address configuration for the PRX and PTX. On the PRX the RX_ADDR_Px, defined as the pipe address, must be unique. On the PTX the TX_ADDR must be the same as the RX_ADDR_P0 and as the pipe address for the designated pipe.

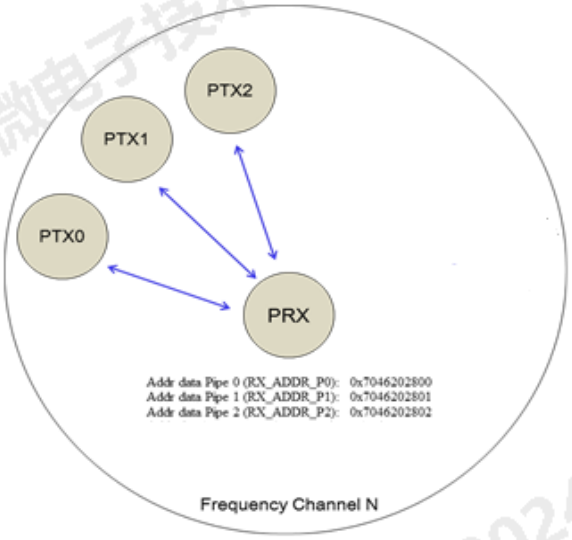


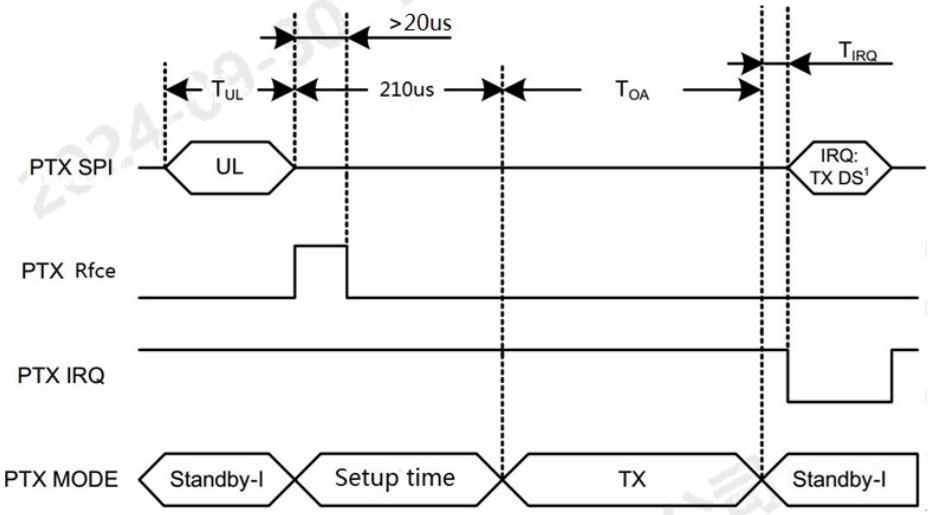
Figure 3.12 Example of data pipe addressing in Multi-Slave

Only when a data pipe receives a complete packet can other data pipes begin to receive data. When multiple PTXs are transmitting to a PRX, the ARD can be used to skew the auto retransmission so that they only block each other once.

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4.24 Protocol engine timing

This section describes the timing sequence of Protocol engine and how all modes are initiated and operated. The Protocol engine timing is controlled through the Data and Control interface. The RF transceiver can be set to static modes or autonomous modes where the internal state machine controls the events. Each autonomous mode/sequence ends with a RFIRQ interrupt. All the interrupts are indicated as IRQ events in the timing diagrams.



1 IRQ if No Ack is on.

T_{IRQ}=3us @ 1Mbps, @2Mbps

Figure 3.13Transmitting one packet with NO_ACK on

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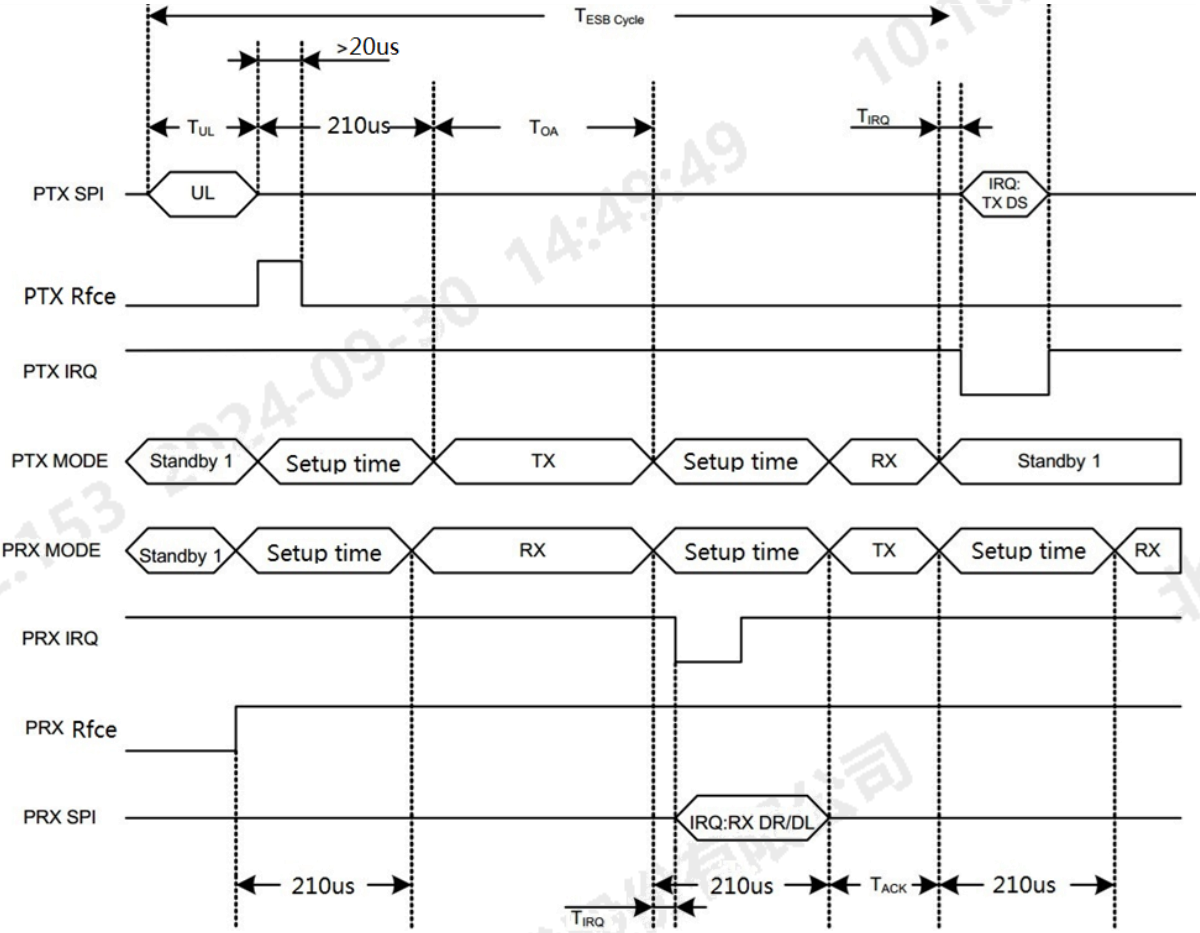


Figure 3.14 Timing of Protocol engine for one packet upload(2Mbps)

In Figure 3.14, the transmission and acknowledgement of a packet is shown. The PRX operation activates RX mode (rfce=1), and the PTX operation is activated in TX mode (rfce=1 for minimum 20µs). After 210µs the transmission starts and finishes after the elapse of T_{OA}.

When the transmission ends the PTX operation automatically switches to RX mode to wait for the ACK packet from the PRX operation. When the PRX operation receives the packet it sets the interrupt for the host MCU and switches to TX mode to send an ACK. After the PTX operation receives the ACK packet it sets the interrupt to the MCU and clears the packet from the TX FIFO.

In Figure 3.15, the PTX timing of a packet transmission is shown when the first ACK packet is lost.

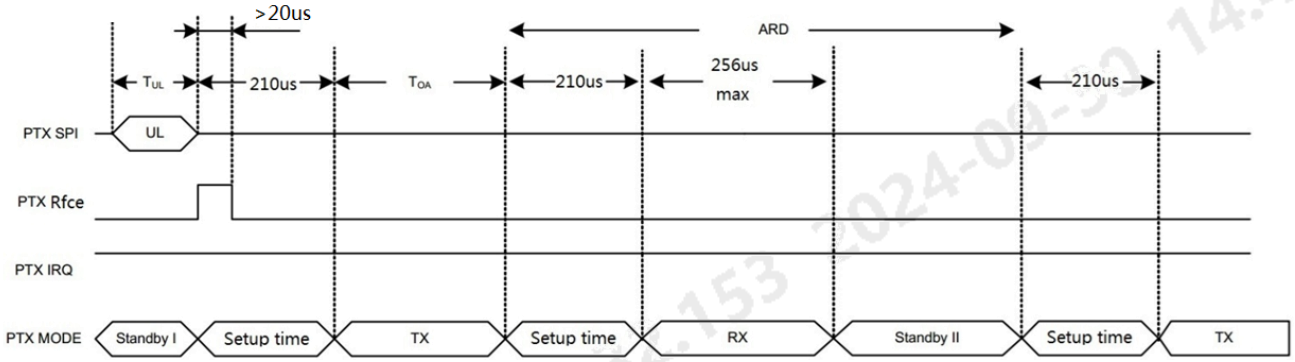


Figure 3.15 Timing of Protocol engine when the first ACK packet is lost (2 Mbps)

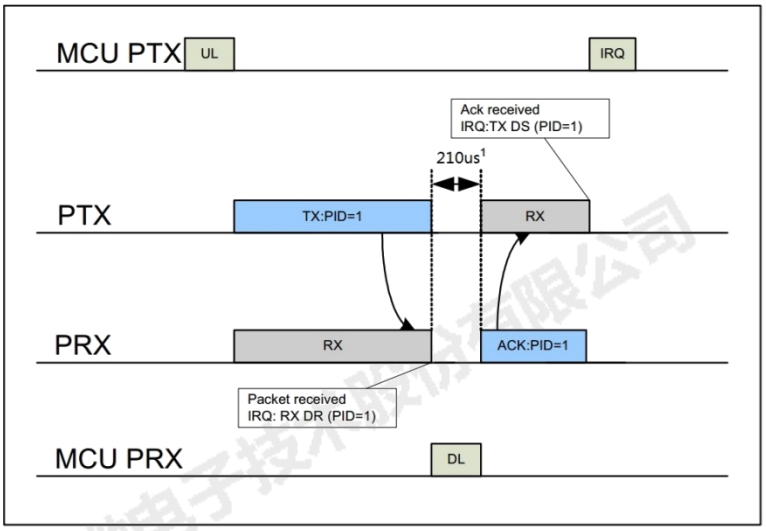
4.25 Protocol engine transaction diagram

This section describes several scenarios for the Protocol engine automatic transaction handling. The call outs in this section's figures indicate the IRQs and other events. For MCU activity the event may be placed at a different timeframe.

Note: The figures in this section indicate the earliest possible download (DL) of the packet to the MCU and the latest possible upload (UL) of payload to the transmitter.

4.26 Single transaction with ACK packet and interrupts

In Fig.3.16, the basic auto acknowledgement is shown. After the packet is transmitted by the PTX and received by the PRX the ACK packet is transmitted from the PRX to the PTX. The RX_DR IRQ is asserted after the packet is received by the PRX, whereas theTX_DS IRQ is asserted when the packet is acknowledged and the ACK packet is received by the PTX.



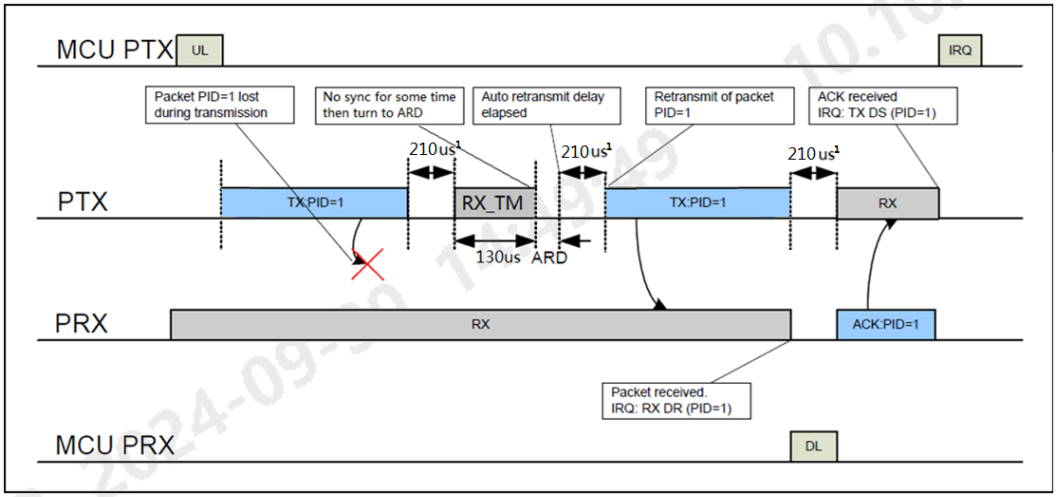
1 Radio Turn Around delay

Fig.3.16TX/RX cycles with ACK and the according interrupts

4.27 Single transaction with a lost packet

Figure 3.17 is a scenario where a retransmission is needed due to loss of the first packet transmits. After the packet is transmitted, the PTX enters RX mode to receive the ACK packet. After the first transmission, the PTX waits a specified time (including setup time, RX_TM and ARD) for the ACK packet, if it is not in the specific time slot the PTX retransmits the packet as shown inFigure 3.17. PTX will turn to RX mode after 210us setup time when packet is transmitted, after 130us RX timeout (RX_TM is RX timeout for PTX, it can be set shorter), then PTX turn to ARD (can be set to 0us, 256us, 512us to 3840us).

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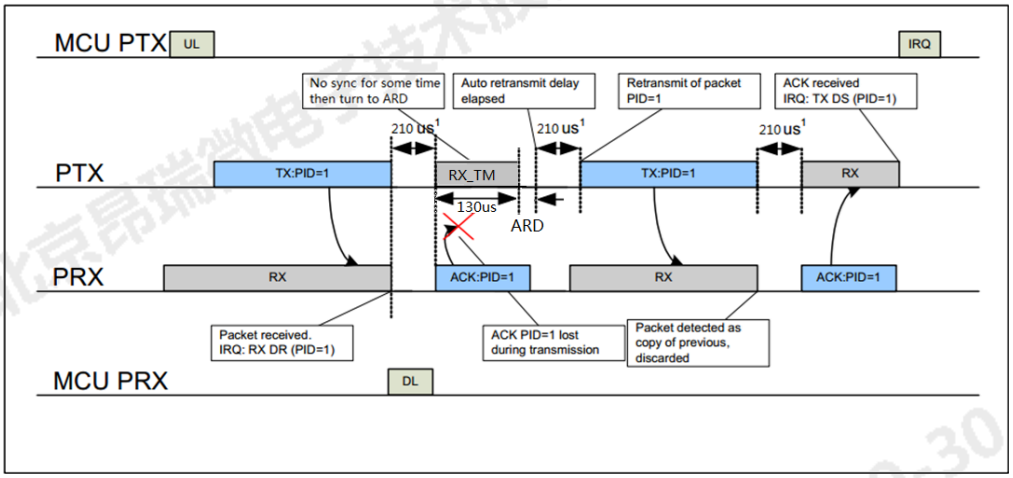
1 Radio Turn Around delay

Figure 3.17TX/RX cycles with ACK and the according interrupts when the first packet transmit fails

When an address is detected the PTX stays in RX mode until the packet is received. When the retransmitted packet is received by the PRX (see Figure 3.17). The RX_DR IRQ is asserted and an ACK is transmitted back to the PTX. When the ACK is received by the PTX, the TX_DS IRQ is asserted.

4.28 Single transaction with a lost ACK packet

Figure 3.18 is a scenario where a retransmission is needed after a loss of the ACK packet. The corresponding interrupts are also indicated.



1 Radio Turn Around delay

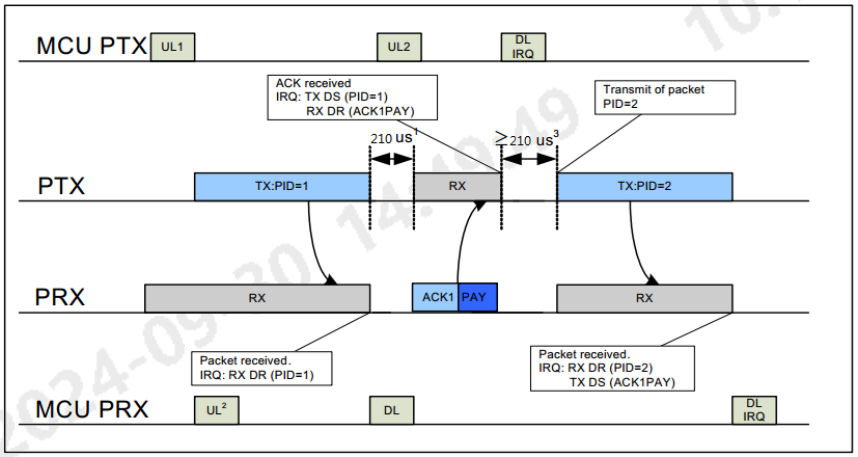
Fig.3.18 TX/RX cycles with ACK and the according interrupts when the ACK packet fails

4.29 Single transaction with ACK payload packet

Figure 3.19 is a scenario of the basic auto acknowledgement with payload. After the packet is transmitted by the PTX and received by the PRX the ACK packet with payload is transmitted from the PRX to the PTX. The RX_DR IRQ is asserted after the packet is received by the PRX, whereas on the PTX side the TX_DS IRQ is asserted when the ACK packet is received by the PTX. On the PRX side, the

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TX_DS IRQ for the ACK packet payload is asserted after a new packet from PTX is received.



1 Radio Turn Around delay

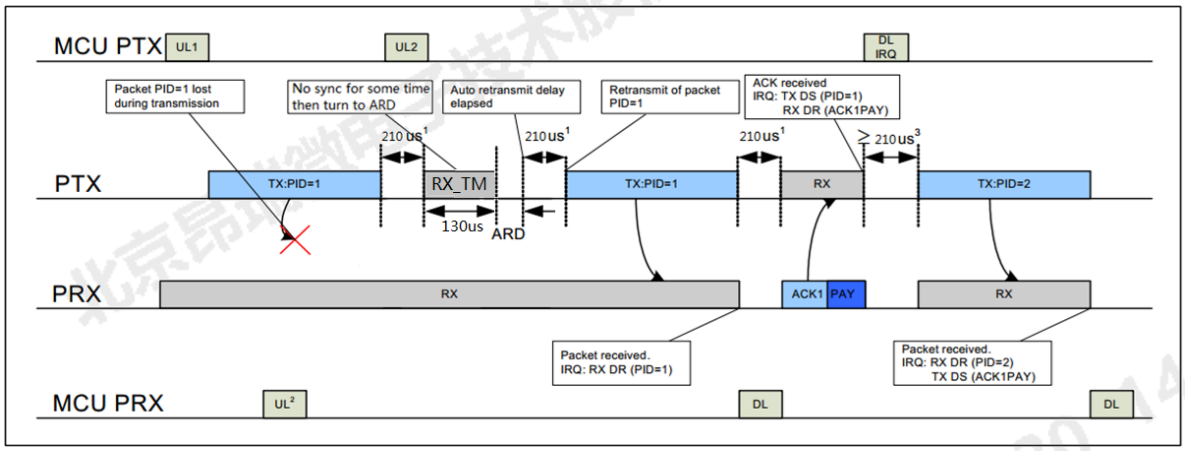
2 Uploading payload for Ack Packet

3 Delay defined by MCU on PTX side, >=210us

Fig.3.19TX/RX cycles with ACK Payload and the according interrupts

4.30 Single transaction with ACK payload packet and lost packet

Figure 3.20 is a scenario where the first packet is lost and a retransmission is needed before the RX_DR IRQ on the PRX side is asserted. For the PTX both the TX_DS and RX_DR IRQ are asserted after the ACK packet is received. After the second packet (PID=2) is received on the PRX side both the RX_DR (PID=2) and TX_DS (ACK packet payload) IRQ are asserted.



1 Radio Turn Around delay

2 Uploading payload for Ack Packet

3 Delay defined by MCU on PTX side, >=210us

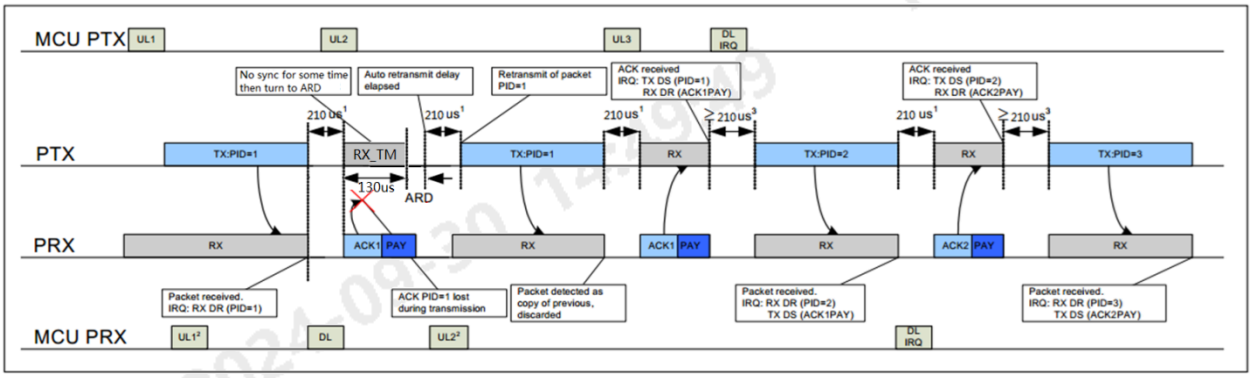
Fig.3.20TX/RX cycles and the according interrupt when the packet transmission fails

4.31 Two transactions with ACK payload packet and the first ACK packet lost

In Fig 3.21 the ACK packet is lost and a retransmission is needed before the TX_DS IRQ is asserted, but the RX_DR IRQ is asserted immediately. The retransmission of the packet (PID=1) results in a discarded packet. For the PTX both the TX_DS and RX_DR IRQ are

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asserted after the second transmission of ACK, which is received. After the second packet (PID=2) is received on the PRX both the RX_DR (PID=2) and TX_DS (ACK1PAY) IRQ is asserted. The callouts explains the different events and interrupts.



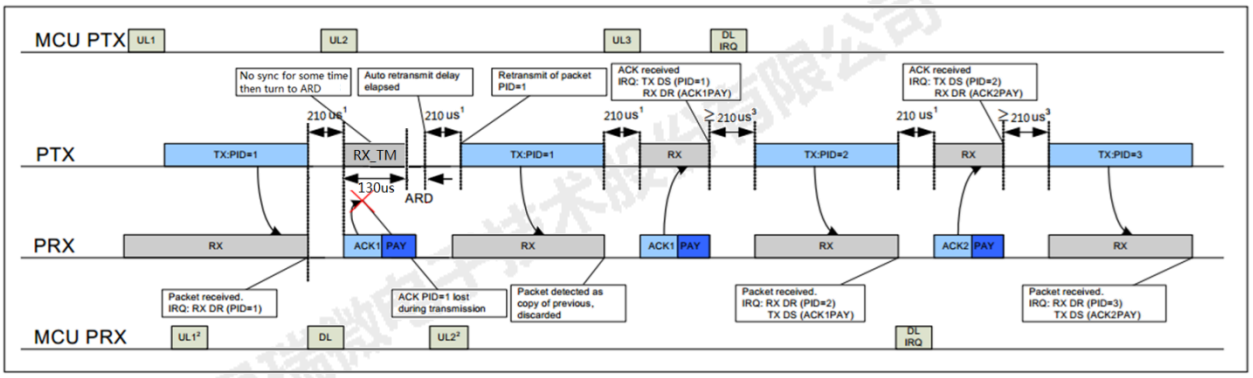
1 Radio Turn Around delay

2 Uploading payload for Ack Packet

3 Delay defined by MCU on PTX side, >=210us

Fig.3.21 TX/RX cycles with ACK Payload and the according interrupts when the ACK packet fails

4.32 Two transactions where max retransmissions is reached



1 Radio Turn Around delay

2 Uploading payload for Ack Packet

3 Delay defined by MCU on PTX side, >=210us

Fig 3.22 TX/RX cycles with ACK Payload and the according interrupts when the transmission fails. ARC is set to 2.

MAX_RT IRQ is asserted if the auto-retransmit counter (ARC_CNT) exceeds the programmed maximum limit (ARC). In Fig 3.22, the packet transmission ends with a MAX_RT IRQ. The payload in TX FIFO is NOT removed and the MCU decides the next step in the protocol. A toggle of the rfce bit in the RFCON register starts a new transmitting sequence of the same packet. The payload can be removed from the TX FIFO using the FLUSH_TX command.

4.33 Data and control interface

The data and control interface gives you access to all the features in the RF transceiver. Compared to the standalone component registers are used instead of port pins. Otherwise the interface is identical to the standalone OM6239 chip.

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4.34 Data FIFO

The data FIFOs store transmitted payloads (TX FIFO) or received payloads that are ready to be clocked out (RX FIFO). The FIFOs are accessible in both PTX mode and PRX mode. The following FIFOs are present in the RF transceiver:

- TX three level, 32 bytes FIFO
- RX three level, 32 bytes FIFO

Both FIFOs have a controller and are accessible by using dedicated DMA. A TX FIFO in PRX can store payloads for ACK packets to one PTX device. If the TX FIFO contains more than one payload to a pipe, payloads are handled using the first in - first out principle. The TX FIFO in a PRX is blocked if all pending payloads are addressed to pipes where the link to the PTX is lost. In this case, the MCU can flush the TX FIFO using the FLUSH_TX command.

The RX FIFO in PRX can contain payloads from up to three different PTX devices and a TX FIFO in PTX can have up to three payloads stored.

You can write to the TX FIFO using these three commands; W_TX_PAYLOAD and W_TX_PAYLOAD_NO_ACK in PTX mode and W_ACK_PAYLOAD in PRX mode. All three commands provide access to the TX_PLD register.

The RX FIFO can be read by the command R_RX_PAYLOAD in PTX and PRX mode. This command provides access to the RX_PLD register.

The payload in TX FIFO in a PTX is not removed if the MAX_RT IRQ is asserted.
You can read if the TX and RX FIFO are full or empty in the FIFO_STATUS register.

4.35 Interrupt

The RF transceiver can send interrupts to the MCU. The interrupt (RFIRQ) is activated when TX_DS, RX_DR or MAX_RT are set high by the state machine in the STATUS register. RFIRQ is deactivated when the MCU writes '1' to the interrupt source bit in the STATUS register. The interrupt mask in the CONFIG register is used to select the IRQ sources that are allowed to activate RFIRQ. By setting one of the mask bits high, the corresponding interrupt source is disabled. By defaults all interrupt sources are enabled.

Note: The 3 bits pipe information in the STATUS register is updated during the RFIRQ high to low transition. The pipe information is unreliable if the STATUS register is read during a RFIRQ high to low transition.

4.36 Register map

You can configure and control the radio by accessing the register map.

4.37 Register map table

All undefined bits in the table below are redundant. They are read out as '0'.
Below is the BLE core address map:

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Address Range	Description
0x4200 0000 to 0x4200 FFFF	BLE core register space

4.38 BLE Core register map

Offset address	Name	Access	Reset value	Description
Base address: 0x42000000				
0x00000	REG_INT	R	0x00000000	
0X00004	REG_INT_RAW	RW	0x00000000	
0X00008	REG_INT_MASK	RW	0x00000000	
0X0000C	REG_CMD	RW	0x00000000	
0X00010	REG_ADDR	RW	0x00000000	
0X00014	REG_RFCE	RW	0x00000000	
0X00018	REG_RX_PW	R	0x00000000	
0X0001C	Reserved	RW	0x00000000	
0X00020	REG_DFIFO	RW	0x00000001	
0X00024	REG_CEPULSE	RW	0x00000000	
0X00028	REG_EVT_ABORT	RW	0x00000000	
0X0002C	Reserved	RW	0x00000000	
0X00030	Reserved	RW	0x00000000	
0X00034	Reserved	RW	0x00000000	
0X00038	Reserved	RW	0x00000000	
0X0003C	Reserved	RW	0x00000000	
0X00040	CONFIG	RW	0x0000008D	
0X00044	EN_AA	RW	0x00000007	
0X00048	EN_RXADDR	RW	0x0000000B	
0X0004C	SETUP_RETR	RW	0x00000003	
0X00050	RF_CH	RW	0x00000032	
0X00054	RF_SETUP	RW	0x00000050	
0X00058	STATUS	R	0x00000006	
0X0005C	RSSI_CTR	RW	0x00008080	
0X00060	2.4G: RX_ADDR_P0 BLE: ACCESS_ADDR	RW	0x41882046	
0X00064	RX_ADDR_P0_H	RW	0x00000070	
0X00068	RX_ADDR_P1	RW	0x000000C2	
0X0006C	2.4G: RX_ADDR_P2 BLE: BLE_CTL	RW	0x000000C3	
0X00070	2.4G: TX_ADDR BLE: TX_ADVA	RW	0x41882046	

0X00074	2.4G: TX_ADDR_H BLE: TX_ADVA_H	RW	0x00007070	
0X00078	2.4G: RX_PW_P0 BLE: TX_HEADER	RW	0x00000000	
0X0007C	2.4G: RX_PW_P1 BLE: TX_HEADER_1	RW	0x00000000	
0X00080	2.4G: RX_PW_P2 BLE: TX_HEADER_2	RW	0x00000000	
0X00084	FIFO_STATUS	R	0x00000011	
0X00088	DNYPD	RW	0x00000000	
0X0008C	FEATURE	RW	0x00000010	
0X00090	SETUP_V	RW	0x00000680	
0X00094	PRE_GURD	RW	0x00000077	
0X00098	RX_ADVA	RW	0x00000000	
0X0009C	RX_ADVA_H	RW	0x00000000	
0X000A0	RX_HEADER	R	0x00000000	
0X000A4	RX_HEADER_1	R	0x00000000	
0X000A8	RX_HEADER_2	R	0x00000000	
0X000AC	RX_STATUS	R	0x00000000	
0X000B0	RX_STATUS_1	R	0x00000000	
0X000B4	RX_STATUS_2	R	0x00000000	
0X000B8	DFIFO_STATUS	R	0x00000000	
0X000BC	SETUP_DELAY	RW	0x000000A9	
0X000C0	CRC_INIT	RW	0x00555555	
0X000C0	LINE			
0X000C8	PLL_CTRL0	RW	0xe1100040	
0X000CC	PLL_CTRL1	RW	0x00104200	
0X000D0	CAL_CTRL	RW	0x28700820	
0X000D4	STATE	RW	0x00000000	
0X000D8	CHAN	RW	0x00000000	
0X000DC	FDEV	RW	0x00000020	
0X000E0	DAC_RANGE	RW	0x00000000	
0X000E4	CTUNING	RW	0x00000000	
0X000E8	FTUNING	RW	0x00000000	
0X000EC	RX_CTRL	RW	0x23293436	
0X000F0	RX_RF_GAIN_MAX	RW	0x00000052	
0X000F4	FAGC_CTRL_1	RW	0x0905a001	
0X000F8	DOC_DACI	RW	0x00000000	
0X000FC	DOC_DACQ	RW	0x00000000	
0X00100	AGC_CTRL	RW	0x0009b2cf	
0X00104	AGC_GAIN	RW	0x1cebc101	

0X00108	RF_IVGEN	RW	0x0081041f	
0X0010C	TEST_PKDET	RW	0x00010020	
0X00110	AGC_GAIN_LOCK	RW	0x00000000	
0X00114	WHITELIST0	RW	0x00000000	
0X00118	WHITELIST0_H	RW	0x00000000	
0X0011C	WHITELIST1	RW	0x00000000	
0X00120	WHITELIST1_H	RW	0x00000000	
0X00124	WHITELIST_CR	RW	0x00000000	
0X00128	OBSERVE_TX	R	0x00000000	
0x0012C	RX_CFO_EST	R	0x00000000	
0x00134	W_PLL_LCK_2	RW	0x00003228	

Register Definition

REG_INT, 0x0

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reversed							RF_CMD_DONE
7'h0							1'b0
R							RW

Description of word:

Bit	Value	Symbol	Description	
31:1	0	Reserved		
0	1'b0	Rf_cmd_done	Indicate if rf_command is done	
			0	Rf_command is not done
			1	RF_command_is done

REG_INT_RAW, 0x04

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reversed							RF_RAW_INTR_STATUS
7'h0							1'b0
R							RW

Description of word:

Bit	Value	Symbol	Description	
31:1	0	Reserved		
0	1'b0	Rf_raw_intr_status	RF command done interruption (internal interrupt value before mask)	
			0	
			1	Set when the command is complete. Writing 1 to clear the interrupt Status

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REG_INT_MASK, 0x08

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reversed							RF_RAW_INTR_STATUS
7'h0							1'b0
R							RW

Description of word:

Bit	Value	Symbol	Description	
31:1	0	Reserved		
0	1'b0	Rf_raw_intr_status	RF command done interruption (internal interrupt value before mask)	
			0	
			1	Set when the command is complete. Writing 1 to clear the interrupt Status

REG_CMD, 0xC

Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
DATA_BYTES							
8'h0							
RW							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
Reversed							
8'h0							
R							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
COMMAND							
8'h0							
RW							

Description of word

Bit	Value	Symbol	Description
23:16	8'h0	Data_bytes	This field specifies the number of data bytes to transfer. Valid values are 1-63.
15:8	8'h0	Reserved	
7:0	8'h0	Command	This field is self-clearing after the transition has completed.

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			0x61	R_RX_PAYLOAD-Read RX-payload. A read operation always starts at byte 0. Payload is deleted from FIFO after it is read. Used in RX mode.
			0xA0	W_TX_PAYLOAD-Write TX-payload. A write operation always starts at byte 0 used in TX payload.
			0xE1	FLUSH_TX -Flush TX FIFO
			0xE2	FLUSH_RX -Flush RX FIFO
			0xE3	REUSE_TX_PL-Used for a PTX operation; Reuse last transmitted payload. TX payload reuse is active until W_TX_PAYLOAD or FLUSH TX is executed. TX payload reuse must not be activated or deactivated during package transmission.
			1010_1PPP	W_ACK_PAYLOAD-Used in RX mode. Write Payload to be transmitted together with ACK packet on PIPE PPP. (PPP valid in the range from 000 to 101). Maximum three ACK packet payloads can be pending. Payloads with same PPP are handled using first in - first out principle. Write payload. A write operation always starts at byte 0.
			0xB0	W_TX_PAYLOAD_NO_ACK-Used in TX mode. Disables AUTOACK on this specific packet specific packet.

REG_ADDR, 0x10

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
ADDRESS							
8'h0							
RW							
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
ADDRESS							
8'h0							
RW							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
ADDRESS							
8'h0							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
ADDRESS							
8'h0							
RW							

Description of word

Bit	Value	Symbol	Description
31:0	32'h0	ADDRESS	This register holds the system memory address for data transfer. It is incremented by 1 as data is read from system memory (in the case of a Write command), or as data is written to system memory (in the case of a Read command).

REG_RFCE, 0x14

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reversed							RFCE
7'h0							1'b0
R							RW

Description of word

Bit	Value	Symbol	Description	
31:1	0	Reserved		
0	1'b0	Rfce	Rf ce	
			0	Ce low
			1	Ce high

REG_RX_PW, 0x18

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reversed		RX_PAYLOAD_LENGTH					
2'h0		6'h0					
		R					

Description of word

Bit	Value	Symbol	Description
31:6	26'h0	Reserved	
5:0	6'h0	Rx_payload_length	Rx payload length

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REG_DFIFO, 0x20

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reversed							Dfifo_en
7'h0							1'b1
R							RW

Description of word

Bit	Value	Symbol	Description	
31:1	0	Reserved		
0	1'b1	Dfifo_en	Dfifo enable	
			0	1 64x8 FIFO
			1	3 32x8 FIFOs

REG_CEPULSE, 0x24

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reversed							RFCE_PULSE
7'h0							1'b0
R							RW

Description of word

Bit	Value	Symbol	Description
31:1	0	Reserved	
0	1'b0	Rfce_pulse	Soft set, hardware clear

REG_EVT_ABORT, 0x28

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reversed							EVENT_ABORT_EN
7'h0							1'b0
R							RW

Description of word

.

Bit	Value	Symbol	Description
31:1	0	Reserved	
0	1'b0	Event_abort_en	BLE connection abort enable

CONFIG, 0x40

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
MASK_EVT_DS/GURD_EN	MASK_RX_DR	MASK_TX_DS	MASK_SYNC_DS	EN_CRC	MASK_RX_TIMEOUT_INT/MASK_RSSI_INT	Reversed	PRIM_RX
1	0	0	0	1	1	0	1
RW	RW	RW	W	RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description	
7	1	BLE:MASK_EVT_DS 2.4G:GURD_EN	BLE:Mask event interrupt by Event_Done 2.4G:guard phase enable	
			0	BLE:Reflect Event_Done as active interrupt on the IRQ 2.4G: disable guard phase
			1	BLE:Interrupt not reflected on the IRQ 2.4G: enable guard phase
6	0	MASK_RX_DR	Mask interrupt caused by RX_DR	
			0	Reflect RX_DR as active interrupt on the IRQ
			1	Interrupt not reflected on the IRQ
5	0	MASK_TX_DS	Mask interrupt caused by TX_DS	
			0	Reflect TX_DS as active interrupt on the IRQ
			1	Interrupt not reflected on the IRQ
4	0	BLE:MASK_SYNC_DS 2,4G:MASK_MAX_TR	BLE: Mask interrupt caused by sync detector 2.4G:Mask interrupt caused by max transmit	
			0	BLE:Reflect SYNC_DS as active interrupt on the IRQ 2.4G:reflect MAX_TR as active interrupt on the IRQ
			1	BLE:Interrupt not reflected on the IRQ 2.4G:interrupt not reflected on the IRQ
3	1	EN_CRC	Enable CRC. Forced high if one of the bits in the EN_AA is high	
			0	Disable CRC
			1	Enable CRC
2	1	BLE:MASK_RX_TIMEOUT	BLE:Mask interrupt caused by RX TIMEOUT 2.4G: Mask interrupt caused by RSSI_INT	

		_INT 2.4G:MASK_RSSI_INT	0	Reflect RX_TIMEOUT_INT as active interrupt on the IRQ
			1	Interrupt not reflected on the IRQ
1	0	Reserved		
0	1	PRIM_RX	RX/TX control	
			0	PTX
			1	PRX

EN_AA, 0x44

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reversed					ENAA_P2	ENAA_P1	ENAA_P0
0					1	1	1
RW					RW	RW	RW

Description of Word

Bit	Value	Symbol	Description	
7:3	0	Reserved		
2	1	ENAA_P2	Enable auto acknowledgement data pipe 2	
			0	Disable
			1	Enable
1	1	ENAA_P1	Enable auto acknowledgement data pipe 1	
			0	Disable
			1	Enable
0	1	ENAA_P0	Enable auto acknowledgement data pipe 0	
			0	Disable
			1	Enable

EN_RXADDR, 0x48

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reversed				scramble_en	ERX_P2	ERX_P1	ERX_P0
0				1	0	1	1
RW				RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description
7:4	0	Reserved	

3	1	scramble_en	Scramble enable	
			0	Disable
			1	Enable
2	0	ERX_P2	Enable data pipe 2	
			0	Disable
			1	Enable
1	1	ERX_P1	Enable data pipe 1	
			0	Disable
			1	Enable
0	1	ERX_P0	Enable data pipe 0	
			0	Disable
			1	Enable

SETUP_RETR, 0x4C

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
ARD				ARC			
4’b0				4’b11			
RW				RW			

Description of Word

Bit	Value	Symbol	Description	
7:4	4’b0	ARD	Auto Retransmit Delay	
			4’hf	Wait 4000μS
			…	…
			4’h1	Wait 500μS
			4’h0	Wait 250μS
3:0	4’b11	ARC	Auto Retransmit Count	
			4’hf	Up to 15 Re-Transmit on fail of AA
			…	…
			4’h1	Up to 1 Re-Transmit on fail of AA
			4’h0	Re-Transmit disabled

RF_CH (RW), 0x50

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reg_Rf_ch							
0x32							
RW							

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Description of Word

Bit	Value	Symbol	Description
7:0	2	Reg_Rf_ch	Sets the frequency channel

RF_SETUP (RW), 0x54

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
CONT_WAVE	PA_PWR[3]	CAL_DONE	CAL_EN	RF_DR_HIGH	Pa_power		
0	1	0	1	0	3'b00		
RW	RW	RW	RW	RW	RW		

Description of Word

Bit	Value	Symbol	Description		
7	0	CONT_WAVE	Enables continuous carrier transmit when high		
			0	Disable	
			1	Enable	
6	1	PA_PWR[3]	PA power select bit 3		
5	0	CAL_DONE	Flag for calibration finishing, after CAL_EN set to 1, CAL_DONE will be reset to 0 0 calibration on 1 calibration done		
4	0	CAL_EN	Calibration enable, 1 for enable; 0 for disable		
3	1	RF_DR_HIGH	Data rate select, 1 for 2Mbps; 0 for 1Mbps, BLE mode should be set to 0		
2:0	3'b000	PA_PWR[2:0]	PA power control, PA_PWR[3:0] with pa_voltage of RF_IVGEN in bank1		
			PA_PWR[3:0]	Pa_voltage(bank1 of RF_IVGEN)	
			1111	0	Output 8 dbm,
			1000	0	Output 5 dbm
			0111	1	Output 4dbm,
			0011	0	Output 0 dbm,
			0001	0	Output -6 dbm
			0001	1	Output -12 dbm
			0000	0	Output -16 dbm
			0000	1	Output -43 dbm

STATUS (RW), 0x58

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
EVENT_DONE	RX_DR	TX_DS	SYNC_DS	RX_TIMEOUT_INT/RSS	RX_P_NO		TX_FULL

.

				I_INT		
0	0	0	0	0	2'b11	0
R	RW	RW	RW	R	R	R

Description of Word

Bit	Value	Symbol	Description	
7	0	EVENT_DONE	Indicate the event is done	
6	0	RX_DR	Data Ready RX FIFO interrupt. Asserted when new data arrives RX FIFO Write 1 to clear bit.	
5	0	TX_DS	Data Sent TX FIFO interrupt. Asserted when packet transmitted on TX. Write 1 to clear bit.	
4	0	SYNC_DS	RX sync detector interrupt, Write 1 to clear bit. If SYNC_DS is asserted it must be cleared to enable further communication.	
3	0	RX_TIMEOUT_INT/RSSI_INT	If ble_en (register FEATURE[6]) is 1, the bit is RX_TIMEOUT_INT, otherwise it is RSSI_INT. cleared after reading.	
2:1	2'b11	RX_P_NO	Data pipe number for the payload available for reading from RX_FIFO	
			11	RX FIFO Empty
			00-10	Data Pipe Number
0	0	TX_FULL	TX FIFO full flag	
			0	Available locations in TX FIFO
			1	TX FIFO full

RSSI_CTR/RPD, 0x5C

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
rssi_est_real_time	rssi_clk_en	rssi_cap_mode	reserved	rssi_smp_num[1:0]		rssi_smp_interval[1:0]	
1	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
rssi_th[7:0]							
8'h80							
RW							

Description of Word

Bit	Value	Symbol	Description
15	1	rssi_est_real_time	RSSI estimation mode 0: real-time output RSSI 1: get maximum RSSI in history
14	0	rssi_clk_en	RSSI module clock enable
13	0	rssi_cap_mode	RSSI capture mode 0: sync mode

			1: threshold mode
12	0	Reserved	
11:10	0	rssi_smp_num	RSSI sampling number 00: 8 01: 16 10: 32 11: 64
9:8	0	rssi_smp_interval	RSSI sampling interval 00: 10us 01: 20us 10: 30us 11: 40us
7:0	8'h80	rssi_th	write: RSSI threshold read: RSSI value

BLE_ACCESS_ADDR (RX_ADDR_P0), 0x60

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
BLE_ACCESS_ADDR							
8'h41							
RW							
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
BLE_ACCESS_ADDR							
8'h88							
RW							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
BLE_ACCESS_ADDR							
8'h20							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
BLE_ACCESS_ADDR							
8'h46							
RW							

Description of Word

Bit	Value	Symbol	Description
31:0	32'h418 82046	BLE_ACCESS_ ADDR	BLE access address

.

Note: 0x40000060 register is ACCESS_ADDR in BLE mode, while in 2.4G mode, it's RX_ADDR_P0.

RX_ADDR_P0_H, 0x64

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
RX_ADDR_P0_H							
8'h70							
RW							

Description of Word

Bit	Value	Symbol	Description
7:0	8'h70	RX_ADDR_P0_H	RX_ADDR_P0_H，2.4G mode

RX_ADDR_P1, 0x68

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
RX_ADDR_P1							
8'hC2							
RW							

Description of Word

Bit	Value	Symbol	Description
7:0	8'hC2	RX_ADDR_P1	RX_ADDR_P1

RX_ADDR_P2, 0x6C

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
RX_ADDR_P2							
8'hC3							
RW							

Description of Word

Bit	Value	Symbol	Description
7:0	8'hC3	RX_ADDR_P3	RX_ADDR_P3

.

BLE_CTL, 0x6C

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
event_abort	md_dsb	nesn_sn_dsb	infiniterx	FORMAT			
1	1	0	0	4'h3			
RW	RW	RW	RW	RW			

Description of Word

Bit	Value	Symbol	Description
7	1	event_abort	Event abort by software
6	1	md_dsb	Md disable
5	0	nesn_sn_dsb	Nesn and sn disable
4	0	infiniterx	Infinite RX
3:0-	4'h3	FORMAT	0011 for BLE broadcast 0100 for BLE master connection 0101 for BLE slave connection 0110 for BLE active scan mode 1000 for BLE passive scan 1001 for BLE TX test 1011 for BLE RX test 1100 for BLE master connection 1101 for BLE slave connection

TX_ADVA（TX_ADDR）,0x70

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
TX_ADVA							
8'h41							
RW							
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
TX_ADVA							
8'h88							
RW							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
TX_ADVA							
8'h20							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
TX_ADVA							
8'h46							
RW							

.

Description of Word

Bit	Value	Symbol	Description
31:0	32'h41882046	TX_ADVA	BLE device address

Note: 0x40000070 register is TX_ADVA in BLE mode, while in 2.4G mode, it's TX_ADDR.

TX_ADVA_H(TX_ADDR_H), 0x74

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
TX_ADVA_H							
8'h70							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
TX_ADVA_H							
8'h70							
RW							

Description of Word

Bit	Value	Symbol	Description
15:0	16'h7070	TX_ADVA_H	BLE device address_H TX_ADD_H

Note: 0x40000074 register is TX_ADVA_H in BLE mode, while in 2.4G mode, it's TX_ADDR_H.

Note: 0x40000070 register is TX_ADVA in BLE mode, while in 2.4G mode, it's TX_ADDR.

TX HEADER_x(data channel; x=0 or 1 or 2), 0x78\0x7C\0x80

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
LENGTH							
0							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Txaclrfu			TXMD	TXSN	TXNESN	TXLLID	
3'h0			1'b0	1'b0	1'b0	2'b0	
R			RW	RW	RW	RW	

.

Description of Word

Bit	Value	Symbol	Description
15:8	2'b00	length	Payload length
7:5	3'h0	txaclrfu	
4	0	Txmd	More data
3	0	Txsn	Sequence Number
2	2'h0	txnesn	Next Expected Sequence Number
1:0	2'h0	TXLLID	The LLID indicates whether the packet is an LL Data PDU or an LL Control PDU. 00b = Reserved 01b = LL Data PDU: Continuation fragment of an L2CAP message, or an Empty PDU. 10b = LL Data PDU: Start of an L2CAP message or a complete L2CAP message with no fragmentation. 11b = LL Control PDU

TX HEADER_x(adv channel, x=0 or 1 or 2), 0x78\0x7C\0x80

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
length							
0							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Rxadd	Txadd	RFU		TYPE			
0	0	0		0			
RW	RW	RW		RW			

Description of Word

Bit	Value	Symbol	Description
15:8	2'b00	length	Payload length
7	0	Rxadd	0 for device RX address 1 for random address
6	0	Txadd	0 for device TX address 1 for random address
5:4	2'h0	RFU	Reserved
3:0	4'h0	TYPE	0000 ADV_IND 0001 ADV_DIRECT_IND 0010 ADV_NONCONN_IND 0100 SCAN_RSP 0110 ADV_SCAN_IND 0111-1111 RESERVED

.

RX_PW_Px(x=0 or 1 or 2), 0x78\0x7C\0x80

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
8’h0							
RW							

Description of Word

Bit	Value	Symbol	Description
7:0	8’h0	Reg_rx_pw_Px	when dynamic length disable, this field is the length of payload in pipe x

FIFO_STATUS, 0x84

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved	TX_REUSE_PL	TX_FULL	TX_EMPTY	Reserved		RX_FULL	RX_EMPTY
0	0	0	1	0		0	1
RW	R	R	R	RW		R	R

Description of Word

Bit	Value	Symbol	Description	
7	0	Reserved		
6	0	TX_REUSE_PL	TX REUSE flag.	
			1	Tx data reused
			0	Tx data not reused
5	0	TX_FULL	TX FIFO full flag.	
			1	TX FIFO full
			0	Available locations in TX FIFO
4	1	TX_EMPTY	TX FIFO empty flag.	
			1	TX FIFO empty
			0	Data in TX FIFO
3:2	2’b00	Reserved	Only '00' allowed	
			0	Keep the current value
			1	Reset to default values
1	0	RX_FULL	RX FIFO full flag.	
			1	RX FIFO full
			0	Available locations in RX FIFO
0	1	RX_EMPTY	RX FIFO empty flag.	
			1	RX FIFO empty

.

			0	Data in RX FIFO

DYNPD, 0x88

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved	Reserved	Reserved	Xn_en	Reserved	DPL_P2	DPL_P1	DPL_P0
0	0	0	0	0	0	0	0
RW		RW	RW	RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description
7	0	Reserved	
6	0	Reserved	
5	0	Reserved	
4	0	Xn_en	Xn mode enable
3	0	Reserved	
2	0	DPL_P2	Enable dynamic payload length data pipe 2. (Requires EN_DPL and ENAA_P2)
1	0	DPL_P1	Enable dynamic payload length data pipe 1. (Requires EN_DPL and ENAA_P1)
0	0	DPL_P0	Enable dynamic payload length data pipe 0. (Requires EN_DPL and ENAA_P0)

FEATURE, 0x8C

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved	Ble_en	Reserved	BP_GAU	Vco_amp_tx_mux	EN_DPL	EN_ACK_PAY	EN_DYN_ACK
0	0	0	1	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description
7	0	Reserved	
6	0	Ble_en	BLE mode enable
5	0	Reserved	
4	1	BP_GAU	Bypass Gaussian filter, BLE mode should not bypass Gaussian filter
3	0	Vco_amp_tx_mux	Vco_amp_tx_mux=0, vco_amp_ctr[3:0] (VCO amplitude control bit) of RX and TX are both from calibration fsm; Vco_amp_tx_mux=1, vco_amp_ctr[3:0] of RX is from calibration fsm, vco_amp_ctr[3:0] of TX is fixed to 0x0c;
2	0	EN_DPL	Enables Dynamic Payload Length
1	0	EN_ACK_PAY	Enables Payload with ACK

.

0	0	EN_DYN_ACK	Enables the W_TX_PAYLOAD_NOACK command

SETUP_VALUE, 0x90

Bit31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit25	Bit24
TX_WAIT_CNT							
8'h00							
RW							
Bit23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit17	Bit16
REG_LNA_WAIT							
8'h00							
RW							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit8
REG_MBG_WAIT							
8'h06							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit2	Bit 1	Bit 0
RX_TM_CNT							
8'h80							
RW							

Description of Word

Bit	Value	Symbol	Description	
31:24	8'h00	TX_WAIT_CNT	Tx_wait_cnt(Used for continuous transmit.)	
			8'hff	256 cycles
			…	…
			1	2cycles
			0	1cycle
23:16	8'h00	REG_LNA_WAIT	Lna wait counter	
			8'hff	256 cycles
			…	…
			1	2cycles
			0	1cycle
15:8	8'h06	REG_MBG_WAIT	Main bandgap wait counter	
			8'hff	255us
			…	…
			1	1 us
			0	0 us
7:0	8'h80	RX_TM_CNT	Rx timeout counter.	
			8'hff	255*16us
			…	…

.

			1	1 *16 us
			0	0 us

PRE_GURD, 0x94

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
SPARE_REG[7:0]							
0							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
TAIL_CTL			GRD_EN	GRD_CNT			
011			1	4'h7			
RW			RW	RW			

Description of Word

Bit	Value	Symbol	Description	
15:8	0	SPARE_REG	Reserved register	
7:5	3'h3	TAIL_CTL	Number of repeat bit after the CRC	
			7	7 repeat tail
			…	…
			1	1 repeat tail
			0	0 No repeat tail
4	1	GRD_EN	Pre-Guard enable	
3:0	4'h7	GRD_CNT	Number of Pre-Guard bit before preamble	
			4'hf	16 bit pre_guard
			…	…
			1	2 bit pre_guard
			0	1 bit pre_guard

RX_ADDA, 0x98

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
RX_ADDA							
8'h0							
R							
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
RX_ADDA							

.

8’h0							
R							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
RX_ADDA							
8’h0							
R							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
RX_ADDA							
8’h0							
R							

Description of Word

Bit	Value	Symbol	Description
31:0	32’h0	RX_ADDA	Peer device address

RX_ADDA_H, 0x9C

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
RX_ADDA_H							
8’h0							
R							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
RX_ADDA_H							
8’h0							
R							

Description of Word

Bit	Value	Symbol	Description
15:0	16’h0	RX_ADDA_H	Peer device address

RX HEADER_x(BLE, x=0 or 1 or 2), 0xA0\0xA4\0xA8

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
length							
0							
R							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0

.

Rxadd	Txadd	RFU	TYPE
0	0	0	0
R	R	R	R

Description of Word

Bit	Value	Symbol	Description
15:8	8'h00	length	Payload length
7	0	Rxadd	0 for device RX address 1 for random address
6	0	Txadd	0 for device TX address 1 for random address
5:4	2'h0	RFU	Reserved
3:0	4'h0	TYPE	0011 SCAN_REQ 0101 CONNECT_REQ 0111-1111 RESERVED

RX_STATUS_x (BLE, x= 0 or 1 or 2), 0xAC\0xB0\0xB4

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Cs_new_lasempty	New_lastempty	snerr	nesnerr	crcerr	lenerr	typeerr	synerr
0	0	0	0	0	0	0	0
R	R	R	R	R	R	R	R

Description of Word

Bit	Value	Symbol	Description
7	0	Cs_new_lastempty	
6	0	lastempty	Last packet is empty
5	0	snerr	Sn error, if rxsn is not equal with the txnesn, the duplication detect
4	0	nesnerr	Nesn error, if rxnesn equals txsn, then NACK detect.
3	0	crcerr	Crc error
2	0	lenerr	Lenth error. Advertiser: If rxtype is SCAN_REQ,packet's length is not longer than 12 bytes; If rxtype is CONNECT_REQ, packet's length is no longer than 34 bytes Scanner or rf_test The packet's length is no longer than 37 bytes Connect data packet's length is no longer than 31 bytes

1	0	typeerr	Type error. The tx type of device should be matched with the rxtype received Advertiser: Txtype is ADV_IND, rxtype should be SCAN_REQ or CONNECT_REQ； Txtype is ADV_SCAN_IND, rxtype should be SCAN_REQ； Txtype is ADV_DIRECT_IND, rxtype should be CONNECT_REQ； Scanner: Rxtype should be ADV_IND, ADV_DIRE_IND, ADV_SCAN_IND, ADV_NOCON_IND； If device is passive scanner, after transmit scan_req, SCAN_RSP can be received
0	0	synerr	Sync error, when rx time out, syncerr is set.

DFIFO_STATUS(BLE), 0xB8

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Res	Res	Txfifo2_status	Txfifo1_status	Txfifo0_status	Rxfifo2_status	Rxfifo1_status	Rxfifo0_status
0	0	0	0	0	0	0	0

Bit	Value	Symbol	Description	
5	0	Txfifo2_status	TX FIFO2 status	
			1	Full
			0	empty
4	0	Txfifo1_status	TX FIFO1 status	
			1	Full
			0	empty
3	0	Txfifo0_status	TX FIFO 0 Status	
			1	full
			0	empty
2	0	Rxfifo2_status	RX FIFO 2 Status	
			1	full
			0	empty
1	0	Rxfifo1_status	RX FIFO 0 Status	
			1	full
			0	empty
0	0	Rxfifo0_status	RX FIFO 0 Status. when DFIFO_EN=1，RX FIFO0 status indicate the state of the FIFO.	
			1	Full, you have to use the next FIFO
			0	empty

BLE_SETUP_DELAY, 0xBC

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Ble_rx_delay[3:0]				Ble_tx_delay[3:0]			
4'ha				4'h9			
RW				RW			

Description of Word

Bit	Value	Symbol	Description
7:4	4'ha	Ble_rx_delay (rx2tx)	0000 no delay 0001 1us delay 0010 2us delay 0011 3us delay 0100 4us delay 0101 5us delay 0110 6us delay 0111 7us delay 1000 0us delay 1001 -1us delay 1010 -2us delay 1011 -3us delay 1100 -4us delay 1101 -5us delay 1110 -6us delay 1111 -7us delay
3:0	4'h9	Ble_tx_delay (tx to rx)	0000 no delay 0001 1us delay 0010 2us delay 0011 3us delay 0100 4us delay 0101 5us delay 0110 6us delay 0111 7us delay 1000 0us delay 1001 -1us delay 1010 -2us delay 1011 -3us delay 1100 -4us delay 1101 -5us delay 1110 -6us delay 1111 -7us delay

.

BLE_CRC_INIT, 0xC0

Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
Ble_crc_init[23:16]							
8'h55							
RW							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
Ble_crc_init[15:8]							
8'h55							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Ble_crc_init[7:0]							
8'h55							
RW							

Description of Word

Bit	Value	Symbol	Description
23:0	24'h555555	Ble_crc_init	

LINE, 0xC4

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
CID							
R							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
CID							
R							

Description of Word

Bit	Value	Symbol	Description
15:0	Can be configured	CID	Chip ID

PLL_CTL0, 0xC8

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
SDM_EN	Pll_dac_en	PLL_RSTN_	CRY_PD_RE	CRY_PD_M	PA_RAMP_	PLL_FOFFSET_SEL	

.

		PFD	G	N	DOWN_EN		
1	1	1	0	0	0	01	
RW	RW	RW	RW	RW	RW	RW	
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
DAC_CAL-EN_REG	DAC_CAL_EN_MN	DAC_CALMODE_REG	BP_RC_BP	DOC_CAL_EN_REG	DOC_CAL_EN_MN_	DOC_DAC_MN	I_Q_RVS
0	0	0	1	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
PLL_TEST_EN	pd_pkdet_reg	pd_pkdet_mn	PD_PLL_REG	PD_PLL_MN	PLL_TX_EN_REG	PLL_TX_EN_MN	BYPASS_VCO_RES
0	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
SDM_DITH_IN	SDM_DITH_EN	DAC_RANGE_MN	SYNC_DET_DIS	AFC_EN_REG	AFC_EN_MN	CTUNING_MN	FTUNING_MN
0	1	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description	
31	1	SDM_EN	SDM enable	
			1	Enable the SDM
			0	Disable the SDM
30	1	Pll_dac_en	Output DAC_EN to analog	
			0	Output 0
			1	Output 1
29	1	PLL_RSTN_PFD	PLL_RSTN_PFD control	
			1	Set 1
			0	Set 0
28	0	CRY_PD_REG	Control the Power down of Crystal	
			1	Power down the crystal
			0	Power up the crystal
27	0	CRY_PD_MN	Select the source of crystal power down	
			1	CRY_PD = CRY_PD_REG
			0	CRY_PD = CRY_PD_FSM
26	0	PA_RAMP_DOWN_EN		
25:24	01	PLL_FOFFSET_SEL	PLL_FOFFSET_SEL control	
23	0	DAC_CAL_EN	DAC calibration enable	

		_REG	1	Enable the DAC calibration
			0	Disable the DAC calibration
22	0	DAC_CAL_EN_MN	Select the source of DAC_CAL_EN	
			1	DAC_CAL_EN = DAC_CAL_REG
			0	DAC_CAL_EN = DAC_CAL_FSM
21	0	DAC_CALMODE_REG	Control the VCO gain	
			1	Normal gain
			0	5 times of the normal gain
20	1	BP_RC_BP	Bypass RC_BP phase in the FSM	
			1	Bypass RC_BP phase
			0	No bypass RC_BP phase
19	0	DOC_CAL_EN_REG	Enable the DOC calibration	
			1	Enable the DOC calibration
			0	Disable the DOC calibration
18	0	DOC_CAL_EN_MN	Select the source of DOC_CAL_EN source	
			1	DOC_CAL_EN = DOC_CAL_EN_REG
			0	DOC_CAL_EN = DOC_CAL_EN_FSM
17	0	DOC_DAC_MN	Select the source of DOC source	
			1	DOC_DACI = DOC_DACI_REG DOC_DACQ=DOC_DACQ_REG
			0	DOC_DACI = DOC_DACI_FSM DOC_DACQ = DOC_DACQ_FSM
16	0	I_q_rvs	Only 0 allowed	
			0	Keep the current value
			1	Reset to default values
15	0	PLL_TEST_EN	Enable the PLL in test mode	
			1	PLL A_CNT and B_CNT come from register
			0	PLL A_CNT and B_CNT come from SDM
14	0	pd_pkdet_reg	Manual mode of pd_pkdet register	
13	0	pd_pkdet_mn	Manual mode of pd_pkdet enable	
12	0	PD_PLL_REG	PLL power down control	
			1	Power down PLL
			0	Power up PLL
11	0	PD_PLL_MN	Select the source of PLL Power down	
			1	PD_PLL = PD_PLL_REG
			0	PD_PLL = PD_PLL_FSM
10	0	PLL_TX_EN_REG	PLL TX mode	
			1	PLL in TX mode
			0	PLL in RX mode
9	0	PLL_TX_EN_MN	Select the source of PLL_TX_EN	
			1	PLL_TX_EN = PLL_TX_EN_REG
			0	PLL_TX_EN = PLL_TX_EN_FSM

8	0	BYPASS_VCO_RES	Bypass VCO tail resistor for max amplitude	
7	0	SDM_DITH_IN	SDM dither in value	
			1	Value 1
			0	Value 0
6	1	SDM_DITH_EN	SDM dither enable	
			1	Enable the SDM dither
			0	Disable the SDM dither
5	0	DAC_RANG_MN	Select the source of DAC_RANG	
			1	DAC_RANG = DAC_RANG_REG
			0	DAC_RANG = DAC_RANG_FSM
4	0	SYNC_DET_DIS	Disable AGC when sync detected	
3	0	AFC_EN_REG	AFC enable	
			1	Enable the AFC
			0	Disable the AFC
2	0	AFC_EN_MN	Select the source of AFC_EN	
			1	AFC_EN = AFC_EN_REG
			0	AFC_EN = AFC_EN_FSM
1	0	CTUNING_MN	Select the source of CTUNING	
			1	CTUNING = CTUNING_REG
			0	CTUNING = CTUNING_FSM
0	0	FTUNING_MN	Select the source of FTUNING	
			1	FTUNING = FTUNING_REG
			0	FTUNING = FTUNING_FSM

PLL_CTL1, 0xCC

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
Reserve						PA_RAMP_DOWN_CTL	
‘h0						2’b01	
RW						RW	
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
REG_TX_PA_WAIT							
8’h10							
RW							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
Tx_PLL_WAIT							
8’h42							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0

REG_AFC_WAIT
8’h00
RW

Description of Word

Bit	Value	Symbol	Description	
25:24	2’b01	PA_RAMP_DOWN_CTL		
23:16	8’h10	REG_TX_PA_WAIT	The time between power up PA to transmit data	
			8’hff	255 cycle
			…	…
			1	1 cycle
			0	0 cycle
15:8	8’h42	Tx_PLL_WAIT	Pll lock wait time in Tx mode	
			8’hff	255 us
			…	…
			1	1 us
			0	0 us
7:0	8’h00	REG_AFC_WAIT	The time between RC done and AFC start	
			8’hff	255 us
			…	…
			1	1 us
			0	0 us

CAL_CTL, 0xD0

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
Rx_pll_wait							
8’h28							
RW							
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
Reserved	afc_w_sel		Bp_calib	Bp_dc	Bp_dac	Bp_afc	Bp_rc
0	11		0	0	0	0	0
RW	RW		RW	RW	RW	RW	RW
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
bp_vco_ldo	Psudo_rnd	Bp_rx_addr	bp_vco_amp	Bp_cp_diox	Vco_ldo_cal_reg		
0	0	0	0	1	0		
WR	RW	RW	RW	RW	RW	RW	RW
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Vco_ldo_cal_mn	Rc_cal_ctr_mn	Pll_rst_cnt	Rc_cal_ctl_reg				

0	0	1	0
RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description	
31:24	8'h28	Rx_pll_wait	Pll lock wait time in rx mode Force_cal : before every transmit and receive do calibration	
23	0	Reserved		
22:21	11	afc_w_sel	Select the AFC wait reference counter	
			0	512
			1	255
			2	127
			3	63
20	0	Bp_calib	Bypass DC and AFC phase when channel is not changed	
			0	disable
			1	When channel isnot changed, bypass AFC and DC
19	0	Bp_dc	Bypass DC calibration phase	
			0	Disable
			1	Enable
18	0	Bp_dac	Bypass DAC calibration phase	
			0	Disable
			1	Enable
17	0	Bp_afc	Bypass AFC calibration phase	
			0	Disable
			1	Enable
16	0	Bp_rc	Bypass RC calibration phase	
			0	Disable
			1	Enable
15	0	bp_vco_ldo	Bypass VCO_LDO calibration phase	
			0	Disable
			1	Enable
14	0	Psudo_rnd	Transmit Random data	
			0	Disable
			1	Enable
13	0	Bp_rx_addr	Bypass the RX_ADDR phase in the main FSM	
12	0	bp_vco_amp	Bypass VCO amplitude calibration phase	
			0	Disable
			1	Enable
11	1	Bp_cp_diox	Bpassdio	
			0	Disable

			1	Enable
10:8	0	Vco_ldo_cal_reg	Vco_ldomamual set value	
7	0	Vco_ldo_cal_mn	Vcoldo calibration select	
			0	Vco_ldo_cal = vco_ldo_cal_fsm
			1	Vco_ldo_cal = vco_ldo_cal_reg
6	0	Rc_cal_ctr_mn	Rc calibration select	
			0	Rc_cal_ctr = rc_cal_ctr_fsm
			1	Rc_cal_ctr = rc_cal_ctr_reg
5	1	Pll_rst_cnt	Pll_rst_cnt	
			0	Output 0
			1	Output 1
4:0	0	Rc_cal_ctr_reg	Rc calibration register	

STATE, 0xD4

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
Reserved				CAL_ST_CS			
0				0			
RW				R			
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved		STATE_CS					
0		0					
RW		R					

Description of Word

Bit	Value	Symbol	Description
15:12	0	Reserved	
11:8	4'b0	Cal_st_cs	Describe the state of calibration state machine
7:6	0	Reserved	Only 0 allowed
			0 Keep the current value
			1 Reset to default values
5:0	6'b0	State_cs	Describe the state of main state machine

CHAN, 0xD8

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
reseved	Chan_mn	CHAN_FRAC_REG					
0	0	0					

RW	RW	RW					
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
CHAN_FRAC_REG							
0							
RW							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
CHAN_FRAC_REG							CHAN_INT_REG
0							0
RW							RW
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
CHAN_INT_REG							
0							
RW							

Description of Word

Bit	Value	Symbol	Description	
30	1'b0	Chan_mn	Channel int and frac part select	
			0	Come from the calculation from rf_ch
			1	Come from register
29:9	0	CHAN_FRAC_REG	PLL frequency fragment pat	
8:0	0	CHAN_INT_REG	PLL frequency integer pat	

FDEV,0xDC

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved	FDEV						
0	0x20						
R	RW						

Description of Word

Bit	Value	Symbol	Description
7	0	Reserved	
6:0	7'h20	FDEV	The max offset of the frequency

DAC_RANGE,0xE0

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0

.

Reserved	DAC_RANGE_REG
0	0
R	RW

Description of Word

Bit	Value	Symbol	Description	
7:6	0	Reserved	Only 0 allowed	
			0	Keep the current value
			1	Reset to default values
5:0	6'b0	DAC_RANGE_REG	DAC calibration Range control	

CTUNING, 0xE4

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
Reserved			CTUNING_REG_RX				
0			0				
R			RW				
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved			CTUNING_REG_TX				
0			0				
R			RW				

Description of Word

Bit	Value	Symbol	Description	
15:13	0	Reserved	Only 0 allowed	
			0	Keep the current value
			1	Reset to default values
12:8	6'b0	CTUNING_REG_RX	AFC coarse tuning register control	
7:5	0	Reserved	Only 0 allowed	
			0	Keep the current value
			1	Reset to default values
4:0	6'b0	CTUNING_REG_TX	AFC coarse tuning register control	

FTUNING, 0xE8

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
Reserved					FTUNING_REG_RX		

0					0		
R					RW		
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved					FTUNING_REG_TX		
0					0		
R					RW		

Description of Word

Bit	Value	Symbol	Description	
15:11	0	Reserved	Only 0 allowed	
			0	Keep the current value
			1	Reset to default values
10:8	3'b0	FTUNING_REG_RX	AFC fine tuning register control	
7:3	0	Reserved	Only 0 allowed	
			0	Keep the current value
			1	Reset to default values
2:0	3'b0	FTUNING_REG_TX	AFC fine tuning register control	

RX_CTRL, 0xEC

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
Reserved		en_ts_mag	en_ts_ext	en_ts_inner	ts_mag_lim		
2'h0		1	0	0	3'h3		
RW		RW	RW	RW	RW		
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
ts_k			ts_est_dur			ts_en_gain_cond	ts_mode
3'h1			3'h2			1'h0	1'h1
RW			RW			RW	RW
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
sbe_max_th			xcorr_th[6:2]				
3'h1			5'h14				
RW			RW				
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
xcorr_th[1:0]		en_dc_removal	en_sbe	rssi_k		h_idx	
2'h0		1	1	2'h1		2'h2	
RW		RW	RW	RW		RW	

Description of Word

Bit	Value	Symbol	Description
31:30	0	Reserved	
29	1	en_ts_mag	tone suppression magnitude estimation function enable
28	0	en_ts_ext	0:tone_suppression is controlled by en_ts_innner 1:tone_suppression is controlled by fsm
27	0	en_ts_inner	tone suppression enable
26:24	3'h3	ts_mag_lim	tone suppression magnitude limitation 000: 2^-2 001: 2^-3 010: 2^-4 011: 2^-5 100: 2^-6 101: 2^-7 110: 2^-8 111: 2^-9
23:21	3'h1	ts_k	tone suppression coefficient 000: 2^-11 001: 2^-10 010: 2^-9 011: 2^-8 100: 2^-7 101: 2^-6 110: 2^-5 111: 2^-4
20:18	3'h2	ts_est_dur	limit for ts estimation counter 000: 5us 001: 10us 010: 20us 011: 40us others: 80us
17	0	ts_en_gain_cond	TS gain condition function enable 0: active TS at any gain 1: active TS only when maximum gain
16	1	ts_mode	tone suppression mode 0: est & cmp 1: HPF
15:13	3'b001	sbe_max_th	Sync address error tolerance
12:6	7'h50	xcorr_th	Sync threshold
5	1	en_dc_removal	Dc remove enable
4	1	en_sbe	Sbe enable
3:2	1	rssi_k	RSSI estimation coefficient

.

1:0	2'b10	h_idx	modualtion index: when 1: 0.32, when 2: 0.50;

RX_RF_GAIN_MAX, 0XF0

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved	rx_rf_gain_max[6:0]						
0	7'h52						
RW	RW						

Description of Word

Bit	Value	Symbol	Description
6:0	7'h52	rx_rf_gain_max	maximum total RFAGC gain (dB)

FAGC_CTRL_1, 0xF4

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit24
reserved			pkdet_vrefc			pkdet_vref2c	
0			3'h2			1	
RW			RW			RW	
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
reserved	agc_dpd_lo_thr[10:4]						
0	7'h5						
RW	RW						
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
agc_dpd_lo_thr[3:0]				Reserved			VCO_AMP_MN
4'ha				0			0
RW				RW			RW
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
VCO_AMP_CTL				VCO_PKREF_CTL			
4'h0				4'h1			
RW				RW			

Description of Word

Bit	Value	Symbol	Description
31:29	0	reserved	

28:26	3'b010	pkdet_vrefc	Pkdetvrefc value
25:24	2'b01	pkdet_vref2c	Pkdet vref2c value
23	0	reserved	
22:12	11'h05a	agc_dpd_lo_thr	Filter gain low threshold, default 90
11:9	0	reserved	
8	0	VCO_AMP_MN	Manual mode of VCO amplitude calibration
7:4	4'h0	VCO_AMP_CTL	VCO amplitude control
3:0	4'h1	VCO_PKREF_CTL	VCO amplitude calibration reference

DOC_DACI, 0xF8

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved	doc_daci						
0	0						
RW	RW						

Description of Word

Bit	Value	Symbol	Description	
7	0	Reserved	Only 0 allowed	
			0	Keep the current value
			1	Reset to default values
6:0	0	doc_daci	Doc calibration daci value	

DOC_DACQ, 0xFC

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved	doc_dacq						
0	0						
RW	RW						

Description of Word

Bit	Value	Symbol	Description	
7	0	Reserved	Only 0 allowed	
			0	Keep the current value
			1	Reset to default values
6:0	0	doc_dacq	Doc calibration dacq value	

.

AGC_CTRL, 0x100

Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
test_pat_en	adp_samp_mode			agc_samp_mode		agc_dpd_mode	agc_dpd_thr_db[5]
0	0	0	0	1	0	0	1
RW	RW	RW	RW	RW	RW	RW	RW
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
agc_dpd_thr_db[4:0]					agc_dpd_thr[10:8]		
1	0	1	1	0	0	1	0
RW	RW	RW	RW	RW	RW	RW	RW
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
agc_dpd_thr[7:0]							
1	1	0	0	1	1	1	1
RW	RW	RW	RW	RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description
23	0	test_pat_en	Test pattern enable
22:20	0	adp_samp_mode	Number of clock cycles delay between every two apd detection: When 0, 0 cycles When 1, 2 cycles When 2, 4 cycles When 3, 8 cycles When 4, 16 cycles When 5, 32 cycles
19:18	2	agc_samp_mode	The number of shift registers which are required for Dpd calculation. When 0, 2 registers When 1, 4 registers When 2, 8 registers When 3, 16 registers
17	0	agc_dpd_mode	'0': mean of sample amplitudes, '1': max of sample amplitudes;
16:11	6'h36	agc_dpd_thr_db	dB value of agc_dpd_thr minus 3 dB, default 54
10:0	11'h2cf	agc_dpd_thr	Filter gain high threshold, default 719

AGC_GAIN, 0x104

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit24
agc_gain_mn	apd_clr_cnt_th					apd_det_cnt_th[4:3]	
0	0	0	1	1	1	0	0
RW	RW	RW	RW	RW	RW	RW	RW

.

Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
apd_det_cnt_th[2:0]			dpd_clr_cnt_th[5:1]				
1	1	1	0	1	0	1	1
RW	RW	RW	RW	RW	RW	RW	RW
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
dpd_clr_cnt_th [0]	reserved		Agc_apd_state_reg[1:0]		Agc_dpd_state_reg[2:0]		
1	1	0	0	0	0	0	1
RW	RW	RW	RW	RW	RW	RW	RW
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Agc_dpd_state_reg[10:3]							
0	0	0	0	0	0	0	1
RW	RW	RW	RW	RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description	
31	0	agc_gain_mn	Agc output gain is get from software or hardware calculation	
			1	From software
			0	From hardware
30:26	5'h7	apd_clr_cnt_th [4:0]	Analog Gsw+Gsetl time interval	
25:21	5'h7	apd_det_cnt_th [4:0]	Apd detection time interval	
20:15	6'h17	dpd_clr_cnt_th [5:0]	Digital Gsw+Gsetl time interval, and add Dpd detection time interval	
14:13	2'b10	reserved		
12:11	0	Agc_apd_state_reg	When agc_gain_mn is 1, the value read from this register is the Lna gain given by the software. When agc_gain_mn is 0, the value read from this register is the Lna gain calculated from the hardware module	
10:0	3'b101	Agc_dpd_state_reg	When agc_gain_mn is 1, the value read from this register is the filter gain given by the software. When agc_gain_mn is 0, the value read from this register is the filter gain calculated from the hardware module	

RF_IVGEN, 0x108

Bit23	Bit 22	Bit 21	Bit 20	Bit19	Bit 18	Bit 17	Bit16
Calib_rx_reg	Tx_0_1_rvs	Pd_lna_reg	Pd_lna_mn	Pd_pa_reg	Pd_pa_mn	Tia_lobias	
1	0	0	0	0	0	0	1
RW	RW	RW	RW	RW	RW	RW	RW
Bit 15	Bit 14	Bit13	Bit12	Bit 11	Bit 10	Bit 9	Bit 8
Pd_adc_ldo_re	Pd_adc_ldo_m	Pd_mix_reg	Pd_mix_mn	Bm_filter[1:0]		Bm_lna	

g	n						
0	0	0	0	01		00	
RW	RW	RW	RW	RW		RW	
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Pa_voltage	Reserved	Reserved	Xtal_cc				
0	0	0	1	1	1	1	1
RW	RW	RW	RW	RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description	
23	1	Calib_rx_reg	Manual calibration TX/RX select	
			1	Manual calibration for RX
			0	Manual calibration for TX
22	0	Tx_0_1_rvs	TX 0/1 reverse	
			1	Reverse the 0/1 when transmit
			0	Not reverse
21	0	Pd_lna_reg	Pd_lna manual value	
20	0	Pd_lna_mn	Pd_lna manual select	
			1	from register
			0	From FSM
19	0	Pd_pa_reg	Pd_pa manual value	
18	0	Pd_pa_mn	Pd_pa manual select	
			1	from register
			0	From FSM
17:16	1	Tia_lobias	Output to analogue	
15	0	Pd_adc_ldo_reg	Pd_adc_ldo manual value	
14	0	Pd_adc_ldo_mn	Pd_adc_ldo manual select	
			1	from register
			0	From FSM
13	0	Pd_mix_reg	Pd_mix manual value	
12	0	Pd_mix_mn	Pd_mix manual select	
			1	from register
			0	From FSM
11:10	2'b01	Bm_filter	Filter bias current control	
9:8	2'b00	Bm_lna	Output to analogue	
7	0	Pa_voltage	Pa voltage output to analogue	
6	0	Reserved		
5	0	reserved		
4:0	5'b11111	Xtal_cc	Output to analogue	

TEST_PKDET, 0x10C

Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
test_en	Reserved			Reserved		pll_icp_sel	
0	0			0	0	0	1
RW	RW			RW	RW	RW	RW
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
test_mode			test_point_sel1			test_point_sel0	
0	0	0	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
xtal_resc		bm_xtal	Reserved	pll_vdiv2_sel		pd_vco_pkdet_mn	pd_vco_pkdet_reg
0	0	1	0	0	0	0	0
RW	RW	RW	RW	RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description
23	0	test_en	Test mode enable
22:20	0	reserved	
19:18	2'b01	reserved	
17:16	2'b01	pll_icp_sel	Charge pump current control
15:14	3'b000	test_mode	Test mode select
12:10	3'b000	test_point_sel1	Test point select
9:8	2'b00	test_point_sel0	Test point select
7:6	2'b00	xtal_resc	Resistor value control of xtal bias
5	1	bm_xtal	Xtal bias current control
4	0	Reserved	
3:2	2'b00	pll_vdiv2_sel	Div2 dc bias control
1	0	pd_vco_pkdet_mn	Manual mode enable of pd_vco_pkdet
0	0	pd_vco_pkdet_reg	Manual mode register of pd_vco_pkdet

AGC_GAIN_LOCK, 0x110

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
							agc_gain_fsync_lock
0	0	0	0	0	0	0	0

.

RW	RW	RW	RW	RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description
0	0	agc_gain_fsync_lock	0: RF_GAIN will not be locked after sync. 1: RF_GAIN will be locked after sync.

WHITELIST0, 0x114

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
WHITE_LIST_0							
8'h0							
RW							
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
WHITE_LIST_0							
8'h0							
RW							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
WHITE_LIST_0							
8'h0							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
WHITE_LIST_0							
8'h0							
RW							

Description of Word

Bit	Value	Symbol	Description
31:0	32'h0	WHITE_LIST_0	White list

WHITELIST0_H, 0x118

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
WHITE_LIST_0_H							
8'h0							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
WHITE_LIST_0_H							
8'h0							

.

RW

Description of Word

Bit	Value	Symbol	Description
15:0	16'h0	WHITE_LIST_0_H	White list[47:32]

WHITELIST1, 0x11C

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24
WHITE_LIST_1							
8'h0							
RW							
Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
WHITE_LIST_1							
8'h0							
RW							
Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
WHITE_LIST_1							
8'h0							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
WHITE_LIST_1							
8'h0							
RW							

Description of Word

Bit	Value	Symbol	Description
31:0	32'h0	WHITE_LIST_1	White list

WHITELIST1_H, 0x120

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
WHITE_LIST_1_H							
8'h0							
RW							

.

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
WHITE_LIST_1_H							
8'h0							
RW							

Description of Word

Bit	Value	Symbol	Description
15:0	16'h0	WHITE_LIST_1_H	White list1[47:32]

WHITELIST_CR, 0x124

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved							
RW							
0							

Description of Word

Bit	Value	Symbol	Description
7:1	0	Reserved	
0	1'b0	WHITELIST_CR	1: enable white list function

OBSERVE_TX, 0x128

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Plos_cnt				Arc_cnt			
R				R			
4'h0				4'h0			

Description of Word

Bit	Value	Symbol	Description
7:4	4'h0	PLOS_CNT	The PLOS_CNT counts the total number of retransmissions since the last channel change. You reset PLOS_CNT by writing to the RF_CH register
3:0	4'h0	ARC_CNT	The ARC_CNT counts the number of retransmissions for the current transaction. You reset ARC_CNT by initiating a new transaction.

RX_CFO_EST, 0x12C

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
Reserved					rx_cfo_est[10:8]		
0					3'h0		
RW					R		
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
rx_cfo_est[7:0]							
8'h0							
R							

Description of Word

Bit	Value	Symbol	Description
15:11	0	Reserved	
10:0	0	rx_cfo_est	indicate the percentage of carrier frequency offset(CFO) relative to the base rate, <s,1,9>

W_PLL_LCK_2, 0x138

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8
W_tx_pll_lck2							
8'h32							
RW							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
W_rx_pll_lck2							
8'h28							
RW							

Description of Word

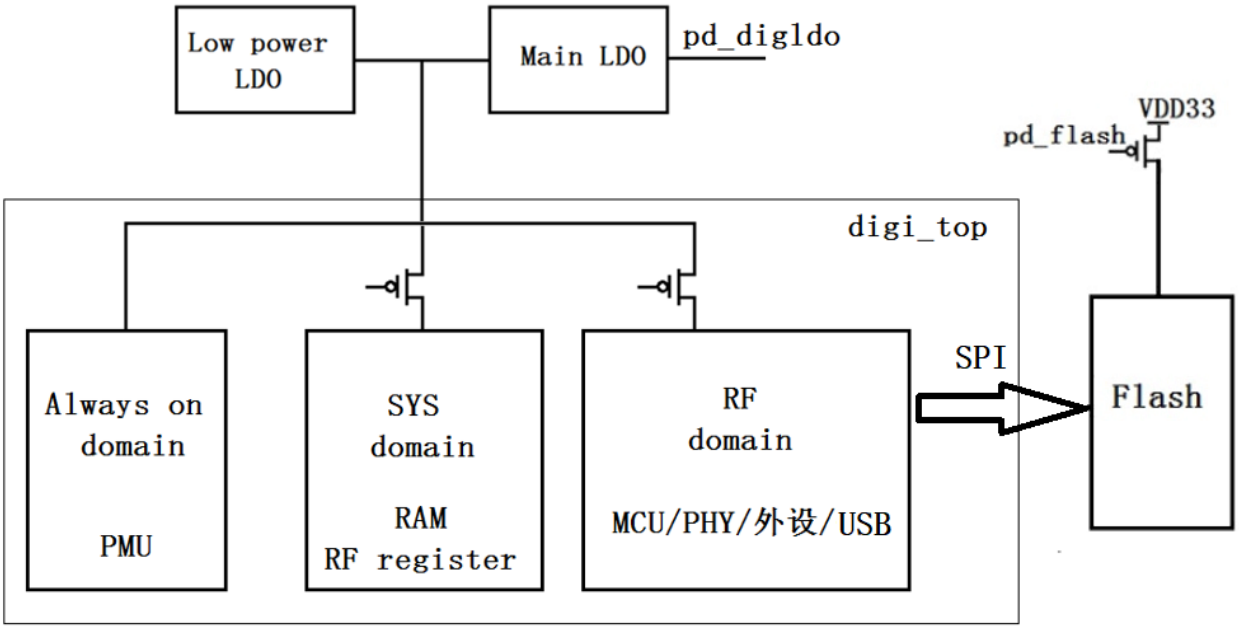
Bit	Value	Symbol	Description
15:8	0	W_tx_pll_lck2	Pll lock wait time in tx mode, bp_calib=1: before every transmit and receive do calibration
7:0	0	W_rx_pll_lck_2	Pll lock wait time in rx mode bp_calib=1: before every transmit and receive do calibration

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5 Power management

5.1 Power domain

There are three power domains in the chip.



Always on domain is always on when the chip is power up. When the chip is in deep sleep mode, all the other two domains include Retention domain and SYS domain are powered down.

When the chip is in register retention and standby mode, Retention domain is kept power up while only SYS domain is powered down.

When the chip is in WFI or active mode, all the three power domains are keep power up.

Mode	Always on domain	SYS domain	RF domain	Main LDO
Active	ON	ON	ON	ON
Sleep	ON	ON	ON	ON
Stop0	ON	ON	ON	ON
Stop1	ON	ON	ON	OFF
Standby	ON	ON	OFF	OFF
Register retention	ON	ON	OFF	OFF
Deep sleep	ON	OFF	OFF	OFF

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5.2 Mode of operation

By default, the microcontroller is in Run mode after a system or a power Reset. Several low power modes are available to save power when the CPU does not need to be kept running, for example when waiting for an external event. It is up to the user to select the mode that gives the best compromise between low-power consumption, short startup time and available wakeup sources.

Mode name	**Entry**	**Wakeup source**	**Effect on clocks**	**Voltage regulators**	
				Main LDO	Low LDO
Sleep (Sleep-now or Sleep-on-exit)	WFI or Return from ISR	Any interrupt	CPU clock OFF no effect on other clocks or analog clock sources	ON	ON
	WFE	Wakeup event		ON	ON
Stop0 mode	LPMS=”001” + SLEEPDEEP bit + WFI or Return from ISR or WFE	WKUP pin edge, RTC event, WDT reset, USB interrupt and RF interrupt	All clocks OFF except XTAL and PLL	ON	ON
Stop1 mode	LPMS=”002” + SLEEPDEEP bit + WFI or Return from ISR or WFE	WKUP pin edge, RTC event, WDT reset	All clocks OFF except XTAL	OFF	ON
Standby	LPMS=”003” + SLEEPDEEP bit + WFI or Return from ISR or WFE	WKUP pin edge, RTC event, WDT reset	All clocks OFF except XTAL	OFF	ON
Register retention	LPMS=”004” + SLEEPDEEP bit + WFI or Return from ISR or WFE	WKUP pin edge	All clocks OFF	OFF	ON
Deep sleep	LPMS=”005” + SLEEPDEEP bit + WFI or Return from ISR or WFE	WKUP pin edge	All clocks OFF	OFF	ON

5.3 Power control Register address map

Below is the power control register address map:

Address Range	Description
0x4000 0000 to 0x4000 00FF	Power control register space

5.4 Power control register map

Offset address	Name	Access	Reset value	Description
Base address: 0x40000000				
0x00000	BOOT_MODE_REG	RW	Boot_pin=0 0x00000000 Boot_pin=1 0x00000001	
0X00004	CLK32K_SEL_REG	RW	0x00000000	
0X00008	CLKLP_CTRL_REG	RW	0x00000001	
0X0000C	ANA_CTRL_REG	RW	0x00000020	
0X00010	LPMS_REG	RW	0x00000000	
0X00014	WKINT_CTRL_REG	RW	0x00000000	
0X00018	PLL_CTRL_REG	RW	0x00000000	
0X0001C	PLL_CTRL1_REG	RW	0x0000002D	
0X00024	DELAY_CTRL_REG	RW	0x00000064	
0X00028	PINMUX_SEL	RW	0x00000000	
0X0002C	RF_DEBUG_MODE	RW	0x00000000	
0X00030	FLASH_PWR_CMD	RW	0x0000ABB9	
0X00038	Reserved	RW	0x00000000	
0X00040	PAD_WK_CTRL_REG	RW	0x00000000	
0x00044	PAD_WK_CTRL1_REG	RW	0x00000000	
0X00048	PAD_BUF_EN_REG	RW	Boot_pin=0 0xFFFF5FFF Boot_pin=1 0xFFFFFFFF	
0X0004c	PAD_PD_CTRL_REG	RW	Boot_pin=0 0xFFFF2FFF Boot_pin=1 0xFFFF7FFF	
0X00050	PAD_PU_CTRL_REG	RW	0x00000000	
0x00054	PAD_PU_CTRL1_REG	RW	0x00000000	

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0X00058	PAD_PAD_DRV_CTRL	RW	0x555555a5	
0x0005c	PAD_PAD_DRV_CTRL1	RW	0x55555555	
0X00060	Reserved	RW	0x00000000	
0X00064	Reserved		0x00000001	
0X00068	Reserved		0x00000000	
0X0006c	SOFT_REV1_REG	RW	0x00000000	
0X00070	SOFT_REV2_REG	RW	0x00000000	
0X00074	SOFT_REV3_REG	RW	0x00000000	
0X00078	SOFT_REV4_REG	RW	0x00000000	
0x0007c	SOFT_REV5_REG	RW	0x00000000	
0x00080	SOFT_REV6_REG	RW	0x00000000	
0x00084	SOFT_REV7_REG	RW	0x00000000	
0x00088	SOFT_REV8_REG	RW	0x00000000	

5.5 Power control register description

field	name	Access	Description
BOOT_MODE_REG			
31:16	vectkey	W	If write this register, vectkey must be 0xAAAA
15:2	Reserved	NA	
1:0	boot_mode	RW	00: boot from flash 01: boot from SRAM 10/11: boot from flash

field	name	Access	Description
CLK32K_SEL_REG			
31:1	Reserved	NA	
0	clk32_sel	RW	0: clk32k is coming from rtc32k 1: clk32k is coming from rtc2m

field	name	Access	Description
CLKLP_CTRL_REG			
31:6	Reserved	NA	
4	Rtc2m_en	RW	0: rtc2m clock disable 1: rtc2m clock enable
3	Reserved	NA	
2	rtc32k_en	RW	0: rtc32k clock disable 1: rtc32k clock enable
1:0	Reserved	NA	

field	name	Access	Description
	ANA_CTRL_REG		
31:6	Reserved	NA	
5:4	Reserved	NA	
3	Reserved	NA	
2	bp_pd_rc2m	RW	0: rc2m power down 1: keep rc2m on
1	bp_pd_ldo	RW	1: Keep main LDO on in deep sleep and register retention mode 0: Switch main LDO to low power LDO in deep sleep and register retention mode(default)
0	bp_xtal_pd	RW	1: Keep XTAL on in deep sleep and register retention mode 0: XTAL power down in deep sleep and register retention mode(default)

field	name	Access	Description
	LPMS_REG		
31:8	Reserved	NA	
7	Reserved	NA	
6	Reserved	RW	
5	sf_pwr_ctrl	RW	0: Flash Power Switch OFF 1: Flash Power Switch ON
4	Reserved	NA	
3	Reserved	NA	
2:0	lpms	RW	Low power mode 3'h000: sleep 3'b001: stop0 mode 3'b010: Stop1 mode 3'b011: Standby 3'b100: Register retention 3'b101: Deep sleep 3'b110: Reserved 3'b111: Reserved

field	name	Access	Description
	WKINT_CTRL_REG		
31:10	Reserved	NA	
9:1	wakeup_source	R	9'b000000001: PIN wake up 9'b000000010: RTC wake up 9'b000000110: RTC tick wake up 9'b000001010: RTC alarm wake up 9'b000010010: RTC alarm1 wake up

			9'b000100000: Watchdog wake up 9'b010000000: encoder wake up 9'b1000000000: seser scan wake up
0	wk_int_clr	W	Write 1 to clear pin wake up interrupt

field	name	Access	Description
	PLL_CTRL_REG		
31:1	Reserved	NA	
1	Pll_clk_rdy	R	0: pll_clk is not ready 1: pll_clk is ready
0	pll_en	RW	0: PLL disable 1: PLL enable

field	name	Access	Description
	PLL_CTRL1_REG		
31:6	Reserved	NA	
5	pll_48m_rstn	RW	
4:2	pll_48m_icp	RW	
1	pd_pll_48m_mn	RW	
0	pd_pll_48m_reg	RW	

field	name	Access	Description
	PINMUX_SEL		
31:16	KEY	W	Must be 16'h5a5a when write this register
15:6	Reserved		
5:4	Ledc_sel	RW	00: select P0.1 01: select P0.5 10: select P2.1 11: select P3.3
1:0	Uart_sel	RW	00: select P0.4 P0.6 01: select P1.1. P1.2 10: select P2.2 P2.5

field	name	Access	Description
	DELAY_CTRL_REG		
31:11	Reserved	NA	
10:8			The time is requited for flash power up before system start working. 3'b000: wait 10us 3'b001: wait 15us 3'b010: wait 20us 3'b011: wait 25us

			3'b100: wait 30us 3'b101: wait 35us 3'b110: wait 40us 3'b111: wait 45us
7:6	pll_wait		The time is requited for PLL power up 2'b00: wait 50us 2'b01: wait 100us 2'b10: wait 150us 2'b11: wait 200us
5:4			The time is required for waiting after system /RF power domain power up 2'b00: wait 2us 2'b01: wait 5us 2'b10: wait 10us 2'b11: wait 15us
2:0		RW	The time is requited for 3.3V power up before system start working. 3'b000: wait 10us 3'b001: wait 20us 3'b010: wait 30us 3'b011: wait 50us 3'b100: wait 100us 3'b101: wait 150us 3'b110: wait 200us

field	name	Access	Description
PAD_WK_CTRL_REG			
31:30	P17_trig	RW	pin wake up control, P17_trig[1] for wake up enable, high active; P17_trig[0] for rise/fall edge detect,0 for rise,1 for fall
29:28	P16_trig	RW	pin wake up control, P16_trig[1] for wake up enable, high active; P16_trig[0] for rise/fall edge detect,0 for rise,1 for fall
27:26	P15_trig	RW	pin wake up control, P15_trig[1] for wake up enable, high active; P15_trig[0] for rise/fall edge detect,0 for rise,1 for fall
25:24	P14_trig	RW	pin wake up control, P14_trig[1] for wake up enable, high active; P14_trig[0] for rise/fall edge detect,0 for rise,1 for fall
23:22	P13_trig	RW	pin wake up control, P13_trig[1] for wake up enable, high active; P13_trig[0] for rise/fall edge detect,0 for rise,1 for fall
21:20	P12_trig	RW	pin wake up control, P12_trig[1] for wake up enable, high active; P12_trig[0] for rise/fall edge detect,0 for rise,1 for fall
19:18	P11_trig	RW	pin wake up control, P11_trig[1] for wake up enable, high active; P11_trig[0] for rise/fall edge detect,0 for rise,1 for fall
17:16	P10_trig	RW	pin wake up control, P10_trig[1] for wake up enable, high active; P10_trig[0] for rise/fall edge detect,0 for rise,1 for fall

15:14	P07_trig	RW	pin wake up control, P07_trig[1] for wake up enable, high active; P07_trig[0] for rise/fall edge detect,0 for rise,1 for fall
13:12	P06_trig	RW	pin wake up control, P06_trig[1] for wake up enable, high active; P06_trig[0] for rise/fall edge detect,0 for rise,1 for fall
11:10	P05_trig	RW	pin wake up control, P05_trig[1] for wake up enable, high active; P05_trig[0] for rise/fall edge detect,0 for rise,1 for fall
9:8	P04_trig	RW	pin wake up control, P04_trig[1] for wake up enable, high active; P04_trig[0] for rise/fall edge detect,0 for rise,1 for fall
7:6	P03_trig	RW	pin wake up control, P03_trig[1] for wake up enable, high active; P03_trig[0] for rise/fall edge detect,0 for rise,1 for fall
5:4	P02_trig	RW	pin wake up control, P02_trig[1] for wake up enable, high active; P02_trig[0] for rise/fall edge detect,0 for rise,1 for fall
3:2	P01_trig	RW	pin wake up control, P01_trig[1] for wake up enable, high active; P01_trig[0] for rise/fall edge detect,0 for rise,1 for fall
1:0	P00_trig	RW	pin wake up control, P00_trig[1] for wake up enable, high active; P00_trig[0] for rise/fall edge detect,0 for rise,1 for fall

field	name	Access	Description
PAD_BUF_EN_REG			
31:16			
15	P17_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
14	P16_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
13	P15_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
12	P14_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
11	P13_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
10	P12_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
9	P11_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
8	P10_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
7	P07_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
6	P06_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
5	P05_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on

4	P04_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
3	P03_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
2	P02_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
1	P01_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on
0	P00_buf_en	RW	0: Digital input buffer off 1: Digital input buffer on

field	name	Access	Description
PAD_PD_CTRL_REG			
31:16			
15	P17_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
14	P16_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
13	P15_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
12	P14_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
11	P13_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
10	P12_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
9	P11_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
8	P10_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
7	P07_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
6	P06_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
5	P05_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
4	P04_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
3	P03_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
2	P02_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected

1	P01_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected
0	P00_pd_en	RW	0: No pull down resistor connected 1: Pull down resistor connected

field	name	Access	Description
	PAD_PU_CTRL_REG		
31:30	P17_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
29:28	P16_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
27:26	P15_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
25:24	P14_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
23:22	P13_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
21:20	P12_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
19:18	P11_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
17:16	P10_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
15:14	P07_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
13:12	P06_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
11:10	P05_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
9:8	P04_pu_en	RW	00: no pull up resistors

			01: pull up 30k resistor connected 10: pull up 150k resistor connected
7:6	P03_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
5:4	P02_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
3:2	P01_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected
1:0	P00_pu_en	RW	00: no pull up resistors 01: pull up 30k resistor connected 10: pull up 150k resistor connected

field	name	Access	Description
	PAD_PU_CTRL_REG		
31:30	P17_drv	RW	00: 01: 10: 11:
29:28	P16_drv	RW	00: 01: 10: 11:
27:26	P15_drv	RW	00: 01: 10: 11:
25:24	P14_drv	RW	00: 01: 10: 11:
23:22	P13_drv	RW	00: 01: 10: 11:
21:20	P12_drv	RW	00: 01: 10: 11:
19:18	P11_drv	RW	00:

			01: 10: 11:
17:16	P10_drv	RW	00: 01: 10: 11:
15:14	P07_drv	RW	00: 01: 10: 11:
13:12	P06_drv	RW	00: 01: 10: 11:
11:10	P05_drv	RW	00: 01: 10: 11:
9:8	P04_drv	RW	00: 01: 10: 11:
7:6	P03_drv	RW	00: 01: 10: 11:
5:4	P02_drv	RW	00: 01: 10: 11:
3:2	P01_drv	RW	00: 01: 10: 11:
1:0	P00_drv	RW	00: 01: 10: 11:

5.6 POF address map

Address Range	Description
0x4000 0900 to 0x4000 09FF	POF register space

5.7 POF register map

Offset address	Name	Access	Reset value	Description
Base address: 0x40000900				
0X00040	POFCON	RW	0x000000D0	
0X00044	POFMSK	RW	0x00000016	

5.8 POF register description

Field	Reset value	name	Access	Description
0x40	**POFCON**			
31:9	23'h0	Reserved	NA	
8:4	5'b10111	pof_thr	RW	If power is lower than pof threshold, pof_warn will be active
3	1'b0	pof_en	RW	POF enable, high active
2:1	2'b0	lvd_in_sel		select LVD PIN 00: VDET PAD 01: 10: 11:
0	1'b0	pof_warn	RO	power fail warning, low active

field	name	Access	Description
0x44	POFMSK		
31:8	Reserved	N/A	
7:0	pof_mst_cnt	RW	

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6 Clock and Reset management

6.1 Reset

System reset

A system reset sets all registers to their reset values except the reset flags in the clock controller RESET register.

A system reset is generated when one of the following events occurs:

1. Window watchdog end of count condition (WDT reset)
2. Software reset (MCU SW reset)
3. Low-power management reset

The reset source can be identified by checking the reset flags in the RESET register.

Software reset

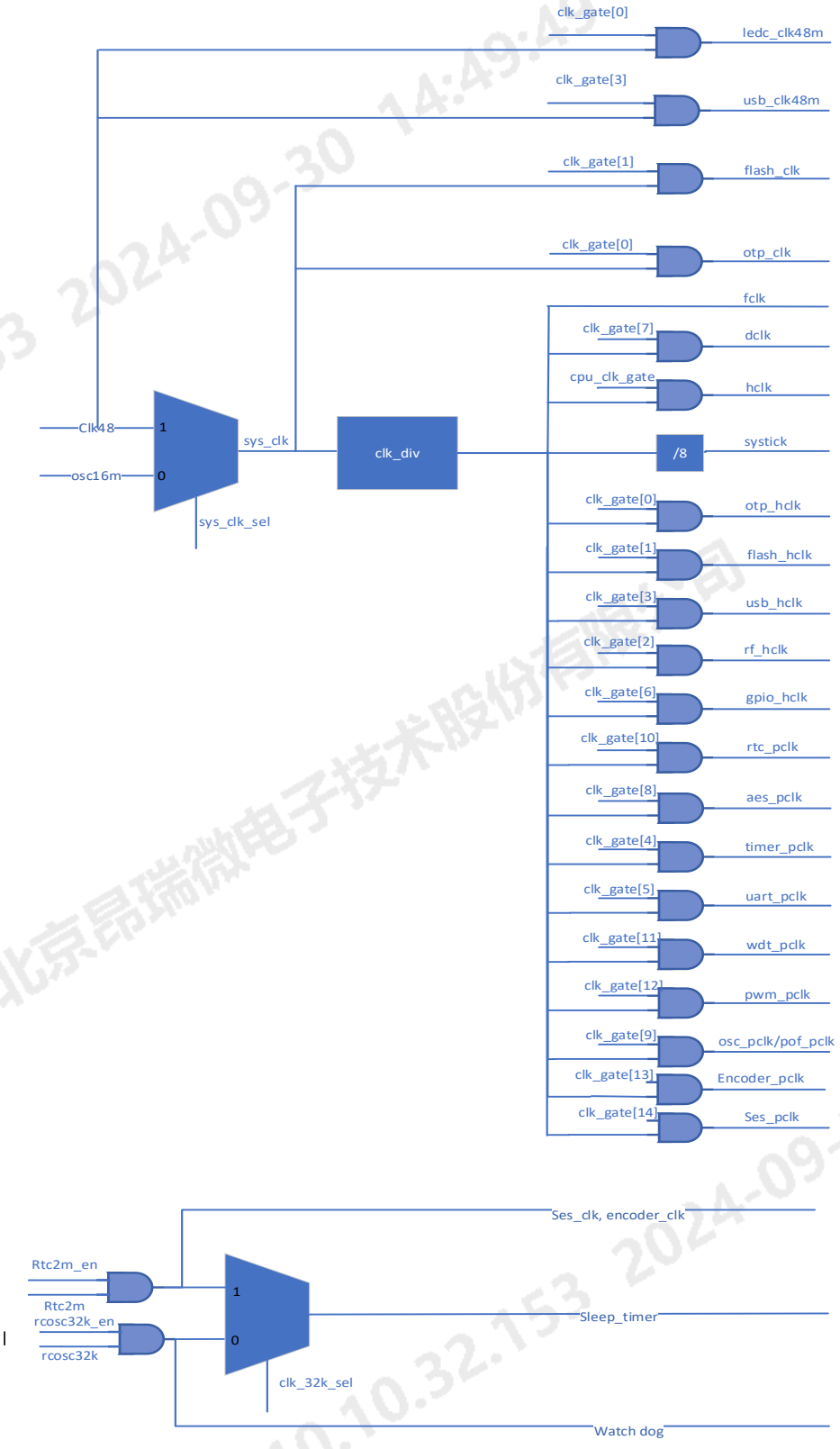
The SYSRESETREQ bit in MCU Application Interrupt and Reset Control Register must be set to force software reset on the device.

6.2 SWD disable

When system power up, hardware will automatic read flash 0x37 content, when it read 0xab. Hardware will disable SWD debug function.

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6.3 Clock Tree



6.4 Reset and Clock control address map

Below is the Reset and Clock control address map:

Address Range	Description
0x4000 0600 to 0x4000 06FF	Reset and Clock Control register space

6.5 Reset and Clock control register map

Offset address	Name	Access	Reset value	Description
Base address : 0x40000600				
0x00000	CLKCTRL	RW	0x00000000	
0X00004	CLKGATE	RW	0x00000482	
0X00008	RSTCTRL	RW	0x00000000	
0X0000C	M0CALIB	RW	0x00000000	
0X00010	SYSCLK_SEL	RW	0x00000000	
0X00014	Reserved	RW	0x00000003	
0X00018	FLASH_CLKCTRL	RW	0x00000500	
0X0001C	FLASH_READ_OPCODE	RW	Boot_pin=0 0x0000003B Boot_pin=1 0x00000000	
0X00020	FLASH_OFFSET_ADDR	RW	0x00000000	
0X00024	USB_CTRL	RW	0x00000300	
0X00028	USB_WK_INT_CLR	RW	0x00000000	
0X0002C	USB_WK_SIG_SEL	RW	0x00000000	
0X00030	USB_WK_CLK_SEL	RW	0x00000000	

6.6 Reset and Clock control register description

field	name	Access	Description
CLKCTRL			
31:3	Reserved	NA	
2:0	sysclk_div	RW	Set and cleared by software to control AHB clock division factor. 0: SYSCLK not divided 1: SYSCLK divided by 2 2: SYSCLK divided by 4 3: SYSCLK divided by 8 4: SYSCLK divided by 16 5: SYSCLK divided by 32 6: SYSCLK divided by 64

.

			7: SYSCLK divided by 128

field	name	Access	Description
	CLKGATE		
31:15	Reserved	NA	
14	Ses_clk_en	RW	Clock enable for Sensor scan
13	Encoder_clk_en	RW	Clock enable for ENCODER module
12	pwm_clk_en	RW	Clock enable for PWM module
11	wdt_clk_en	RW	Clock enable for WDT module
10	rtc_clk_en	RW	Clock enable for RTC module
9	ana_clk_en	RW	Clock enable for POF module
8	aes_clk_en	RW	Clock enable for AES module
7	swd_clk_en	RW	Clock enable for SWD Debug module
6	gpio_clk_en	RW	Clock enable for GPIO module
5	uart_clk_en	RW	Clock enable for UART module
4	time_clk_en	RW	Clock enable for timer module
3	usb_clk_en	RW	Clock enable for USB module
2	rf_clk_en	RW	Clock enable for RF module
1	flash_clk_en	RW	Clock enable for FLASH controller
0	ledc_clk_en	RW	Clock enable for LEDC controller

field	name	Access	Description
	RSTCTRL		
31:17	Reserved	NA	
16		1'b0	Writing 1 to reset sesor scan module
15		1'b0	Writing 1 to reset encoder module
14		1'b0	Writing 1 to reset NZ module
13		1'b0	Writing 1 to reset PWM module
12		1'b0	Writing 1 to reset UART module
11		1'b0	Writing 1 to reset POF module
10		1'b0	Writing 1 to reset AES module
9		1'b0	Writing 1 to reset GPIO module
8		1'b0	Writing 1 to reset timer module
7		1'b0	Writing 1 to reset FLASH controller
6		1'b0	Writing 1 to reset USB module
5		1'b0	Writing 1 to reset RF module
4		1'b0	Writing 1 to reset RF module except register and TX FIFO
3	Reserved	NA	
2:0		3'h0	000: power on reset 001: Watchdog reset 010: soft reset

			011: deep sleep reset 100: standby reset

field	name	Access	Description
	M0CALIB		
31:26	Reserved	NA	
25		W	1: Indicate HCLK/8 is used as a reference clock source for sys tick
24		W	Refer to Cortex-M0 STCALIB[24]
23:0		W	The SysTick calibration value.

field	name	Access	Description
	SYSCLK_SEL		
31:1	Reserved	NA	
0	Sysclk_sel	RW	0: sysclk is coming from osc16m 1: sysclk is coming from PLL 48m

field	name	Access	Description
	FLASH_CLKCTRL		
31:11	Reserved	NA	
10	fe_dly_sample	RW	Falling edge delayed sampling. 1: sample SPI_DI on falling edge of delayed sampling clock (SPI_CLK_DLY). 0: sample SPI_DI on falling edge of internal sampling clock (SPI_CLK). Enabling this bit will allow higher frequency operation. If set, the dly_sample[13:12] must be set to 1 or greater.
9:8	dly_sample	RW	Delayed Sampling, the number of REF_CLKs after the falling edge of SPI_CLK to sample SPI_DI. Setting these bits will allow higher frequency operation. If fe_dly_sample[14] is set, this field must be set to 1 or greater.
7:0	clock_div	RW	This register is the divider for the system clock and to generate the SPI clock. Only even values should be programmed in order to keep a 50% duty cycle clock. The minimum value of this register is 0.

field	name	Access	Description
	FLASH_READ_OPCODE		
31:16	vectkey	W	If write this register, vectkey must be 0x6666
15:8	Reserved	NA	
7:0	flash_read_opcode	RW	This register holds the OPCODE which is used to read from a

			Serial Flash device when a Transparent Read occurs;

field	name	Access	Description
FLASH_OFFSET_ADDR			
31:24	vectkey	W	If write this register, vectkey must be 0x66
23:0	flash_offset_addr	RW	

field	name	Access	Description
USB_CTRL			
31:6	Reserved	NA	
11:10	Usb_pmos_sr	RW	Default value：1’b00, D+/D- output driver pmos slew rate control
9:8	Usb_nmos_sr	RW	Default value:’b11, D+/D- output driver nmos slew rate control
7	P13_2v_schmit_en	RW	P13 2V input schmit trigger enable, high active
6	p07_2v_schmit_en	RW	P07 2V input schmit trigger enable, high active
5	usb_p_pull_down	RW	
4	usb_p_pull_up	RW	
3	usb_p_pull_up_lo	RW	
2	usb_m_pull_down	RW	
1	usb_m_pull_up	RW	
0	usb_m_pull_up_lo	RW	

field	name	Access	Description
USB_WK_INT_CTR			
31:2	Reserved	NA	
1	usb_wk_int_status	R	USB wake up interrupt status
0	usb_wk_int_clr	W	Write 1 to clear USB wakeup interrupt flag

field	name	Access	Description
USB_WK_SIG_SEL			
31:1	Reserved	NA	
0	usb_wk_sig_sel	RW	USB wakeup source select; 0: D+; 1:diff

field	name	Access	Description
USB_WK_SIG_SEL			
31:1	Reserved	NA	
0	usb_wk_clk_sel	RW	USB wakeup clock select; 0: osc16m; 1:rtc32k

7 SFLASH controller

7.1 Overview

The Serial Peripheral Interface (SPI) is used primarily for a synchronous serial communication of host processor and peripherals. The SPI controller is Motorola SPI-compatible interface. Up to two devices can be connected using four chip selects, data-in (SPI_DI), data-out (SPI_DO) and clock (SPI_CLK) signals are common. SPI interface can also be used to connect to SPI flash devices; commands such as read/write are user configurable.

Following features are supported

• Motorola SPI compatible 4 wire interface up to 48MHz

• SPI Master Mode support only

• DMA read and write support. DMA reads can be maximum size of 511 Bytes

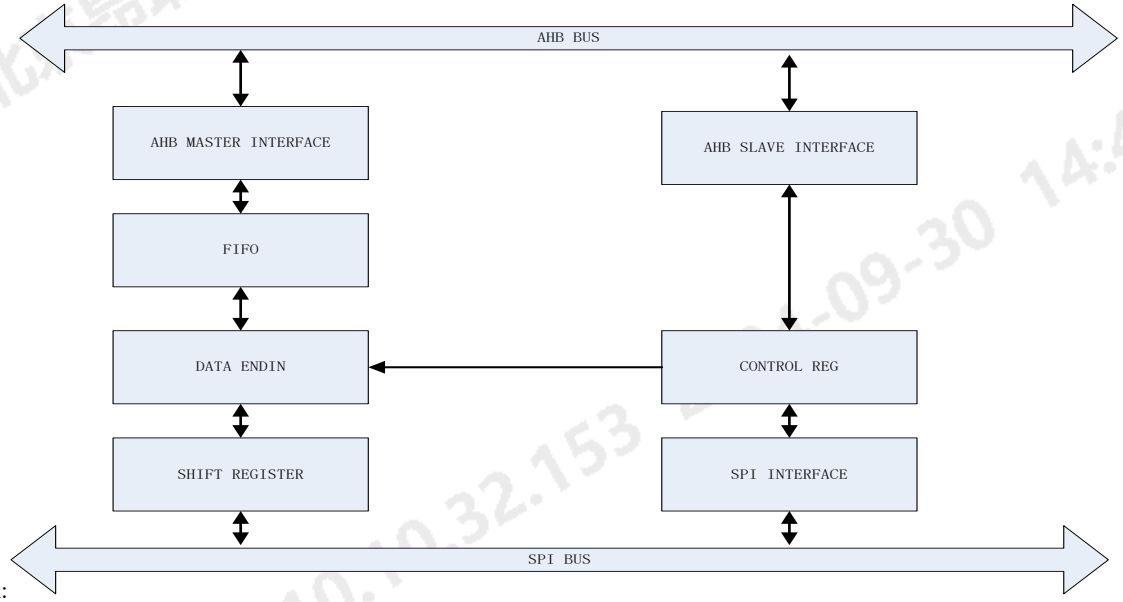
• Operating speed is software configurable

• Transparent read support for flash device

7.2 Module description

The AHB Master I/F transfers data from the SPI FIFO to system memory for SPI reads or from system memory to the SPI FIFO for SPI writes. The CPU uses the AHB slave interface to setup a SPI read or write transactions. The single buffered FIFO is used for data transport through the AHB Master Interface. The same FIFO is used for both SPI reads and SPI writes. When a transaction is completed, an interrupt can be generated, if enabled, to signal the CPU that the requested transaction has completed.

For SPI devices that are less than 32 bits wide, the endianness of the SPI device needs to be considered. Since data is sent MSB-first, when the SPI device is little-endian, this module will internally swap the data bytes (for 8-bit devices) or half-words (for 16-bit devices) before sending and after receiving data from the SPI device. SPI device endianness is register configurable and programmed with the WIDTH field in the SPI Configuration Register. Frame work below indicates the bit order sent for a specific SPI device configuration:



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7.3 SFLASH address map

SPI address is mapped to transparent read space. Below is the SPI address map:

Address Range	Description
0x5200 0000 to 0x52FF FFFF(un-cacheable)	Transparent read space(physical address)
0x0080 0000 to 0x00BF FFFF(cacheable/un-cacheable)	Transparent read space
0x0000 0000 to 0x00FF FFFF(cacheable/un-cacheable)	Remap address space
0x5300 0000 to 0x53FF FFFF	SPI control register space

7.4 SFLASH register map

Offset address	Name	Access	Reset value	Description
Base address: 0x53000000				
0x00000	spi_intr_status	RO	0x00000000	SPI Interrupt Status Register
0X00004	spi_raw_intr_status	RW	0x00000000	SPI Raw Interrupt Status Register
0X00008	spi_intr_mask	RW	0x00000000	SPI Interrupt Mask register
0X0000C	spi_command	RW	0x00000000	SPI Command Register
0X00010	spi_command_data0_reg	RW	0x00000000	SPI command data0 register
0X00014	spi_command_data1_reg	RW	0x00000000	SPI command data1 register
0X00018	spi_address_reg	RW	0x00000000	SPI Address Register
0X00020	spi_configuration_0	RW	0x00000500	SPI Configuration Register (MOVE to CPM register)
0X00024	spi_configuration_1	RW	0x00000000	SPI Configuration Register
0X00028	Spi_wp_hold_reg	RW	0x00000003	SPI WP_HOLD Control Register

7.5 SFLASH register description

field	name	Access	Description
spi_intr_status			
31:1	reseved	NA	
0	spi_cmd_done	RO	SPI command done

field	name	Access	Description
spi_raw_intr_status			
31:1	reserved	NA	
0	spi_raw_intr_status	RW	SPI command done interruption (internal interrupt value before mask). Set when the SPI command is complete. Writing 1 to clear the interrupt Status.

field	name	Access	Description

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	spi_intr_mask		
31:1	reserved	NA	
0	spi_cmd_done_mask	RW	SPI command done interrupt mask. When 1, allows the interrupt to assert the interrupt.

field	name	Access	Description
	spi_command		
28:20	data_bytes	RW	This field specifies the number of data bytes to transfer after the Command Data bits have been sent. Valid values are 0-65535. For a Read command, if this field is not a multiple of 4, zeros will be padded before data is written to system memory. This value is decremented during SPI the transaction until it reaches 0.
19:15	dummy_bits	RW	This field specifies the number of bits of the Dummy Data to send.
13:8	addr_bits		This field specifies the number of bits of the address Data to send.
7:4	cmd_bits		This field specifies the number of bits of the Command Data to send.
2	keep_cs	RW	When this bit is set, the CS will remain asserted after the command is finished.
1:0	command	RW	Command. This field is self-clearing after the SPI transition has completed. 0x0: NOP 0x1: Read. Data is transferred to Memory after the Command Data bits are sent 0x2: Write. Data is transferred to the SPI device after the Command Data bits are sent.

field	name	Access	Description
	spi_command_data0_reg		
31:0	command_data	RW	This is the command data which is sent for the first32 SPI clock cycles, depending on the CMD_BITS field. It is sent MSB first (data is left-shifted out of bit31). This value is maintained during the SPI transaction.

field	name	Access	Description
	spi_command_data1_reg		
31:0	command_data	RW	This is the command data which is sent for the first 32 SPI clock cycles, depending on the CMD_BITS, ADDR_BITS and DUMMY_BITS field. It is sent MSB first (data is left-shifted out of bit31). This value is maintained during the SPI transaction.

field	name	Access	Description
	spi_address_reg		
31:0	address	RW	This register holds the system memory address for data transfer. It is

			incremented by 1 as data is read from system memory (in the case of a Write command), or as data is written to system memory (in the case of a Read command).

field	name	Access	Description
	spi_configuration_0		
31:11	reserved	NA	
10	fe_dly_sample	R	Falling edge delayed sampling. 1: sample SPI_DI onfalling edge of delayed sampling clock (SPI_CLK_DLY). 0: sample SPI_DI on falling edge of internal sampling clock (SPI_CLK). Enabling this bit will allow higher frequency operation. If set, the dly_sample[13:12] must beset to 1 or greater. This is only valid for SPI modes 0 and3. For modes 1 and 2, these bits must be set to 0.
9:8	dly_sample	R	Delayed Sampling, the number of REF_CLKs after the falling edge of SPI_CLK to sample SPI_DI. Setting thesebits will allow higher frequency operation. If fe_dly_sample[14] is set, this field must be set to 1 or greater. This isonly valid for SPI modes 0 and 3. For modes 1 and 2, these bits must be set to 0.
7:0	clock_div	R	This register is the divider for the system clock and to generate the SPI clock. Only even values should be programmed in order to keep a 50% duty cycle clock. The minimum value of this register is 0.

field	name	Access	Description
	spi_configuration_1		
31:10	Reserved	RW	
9:8	Data_with	RW	00: 1 wire 01: 2 wire 10: 4 wire
7:6	Reserved	RW	
5:4	Addr_with	RW	00: 1 wire 01: 2 wire 10: 4 wire
3:2	Reserved	RW	
1:0	cmd_with	RW	00: 1 wire 01: 2 wire 10: 4 wire

7.6 DMA Write Operation

For DMA write operation, only a maximum of PageSize number of bytes supported by the flash can be written at a time. The Page Write

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or Page Program command (8-bits) is sent along with the 24-bit start flash address; this is followed by the required number of data bytes. Thus the number of command bits should be set to 32 and the number of data bytes is set to the required number.

1. Perform a Sector Erase operation if the flash supports only Page Program command; if it supports Page Write then there is no need to do a separate Erase command.
2. Send Write enable command to the flash.
3. Write the SPI Command Data0 register with the Page Program (Or Page Write) command and the 24-bit start address in flash; please remember the command is transmitted MSB first; hence for example if the start address in flash is 0, then write 0x02000000 to do a page program. (SPI Command Data1register is not used for this transaction).
4. Set the DMA Source Address in the SPI Address Register.
5. Write the SPI configuration_1 Register with the following details: With of command Phase = 0, With of Address Phase = 0, With of Data Phase = 0.
6. Write the SPI Command Register with the following details: Number of command bits = 8, Number of Address bits = 24, Number of Dummy bits = 0, Keep_CS=0, Command = Write i.e. 2.
7. Wait until the command completes. One way to do this is by polling the spi_raw_intr_status bit in the SPI Raw Interrupt Status Register. Another option is to use the interrupt.
8. Use the Read-Status-Register command to find out when the Write operation is completed by flash.

7.7 DMA Read Operation

For DMA Read operation, the whole flash can be read using a single command. Normal Read command (0x03) can be used only up-to a certain frequency (example 20 MHz for M25P128); above that frequency the Fast Read command (0x0b) should be used. When the Normal Read command is used the SPI Master sends the 8-bit read command along with a 24-bit start flash address; thus the number of command bits will be 32 for normal read command. When the Fast Read command is used the SPI Master sends the 8-bitFast-read command along with a 24-bit start flash address; the flash device returns a dummy byte before starting to return valid data. Hence the number of command bits for Fast-read command will be 40 bits.

1. Write the SPI Command Data0 register with the Normal Read command (0x03) or the Fast-Read command (0x0b) based on the SPI clock frequency, along with the 24-bit start flash address; please remember the command is transmitted MSB first; hence for example if the start address in flash is 0, then write0x0b000000 to do a fast read. Write 0x00000000 to SPI Command Data1 register for fast read to make sure the MOSI line does not toggle during the dummy byte interval; this is safer.
2. Set the DMA Destination Address in the SPI Address Register.
3. Write the SPI configuration_1 Register with the following details: With of command Phase = 0, With of Address Phase = 0, With of Data Phase = 0.
4. Write the SPI Command Register with the following details: Number of command bits = 8, Number of Address bits = 24, Number of Dummy bits = 0(0x3) or 8(0x0b), Number of data bytes=N (required number), Keep_CS=0, Command = Read i.e. 1.
5. Wait until the command completes. One way to do this is by polling the spi_raw_intr_status bit in the SPI Raw Interrupt Status Register. Another option is to use the interrupt.
5. The data will be available in the system memory starting from address DMA Destination Address.

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8 USB Configuration

The USB device controller is compatible with USB 2.0 Full-speed (12 Mbps) function.

8.1 Feature

- Conforms to 1.1 and 2.0 revision of the USB specification.
- Support 8 endpoints (include endpoint 0).
- EP0 and EP1 endpoints have 16 bytes IN buffer and 16 bytes OUT buffer
- EP2 endpoint have 32 bytes IN buffer and 32 bytes OUT buffer
- EP3 endpoint have 64 bytes IN buffer and 64 bytes OUT buffer
- Serial Interface Engine
  - Supports full speed devices
  - NRZI decoding/encoding
  - Bit stuffing/stripping
  - CRC checking/generation
  - On-chip pull-up resistor with software controlled disconnect
- Automatic data retry mechanism
- Data toggle synchronization mechanism
- Suspend and resume power management functions
- Remote Wakeup function

8.2 Wakeup/suspend control

If the USB has not find activity on the USB for 3ms, it will generate a Suspend interrupt (if enabled).It is up to the software to decide what, if anything, to disable when the USB is in Suspend mode. The USB 48M PLL will stop. When USB detect Resume signaling (by monitoring the DIM and DIP signals) it will enable the 48M PLL and generate Resume interrupt. The en_48M will manual turn the 48M PLL on.

If the USB is in Suspend mode and the software wants to initiate a remote wakeup, it should write to the sw_wu to enable the 48M PLL. It will take 100us to let the PLL stable then software need to set the Resume bit (D2) in Power register to 1. The software should leave this bit set for approximately 10ms (minimum of 2ms, a maximum of 15ms) then reset it to 0. By this time the hub should have taken over driving Resume signaling on the USB. The sw_wu must be clear before USB goes into the suspend mode otherwise the 48M PLL will not stop.

Note: No Resume interrupt will be generated when the software initiates a remote wakeup.

8.3 USB Registers

Note: In the following bit descriptions:

- ‘R’ means that the bit is read only
- ‘RW’ means that the bit can be both read and written
- ‘Set’ means that the bit can only be written to set it

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- ‘R/Set’ means that the bit can be read or set but it can’t be cleared
- ‘Clear’ means that the bit can only be written to clear it
- ‘R/Clear’ means that the bit can be read or cleared but it can’t be set
- ‘Self-clearing’ means the bit will be cleared automatically when the associated action has been executed.

8.4 Function address register– FADDR

FAddr is an 8-bit register that should be written with the function’s 7-bit address. It is then used for decoding the function address in subsequent token packets.

FADDR (RW) 0x000

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
Update	Function Address						
1’b0	7’b0000000						
R	RW						

Description of Word

Bit	Value	Symbol	Description
7	1’b0	Update	Set when FAddr is written. Cleared when the new address takes effect (at the end of the currenttransfer).
6:0	7’h0	FAddr	The function address.

8.5 Power management register-POWER

POWER (RW) 0x001

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
ISO Update	---			Reset	Resume	SuspendMode	EnableSuspend
1’b0	3’b000			1’b0	1’b0	1’b0	1’b0
RW	R			R	RW	R	RW

Description of Word

Bit	Value	Symbol	Description
7	1’b0	ISO Update	When set by the CPU the MUSBFSFC will wait for an SOF token from the time InPktRdy is setbefore sending the packet. If an IN token is received before an SOF token, then a zero length datapacket will be sent. This bit is only used by endpoints performing Isochronous transfers.
6:4	3’b000	---	Unused, always return zero.

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3	1’b0	Reset	This read only bit is set while Reset signaling is present on the bus.
2	1’b0	Resume	Set by the CPU to generate Resume signaling when the function is in Suspend mode. The CPUshould clear this bit after 10ms (a maximum of 15ms) to end Resume signaling.
1	1’b0	SuspendMode	Set by the USB when Suspend mode is entered. Cleared when the CPU reads the interrupt register,or sets the Resume bit of this register.
0	1’b0	EnableSuspend	Set by the CPU to enable entry into Suspend mode when Suspend signaling is received on the bus.

8.6 Interrupt register for Endpoint 0 plus IN Endpoints 1 to 7-INTRIN1

IntrIn1 is an 8-bit read-only register that indicates which of the interrupts for IN Endpoints 1–7 are currently active. It also indicates whether the Endpoint 0 interrupt is currently active. Note: All active interrupts will be cleared when this register is read.

INTRIN1 (R) 0x002

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
EP7	EP6	EP5	EP4	EP3	EP2	EP1	EP0
1’b0	1’b0	1’b0	1’b0	1’b0	1’b0	1’b0	1’b0
R	R	R	R	R	R	R	R

Description of Word

Bit	Value	Symbol	Description
7	1’b0	EP7	IN Endpoint 7 interrupt.
6	1’b0	EP6	IN Endpoint 6 interrupt.
5	1’b0	EP5	IN Endpoint 5 interrupt.
4	1’b0	EP4	IN Endpoint 4 interrupt.
3	1’b0	EP3	IN Endpoint 3 interrupt.
2	1’b0	EP2	IN Endpoint 2 interrupt.
1	1’b0	EP1	IN Endpoint 1 interrupt.
0	1’b0	EP0	IN Endpoint 0 interrupt.

8.7 Interrupt register for IN Endpoints 8 to 15-INTRIN2

IntrIn2 is an 8-bit read-only register that indicates which of the interrupts for IN Endpoints 8–15 are currently active. This register will only be present if the MUSBFSFC is configured for more than 7 IN endpoints. Note: All active interrupts will be cleared when this register is read.

INTRIN2 (R) 0x003

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
EP15	EP14	EP13	EP12	EP11	EP10	EP9	EP8

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1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0
R	R	R	R	R	R	R	R

Description of Word

Bit	Value	Symbol	Description
7	1'b0	EP15	IN Endpoint 14 interrupt.
6	1'b0	EP14	IN Endpoint 13 interrupt.
5	1'b0	EP13	IN Endpoint 13 interrupt.
4	1'b0	EP12	IN Endpoint 12 interrupt.
3	1'b0	EP11	IN Endpoint 11 interrupt.
2	1'b0	EP10	IN Endpoint 10 interrupt.
1	1'b0	EP9	IN Endpoint 9 interrupt.
0	1'b0	EP8	IN Endpoint 8 interrupt.

8.8 Interrupt register for OUT Endpoints 1 to 7-INTROUT1

IntrOut1 is an 8-bit read-only register that indicates which of the interrupts for OUT Endpoints 1–7 are currently active.Note:All active interrupts will be cleared when this register is read.

INTROUT1 (R) 0x004

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
EP7	EP6	EP5	EP4	EP3	EP2	EP1	EP0
1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0
R	R	R	R	R	R	R	R

Description of Word

Bit	Value	Symbol	Description
7	1'b0	EP7	OUT Endpoint 7 interrupt.
6	1'b0	EP6	OUT Endpoint 6 interrupt.
5	1'b0	EP5	OUT Endpoint 5 interrupt.
4	1'b0	EP4	OUT Endpoint 4 interrupt.
3	1'b0	EP3	OUT Endpoint 3 interrupt.
2	1'b0	EP2	OUT Endpoint 2 interrupt.
1	1'b0	EP1	OUT Endpoint 1 interrupt.
0	1'b0	--	Unused, always returns 0

8.9 Interrupt register for OUT Endpoints 8 to 15-INTROUT2

IntrOut2 is an 8-bit read-only register that indicates which of the interrupts for OUT Endpoints 8–15 are currently active. This register

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will only be present if the MUSBFSFC is configured for more than 7 OUT endpoints. Note: All active interrupts will be cleared when this register is read.

INTROUT2 (R) 0x005

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
EP15	EP14	EP13	EP12	EP11	EP10	EP9	EP8
1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0
R	R	R	R	R	R	R	R

Description of Word

Bit	Value	Symbol	Description
7	1'b0	EP15	OUT Endpoint 15 interrupt.
6	1'b0	EP14	OUT Endpoint 14 interrupt.
5	1'b0	EP13	OUT Endpoint 13 interrupt.
4	1'b0	EP12	OUT Endpoint 12 interrupt.
3	1'b0	EP11	OUT Endpoint 11 interrupt.
2	1'b0	EP10	OUT Endpoint 10 interrupt.
1	1'b0	EP9	OUT Endpoint 9 interrupt.
0	1'b0	EP8	OUT Endpoint 8 interrupt.

8.10 Interrupt register for common USB interrupts-INTRUSB

INTRUSB is an 8-bit read-only register that indicates which USB interrupts are currently active. All active interrupts will be cleared when this register is read.

INTRUSB (R) 0x006

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
---				SOF	Reset	Resume	Suspend
4'b0000				1'b0	1'b0	1'b0	1'b0
R				R	R	R	R

Description of Word

Bit	Value	Symbol	Description
7:4	4'b0000	---	Unused, always return 0.
3	1'b0	SOF	Set at the start of each frame.
2	1'b0	Reset	Set when Reset signaling is detected on the bus.
1	1'b0	Resume	Set when Resume signaling is detected on the bus while the MUSBFSFC is in Suspend mode.
0	1'b0	Suspend	Set when Suspend signaling is detected on the bus.

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8.11 Interrupt enable register for INTRIN1-INTRIN1E

IntrIn1E is 8-bit registers that provide interrupt enable bits for the interrupts in IntrIn1.On reset, the bits corresponding to Endpoint 0 and the IN endpoints included in the design are set to 1, while the remaining bits are set to 0.

INTRIN1E (RW) 0x007

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
EP7	EP6	EP5	EP4	EP3	EP2	EP1	EP0
1'b0	1'b0	1'b0	1'b0	1'b1	1'b1	1'b1	1'b1
RW	RW	RW	RW	RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description
7	1'b0	EP7	enable bit for the IN Endpoint 7 interrupt 1:enable 0:disable
6	1'b0	EP6	enable bit for the IN Endpoint 6 interrupt 1:enable 0:disable
5	1'b0	EP5	enable bit for the IN Endpoint 5 interrupt 1:enable 0:disable
4	1'b0	EP4	enable bit for the IN Endpoint 4 interrupt 1:enable 0:disable
3	1'b1	EP3	enable bit for the IN Endpoint 3 interrupt 1:enable 0:disable
2	1'b1	EP2	enable bit for the IN Endpoint 2 interrupt 1:enable 0:disable
1	1'b1	EP1	enable bit for the IN Endpoint 1 interrupt 1:enable 0:disable
0	1'b1	EP0	enable bit for the IN Endpoint 0 interrupt 1:enable 0:disable

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8.12 Interrupt enable register for INTRIN1-INTRIN2E

IntrIn2E is 8-bit registers that provide interrupt enable bits for the interrupts in IntrIn2.On reset, the bits corresponding to Endpoint 0 and the IN endpoints included in the design are set to 1, while the remaining bitsare set to 0.

INTRIN2E (RW) 0x008

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
EP15	EP14	EP13	EP12	EP11	EP10	EP8	EP8
1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0
RW	RW	RW	RW	RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description
7	1'b0	EP15	enable bit for the IN Endpoint 15 interrupt 1:enable 0:disable
6	1'b0	EP14	enable bit for the IN Endpoint 14 interrupt 1:enable 0:disable
5	1'b0	EP13	enable bit for the IN Endpoint 13 interrupt 1:enable 0:disable
4	1'b0	EP12	enable bit for the IN Endpoint 12 interrupt 1:enable 0:disable
3	1'b0	EP11	enable bit for the IN Endpoint 11 interrupt 1:enable 0:disable
2	1'b0	EP10	enable bit for the IN Endpoint 10 interrupt 1:enable 0:disable
1	1'b0	EP9	enable bit for the IN Endpoint 9 interrupt 1:enable 0:disable
0	1'b0	EP8	enable bit for the IN Endpoint 8 interrupt 1:enable 0:disable

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8.13 Interrupt enable register for INTROUT1-INTROUT1E

IntrOut1Eis 8-bit register that provides interrupt enable bits for the interrupts in IntrOut1. On reset, the bits corresponding to the OUT endpoints included in the design are set to 1, while the remaining bitsare set to 0.

INTROUT1E (R) 0x009

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
EP7	EP6	EP5	EP4	EP3	EP2	EP1	---
1'b0	1'b0	1'b0	1'b0	1'b1	1'b1	1'b1	1'b0
RW	RW	RW	RW	RW	RW	RW	R

Description of Word

Bit	Value	Symbol	Description
7	1'b0	EP7	enable bit for the OUT Endpoint 7 interrupt 1:enable 0:disable
6	1'b0	EP6	enable bit for the OUT Endpoint 6 interrupt 1:enable 0:disable
5	1'b0	EP5	enable bit for the OUT Endpoint 5 interrupt 1:enable 0:disable
4	1'b0	EP4	enable bit for the OUT Endpoint 4 interrupt 1:enable 0:disable
3	1'b1	EP3	enable bit for the OUT Endpoint 3 interrupt 1:enable 0:disable
2	1'b1	EP2	enable bit for the OUT Endpoint 2 interrupt 1:enable 0:disable
1	1'b1	EP1	enable bit for the OUT Endpoint 1 interrupt 1:enable 0:disable
0	1'b0	--	Unused, always returns 0

8.14 Interrupt enable register for INTROUT2-INTROUT2E

IntrOut2Eis 8-bit register that provides interrupt enable bits for the interrupts in IntrOut2. On reset, the bits corresponding to the OUT endpoints included in the design are set to 1, while the remaining bitsare set to 0.

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INTROUT2E (RW) 0x00A

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
EP15	EP14	EP13	EP12	EP11	EP10	EP9	EP8
1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0
RW	RW	RW	RW	RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description
7	1'b0	EP15	enable bit for the OUT Endpoint 15 interrupt 1:enable 0:disable
6	1'b0	EP14	enable bit for the OUT Endpoint 14 interrupt 1:enable 0:disable
5	1'b0	EP13	enable bit for the OUT Endpoint 13 interrupt 1:enable 0:disable
4	1'b0	EP12	enable bit for the OUT Endpoint 12 interrupt 1:enable 0:disable
3	1'b0	EP11	enable bit for the OUT Endpoint 11 interrupt 1:enable 0:disable
2	1'b0	EP10	enable bit for the OUT Endpoint 10 interrupt 1:enable 0:disable
1	1'b0	EP9	enable bit for the OUT Endpoint 9 interrupt 1:enable 0:disable
0	1'b0	EP8	enable bit for the OUT Endpoint 8 interrupt 1:enable 0:disable

8.15 Interrupt enable register for INTRUSB-INTRUSBE

INTRUSBE is an 8-bit register that provides interrupt enable bits for each of the interrupts in INTRUSB.

INTRUSBE (R) 0x00B

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0

.

---	SOF	Reset	Resume	Suspend
4'b0000	1'b0	1'b1	1'b1	1'b0
R	RW	RW	RW	RW

Description of Word

Bit	Value	Symbol	Description
7:4	4'b0000	---	Unused, always return 0.
3	1'b0	SOF	enable bit for the interrupt 1:enable 0:disable
2	1'b1	Reset	enable bit for the interrupt 1:enable 0:disable
1	1'b1	Resume	enable bit for the interrupt 1:enable 0:disable
0	1'b0	Suspend	enable bit for the interrupt 1:enable 0:disable

8.16 Frame number bits 0 to 7-FRAME1

Frame1 is an 8-bit read-only register that holds the lower 8 bits of the last received frame number.

FRAME1 (R) 0x00C

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
Frame1							
8'h00							
R							

Description of Word

Bit	Value	Symbol	Description
7:0	8'h00	Frame1	Lower 8 bits of Frame Number.

8.17 Frame number bits 8 to 10-FRAME2

Frame2 is a 3-bit read-only register that holds the upper 3 bits of the last received frame number.

FRAME2 (R) 0x00D

.

Bit 2	Bit 1	Bit 0
Frame2		
3'b000		
R		

Description of Word

Bit	Value	Symbol	Description
2:0	3'b000	Frame2	Upper 3 bits of Frame Number.

8.18 Index register for selecting the endpoint status and control registers-INDEX

Index is a 4-bit register that determines which endpoint control/status registers are accessed at addresses 0xC010h to 0xC017h.Each IN endpoint and each OUT endpoint have their own set of control/status registers. Only one set of IN control/status andone set of OUT control/status registers appear in the memory map at any one time. Before accessing an endpoint'scontrol/status registers, the endpoint number should be written to the Index register to ensure that the correct control/statusregisters appear in the memory map.

INDEX (RW) 0x00E

Bit 3	Bit 2	Bit 1	Bit 0
Index			
4'b0000			
RW			

Description of Word

Bit	Value	Symbol	Description
3:0	4'b0000	Index	Selected Endpoint.

8.19 Control Status register for Endpoint 0-CSR0

CSR0 is an 8-bit register that provides control and status bits for Endpoint 0.CSR0 is used for all control/status of Endpoint0.

CSR0 0x011

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
Serviced SetupEnd (self-clearing)	Serviced OutPktRdy (self-clearing)	SendStall (self-clearing)	SendStall (self-clearing)	DataEnd (self-clearing)	SentStall	InPktRdy (self-clearing)	OutPktRdy
1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0
Set	Set	Set	R	Set	R/Clear	R/Set	R

.

Description of Word

Bit	Value	Symbol	Description
7	1'b0	ServicedSetupEnd	The CPU writes a 1 to this bit to clear the SetupEnd bit. It is cleared automatically.
6	1'b0	ServicedOutPktRdy	The CPU writes a 1 to this bit to clear the OutPktRdy bit. It is cleared automatically.
5	1'b0	SendStall	The CPU writes a 1 to this bit to terminate the current transaction. The STALL handshakewill be transmitted and then this bit will be cleared automatically.
4	1'b0	SendStall	This bit will be set when a control transaction ends before the DataEnd bit has been set. An interrupt will be generated and the FIFO flushed at this time. The bit is cleared by the CPU writing a 1 to the ServicedSetupEnd bit.
3	1'b0	DataEnd	The CPU sets this bit: 1. When setting InPktRdy for the last data packet. 2. When clearing OutPktRdy after unloading the last data packet. 3. When setting InPktRdy for a zero length data packet. It is cleared automatically.
2	1'b0	SentStall	This bit is set when a STALL handshake is transmitted. The CPU should clear this bit.
1	1'b0	InPktRdy	The CPU sets this bit after loading a data packet into the FIFO. It is cleared automaticallywhen the data packet has been transmitted. An interrupt is generated when the bit iscleared.
0	1'b0	OutPktRdy	This bit is set when a data packet has been received. An interrupt is generated when this bitis set. The CPU clears this bit by setting the ServicedOutPktRdy bit.

8.20 Maximum packet size for IN endpoint-INMAXP

InMaxP is an 8-bit register that holds the maximum packet size for transactions through the currently-selected IN endpoint – inunits of 8 bytes, except that a value of 2 sets the maximum packet size to 16.In setting this value, you should note the constraints placed by the USBSpecification on packet sizes for Bulk, Interrupt and Isochronous transactions in Full-speed operations.
There is an InMaxP register for each IN endpoint (except Endpoint 0).

INMAXP (RW) 0x010

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Inmaxp							
8'h00							
RW							

Description of Word

Bit	Value	Symbol	Description
7:0	8'h00	Inmaxp	Maximum Packet Size/transaction.

.

8.21 Control Status register 1 for IN endpoint-INCSR1

InCSR1 is an 8-bit register that provides control and status bits for IN transactions through the currently-selected endpoint.There is an InCSR1 register for each IN endpoint (not including Endpoint 0).

INCSR1 0x011

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
---	ClrDataTog	SentStall	SendStall	FlushFIFO (self-clearing)	UnderRun	FIFO NotEmpty	InPktRdy
1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0
R	Set	R/Clear	RW	Set	R/Clear	R/Clear	R/Set

Description of Word

Bit	Value	Symbol	Description
7	1'b0	---	Unused. Returns zero when read.
6	1'b0	ClrDataTog	The CPU writes a 1 to this bit to reset the endpoint IN data toggle to 0.
5	1'b0	SentStall	This bit is set when a STALL handshake is transmitted. The FIFO is flushed and the InPktRdy bitis cleared (see below). The CPU should clear this bit.
4	1'b0	SendStall	The CPU writes a 1 to this bit to issue a STALL handshake to an IN token. The CPU clears this bitto terminate the stall condition. This bit has no effect if the IN endpoint is in ISO mode.
3	1'b0	FlushFIFO	The CPU writes a 1 to this bit to flush the next packet to be transmitted from the endpoint INFIFO. The FIFO pointer is reset and the InPktRdy bit (below) is cleared. Note:If the FIFOcontains two packets, FlushFIFO will need to be set twice to completely clear the FIFO.
2	1'b0	UnderRun	In ISO mode, this bit is set when a zero length data packet is sent after receiving an IN token with the InPktRdy bit not set. In Bulk/Interrupt mode, this bit is set when a NAK is returned in response to an IN token. The CPU should clear this bit.
1	1'b0	FIFONotEmpty	This bit is set when there is at least 1 packet in the IN FIFO.
0	1'b0	InPktRdy	The CPU sets this bit after loading a data packet into the FIFO. It is cleared automatically when adata packet has been transmitted. An interrupt is generated (if enabled) when the bit is cleared.

8.22 Control Status register 2 for IN endpoint-INCSR2

INCSR2 is an 8-bit register that provides further control bits for IN transactions through the currently-selected endpoint. There isan INCSR2 register for each IN endpoint (not including Endpoint 0).

INCSR2 0x012

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0

.

AutoSet	ISO	Mode	DMAEnab	FrcDataTog	---	---	---
1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0
RW	RW	RW	RW	RW	R	R	R

Description of Word

Bit	Value	Symbol	Description
7	1'b0	AutoSet	If the CPU sets this bit, InPktRdy will be automatically set when data of the maximum packet size(value in InMaxP) is loaded into the IN FIFO. If a packet of less than the maximum packet size is loaded, then InPktRdy will have to be set manually.
6	1'b0	ISO	The CPU sets this bit to enable the IN endpoint for isochronous transfers (ISO mode), and clears itto enable the IN endpoint for bulk/interrupt transfers.
5	1'b0	Mode	The CPU sets this bit to enable the endpoint direction as IN, and clears it to enable the endpoint direction as OUT. Valid only where the same endpoint FIFO is used for both IN and OUT transactions.
4	1'b0	DMAEnab	The CPU sets this bit to enable the DMA request for the IN endpoint.
3	1'b0	FrcDataTog	The CPU sets this bit to force the endpoint's IN data toggle to switch after each data packet is sent regardless of whether an ACK was received. This can be used by interrupt IN endpoints which are used to communicate rate feedback for Isochronous endpoints.
2:0	3'b000	---	Unused, always return 0.

8.23 Maximum packet size for OUT endpoint-OUTMAXP

OutMaxP is an 8-bit register that holds the maximum packet size for transactions through the currently-selected OUT endpoint–in units of 8 bytes, except that a value of 2 sets the maximum packet size to 16.In setting this value, you should note the constraints placed by the USB Specification on packet sizes for Bulk, Interrupt and Isochronous transactions in Full-speed operations.
There is an OutMaxP register for each OUT endpoint (except Endpoint 0).

OUTMAXP (RW) 0x013

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Outmaxp							
8'h00							
RW							

Description of Word

Bit	Value	Symbol	Description
7:0	8'h00	Outmaxp	Maximum Packet Size/transaction.

8.24 Control Status register 1 for OUT endpoint-OUTCSR1

OutCSR1 is an 8-bit register that provides control and status bits for OUT transactions through the currently-selected endpoint. It is reset to 0.

OUTCSR1 0x014

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
ClrDataTog	SentStall	SendStall	FlushFIFO (self-clearing)	DataError	UnderRun	FIFO Full	OutPktRdy
1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0
Set	R/Clear	RW	Set	R	R/Clear	R	R/Clear

Description of Word

Bit	Value	Symbol	Description
7	1'b0	ClrDataTog	The CPU writes a 1 to this bit to reset the endpoint data toggle to 0.
6	1'b0	SentStall	This bit is set when a STALL handshake is transmitted. The CPU should clear this bit.
5	1'b0	SendStall	The CPU writes a 1 to this bit to issue a STALL handshake. The CPU clears this bit to terminate the stall condition. This bit has no effect if the OUT endpoint is in ISO mode.
4	1'b0	FlushFIFO	The CPU writes a 1 to this bit to flush the next packet to be read from the endpoint OUT FIFO. Note: If the FIFO contains two packets, Flush FIFO will need to be set twice to completely clear the FIFO.
3	1'b0	DataError	This bit is set when OutPktRdy is set if the data packet has a CRC or bit-stuff error. it is cleared when OutPktRdy is cleared. The bit is only valid in ISO mode.
2	1'b0	UnderRun	This bit is set if an OUT packet cannot be loaded into the OUT FIFO. The CPU should clear this bit. The bit is only valid in ISO mode.
1	1'b0	FIFOfull	This bit is set when no more packets can be loaded into the OUT FIFO.
0	1'b0	OutPktRdy	This bit is set when a data packet has been received. The CPU should clear this bit when the packet as been unloaded from the OUT FIFO. An interrupt is generated when the bit is set.

8.25 Control Status register 2 for OUT endpoint-OUTCSR2

OutCSR2 is an 8-bit register that provides further control bits for OUT transactions through the currently-selected endpoint. It isreset to 0.

OUTCSR2 0x015

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit0
AutoClear	ISO	DMAEnab	DMAMode	---	---	---	---

.

1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0	1'b0
RW	RW	RW	RW	RW	R	R	R

Description of Word

Bit	Value	Symbol	Description
7	1'b0	AutoClear	If the CPU sets this bit then the OutPktRdy bit will be automatically cleared when a packet of OutMaxP bytes has been unloaded from the OUT FIFO. When packets of less than the maximum packet size are unloaded, OutPktRdy will have to be cleared manually
6	1'b0	ISO	The CPU sets this bit to enable the OUT endpoint for Isochronous transfers, and clears it to enable the OUT endpoint for Bulk/Interrupt transfers
5	1'b0	DMAEnab	The CPU sets this bit to enable the DMA request for the OUT endpoint.
4	1'b0	DMAMode	Two modes of DMA operation are supported: DMA Mode 0 in which a DMA request is generated for all received packets, together with an interrupt (if enabled); and DMA Mode 1 in which a DMA request (but no interrupt) is generated for OUT packets of size OutMaxP bytes and an interrupt(but no DMA request) is generated for OUT packets of any other size. The CPU sets this bit to select DMA Mode 1 and clears this bit to select DMA Mode 0.
3:0	4'b0000	---	Unused, always return 0.

8.26 Number of received bytes in Endpoint 0 FIFO-COUNT0

Count0 is a 7-bit read-only register that indicates the number of received data bytes in the Endpoint 0 FIFO. The value returnedis valid while OutPktRdy (CSR0.bit0) is set.

COUNT0 (R) 0x016

Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Count0						
7'h00						
R						

Description of Word

Bit	Value	Symbol	Description
6:0	7'h00	Count0	Endpoint 0 OUT Count.

8.27 Number of bytes in OUT endpoint FIFO-OUTCOUNT1

OutCount1 is an 8-bit read-only register that holds the lower 8 bits of the number of received data bytes in the packet in the FIFO associated with the currently-selected OUT endpoint. The value returned is valid while OutPktRdy (OutCSR1.bit0) is set.

OUTCOUNTT1 (R) 0x016

.

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Outcounter1							
8'h00							
R							

Description of Word

Bit	Value	Symbol	Description
7:0	8'h00	Outcounter1	Endpoint OUT Count – lower 8 bits

8.28 Number of bytes in OUT endpoint FIFO-OUTCOUNT2

OutCount2 is a 3-bit read-only register that holds the upper 3 bits of the number of received data bytes in the packet in the FIFO associated with the currently-selected OUT endpoint. The value returned is valid while OutPktRdy (OutCSR1.bit0) is set.

OUTCOUNTER2 (R) 0x017

Bit 2	Bit 1	Bit 0
Outcounter2		
3'b000		
R		

Description of Word

Bit	Value	Symbol	Description
2:0	3'b000	Outcounter2	Endpoint OUT Count – upper 3 bits.

8.29 FIFOs for Endpoints 0 to 3-FIFOx

(Addresses 0x020–0x023)

This address range provides 4 addresses for CPU access to the FIFOs for each endpoint. Writing to these addresses loads data into the IN FIFO for the corresponding endpoint. Reading from these addresses unloads data from the OUT FIFO for the corresponding endpoint. The FIFOs are located on byte boundaries (Endpoint 0 at 0x020, Endpoint 1 at 0x021 ... Endpoint 3 at 0x023).

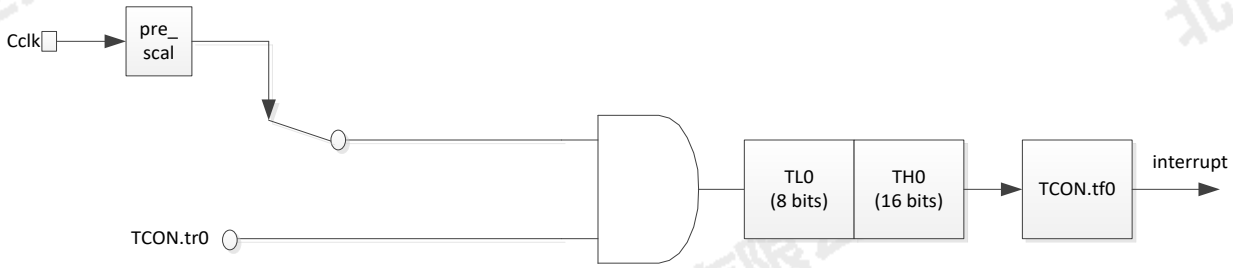
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9 Timer

The OM6239 contain one 32-bit timer/counters, also configure two 16bit timer, In the “timer mode”, timer registers are incremented every system pre-scale clock period, when appropriate timer is enabled.

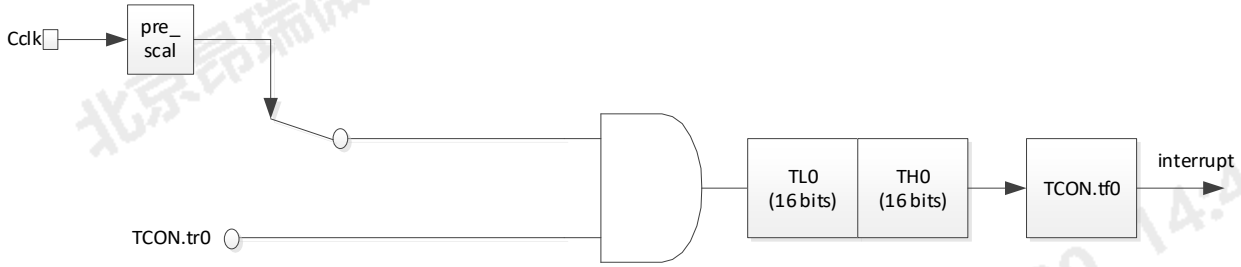
9.1 Mode 0

In this mode, the timer register is configured as a 24-bit register. As the count rolls over from all 1’sto all 0’s, it sets the timer overflow flag TCON.tf0. The overflow flag TCON.tf0 then can be used to request an interrupt.
The 24-bit register consists of all 16 bits of TH0 and the lower 8 bits of TL0. The upper 8 bits of TL0 are indeterminate and should be ignored. Setting the run flag (TCON.tr0) does not clear the registers.



9.2 Mode 1

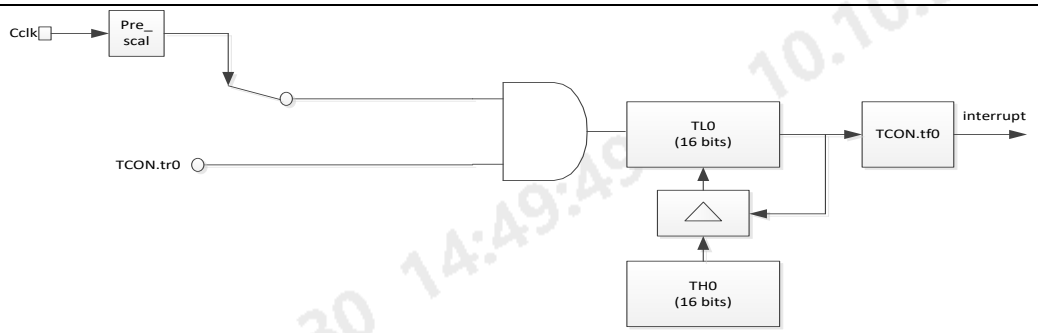
Mode 1 is the same as mode 0, except that the timer register is run with all 32 bits. Mode 1 is shown in figure below.



9.3 Mode 2

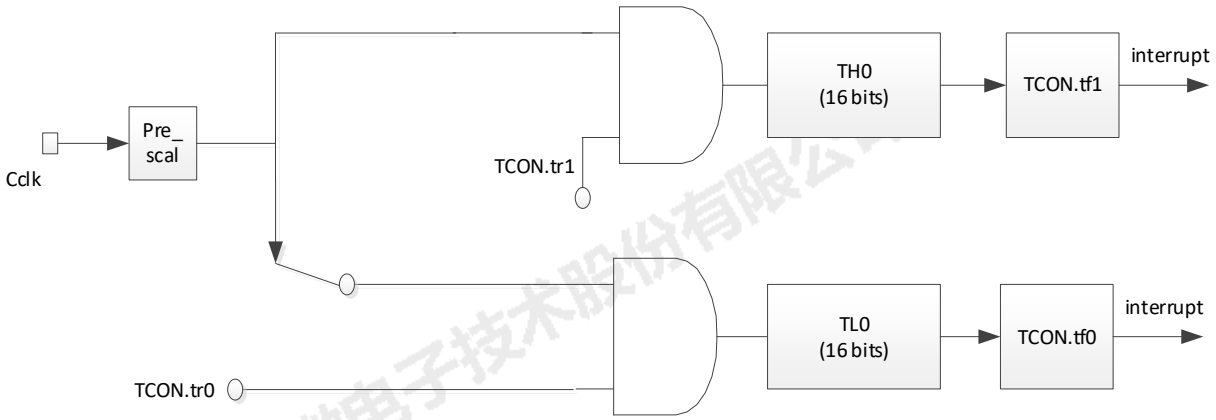
Mode 2 configures the timer register as an 16-bit counter (TL0) with automatic reloads as shown in figure below. Overflow from TL0 not only sets TCON.tf0, but also reloads TL0 with the contents of TH0, which is preset by software. The reload leaves TH0 unchanged.

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9.4 Mode 3

Timer 0 in Mode 3 establishes TL0 and TH0 as two separate counters. The logic for Mode 3 on Timer is shown in figure below. TH0 is locked into a timer function (counting machine cycles) and uses the TCON.tr1 and TCON.tf1 flags interrupt. Mode3 is provided for applications requiring an extra 16-bit timer/counter.



9.5 TIMER Register address map

Below is the Timer register address map:

Address Range	Description
0x4000 0500 to 0x4000 05FF	TIMER register space

9.6 TIMER register map

Offset address	Name	Access	Reset value	Description
Base address : 0x40000500				
0x00000	TINT	RW	0x00000000	
0X00004	TINTMASK	RW	0x00000000	
0X00008	TCON	RW	0x00000000	

.

0X0000C	TMOD	RW	0x00000000	
0X00010	TH0	RW	0x00000000	
0X00014	TL0	RW	0x00000000	
0X00018	TDIV	RW	0x0000000F	

9.7 Timer register description

field	name	Access	Description
	TINT		
31:2	Reserved	RW	
1	tf1		Timer 1 overflow flag. Set by hardware when Timer1 overflows. Cleared by hardware when interrupt is processed.
0	tf0	RW	Timer 0 overflow flag. Set by hardware when Timer 0 overflows. Cleared by hardware when interrupt is processed.

field	name	Access	Description
	TINTMASK		
31:2	Reserved	RW	
1	tf1_mask	RW	When 1, allows the interrupt to assert the interrupt.
0	tf0_mask	RW	When 1, allows the interrupt to assert the interrupt.

field	name	Access	Description
	TCON		
31:2	Reserved	RW	
1	tr1	RW	Timer 1 Run control. If cleared, Timer 1 stops.
0	tr0	RW	Timer 0 Run control. If cleared, Timer 1 stops.

field	name	Access	Description
	TMOD		
31:1	Reserved	RW	
0	tr0	RW	Timer mode 00 – Mode 0: 24-bit counter/timer 01 – Mode 1: 32-bit counter/timer 10 – Mode 2: 16-bit auto-reload timer 11 – Mode 3: two 16-bit timers/counters

field	name	Access	Description
	TH0		
31:16	Reserved	RW	
15:0	th0	RW	higher half word of Timer

field	name	Access	Description
	TL0		
31:16	Reserved	RW	
15:0	tl0	RW	low half word of Timer

field	name	Access	Description
	TDIV		
31:6	Reserved	RW	
5:0	pre_scaler	RW	This register is the divider for the timer clock

.

10 Watchdog

A programmable watchdog timer can be enabled to reset the system if the software hangs.

10.1 Features

- RTC32K
- 24-bit range Watchdog
- Watchdog time range from 31.25us to 524s
- Watchdog can disabled by a power-on reset or watchdog reset, or possibly when the chip enters the Deep Sleep and the Register retention power-saving mode.
- When the microcontroller enters debug mode，the WDT counter either continues to work normally or stops

10.2 Functional description

Watchdog Start Value (WDSV) contains the 24 bits of the Watchdog counter's initial value. Readout of WDSV will not give the current value of the Watchdog counter, but the start value for the counter.

After a reset, the default state of the Watchdog is disabled. The Watchdog is activated when WDSV have been written. The Watchdog counter then counts down from WDSV towards 0. When 0 is reached, the complete microcontroller, as well as the peripherals, is reset. A reset from the Watchdog will have the same effect as a power-on reset. To avoid the reset, the software must reload WDSV sufficiently often. The Watchdog counter is updated with a new start value and restarted every time WDSV is written. If WDSV is loaded with 0x000000, the Watchdog WDSV = 0xffffff is used.

The Watchdog timeout is given by:

$$\frac{WDSV+1}{32000}[s]$$

If the Watchdog has been started, it can only be disabled by a power-on reset, watchdog reset, or possibly when the chip enters the Deep Sleep and the Register retention power-saving mode. Please refer to watchdog control register bit 0.

The generated reset is passed to a reset controller, which in turn generates a reset for the components in the system. The WDT may be reset independently to the other components.

If the Watchdog has been started, it can only be disabled by a power-on reset.

When the microcontroller enters debug mode, the WWDG counter either continues to work normally or stops, depending on DBG_WDT_STOP configuration bit 2 of the Watchdog Control Register (WDT_CR) in DBG module.

Note:

1: If the CLKLF has been disabled by software, the Watchdog counter retains the value before the CLKLF stops and the Watchdog is activated, and when the CLKLF is enabled, the Watchdog counts down from current value towards 0.

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2: Only if the CLKLF is keep on, the Watchdog can be disabled by Watchdog reset or possibly when the chip enters the Deep Sleep and the Register retention power-saving mode.

10.3 Watchdog Register

Below is the Watchdog address map:

Address Range	Description
0x4000 0700 to 0x4000 07FF	Watchdog register space

10.4 Watchdog register map

Offset address	Name	Access	Reset value	Description
Base address: 0x40000700				
0x00000	REG_WDT_CR	RW	0x00000000	Watchdog Control Register
0X00004	REG_WDSV	RW	0x00000000	Watchdog start value Register

10.5 Watchdog register description

field	name	Access	Description
REG_WDT_CR			
31:2	Reserved	NA	
1	DBG_WDT_STOP	RW	0: The watchdog counter clock continues even if the core is halted. 1: The watchdog counter clock is stopped when the core is halted.
0	SLP_WDT_STOP	RW	0: The watchdog counter clock continues even if the core enters sleep power-saving mode. 1: The watchdog counter clock is stopped when the core is enters sleep power-saving mode.

field	name	Access	Description
REG_WDT_CR			
31:2	vectkey	W	If write this register, vectkey must be 0xA5
23:0	WDTSV	RW	Watchdog start value

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11 RTC Wakeup Timer

In order to achieve the lowest possible average current consumption, the processor clock can be stopped under firmware control. Operation can be resumed (wakeup) on external events like toggling of GPIO pins or from the internal RTC wakeup timer or the RFIRQ.

11.1 Features

- RTC32K
- 32bit range RTC wakeup timer
- RTC time range from 31.25us to 134144s
- RTC wakeup timer only disabled by porb or by software
- Compare with WU (IRQ), Resolution: 31.25µs.
- TICK mode
- ALARM mode

11.2 Functional description

RTC can give an interrupt at predefined intervals due to value equality between the timer and a tick register. RTC ensures that the functions the interrupt is used for are awoken prior to the interrupt. A wakeup interrupt and RTC timer interrupt are not the same interrupt, and known which one is interrupted.

The RTC timer is a 32-bit timer counting from zero and upwards at the rate of the 32 kHz clock. When the RTC timer is equal to the concatenation of TICK, an RTC IRQ, also referred to as WU, is generated. There is an uncertainty of one CLKLF period 31.25µs, from when the RTC is started or a new value is given to the RTC TICK value registers and until the IRQ is given.

The RTC counter uses the 32 kHz low frequency clock for the RTC timer, and the 32 kHz source must be enabled when using the RTC.

Note

1: The RTC can be disabled by a power-on reset.

2: The RTC can be disabled by software only if the CLKLF is keep on.

3: If the CLKLF has been disabled by software, the RTC timer retains the value before the CLKLF stops and the RTC is enabled, and when the CLKLF is enabled, the RTC timer counts up from current value.

11.3 _REG_CR offset 0x0

Bits	Access	Reset	FieldName	Description
12	ro	'h1	alarm2_wr_status	Indicator of current status, 1-Allow to write _REG_ALARM2, 0-Don't allow to write _REG_ALARM2

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11	ro	'h1	alarm1_wr_status	Indicator of current status, 1-Allow to write _REG_ALARM1, 0-Don't allow to write _REG_ALARM1
10	ro	'h1	alarm_wr_status	Indicator of current status, 1-Allow to write _REG_ALARM, 0-Don't allow to write _REG_ALARM
9	ro	'h1	tick_wr_status	Indicator of current status, 1-Allow to write _REG_TICK, 0-Don't allow to write _REG_TICK
8	ro	'h1	cr_wr_status	Indicator of current status, 1-Allow to write _REG_CR, 0-Don't allow to write _REG_CR
6	rw	'h0	alarm1_wakeup_en	Alarm1 Wake-Up enable, 1-timer Core can be woken by alarm wake-up, 0-timer Core cannot be woken up by alarm1 wake-up
5	rw	'h0	alarm_wakeup_en	Alarm Wake-Up enable, 1-timer Core can be woken by alarm wake-up, 0-timer Core cannot be woken up by alarm wake-up
4	rw	'h0	tick_wakeup_en	Tick Wake-Up enable, 1-timer Core can be woken by tick wake-up, 0-timer Core cannot be woken up by tick wake-up
3	rw	'h0	alarm1_dis	Disable alarm1 compare, when written with a 1.Resets at 0 when action is performed. No action happens if it is written with 0
2	rw	'h0	alarm_dis	Disable alarm compare, when written with a 1.Resets at 0 when action is performed. No action happens if it is written with 0
1	rw	'h0	tick_dis	Disable tick compare, when written with a 1.Resets at 0 when action is performed. No action happens if it is written with 0
0	rw	'h0	timer_on	timer counter enable, 1-enable timer counter ,0-disable timer counter

11.4 _REG_INTSTAT offset 0x4

Bits	Access	Reset	FieldName	Description
2	ro	'h0	alarm1_int_status	set by hardware
1	ro	'h0	alarm_int_status	set by hardware
0	ro	'h0	tick_int_status	set by hardware

11.5 _REG_INTRAWSTAT offset 0x8

Bits	Access	Reset	FieldName	Description
2	rw	'h0	alarm1_intraw_status	set by hardware, write 1 to clear
1	rw	'h0	alarm_intraw_status	set by hardware, write 1 to clear
0	rw	'h0	tick_intraw_status	set by hardware, write 1 to clear

11.6 _REG_INTMASK offset 0xc

Bits	Access	Reset	FieldName	Description
2	rw	'h0	alarm1_int_mask	mask alarm1 interrupt, 1-not mask, 0-mask
1	rw	'h0	alarm_int_mask	mask alarm interrupt, 1-not mask, 0-mask
0	rw	'h0	tick_int_mask	mask tick interrupt, 1-not mask, 0-mask

11.7 _REG_TICK offset 0x10

Bits	Access	Reset	FieldName	Description
31:0	rw	'h0	tick	set tick value and enable tick compare

11.8 _REG_ALARM offset 0x14

Bits	Access	Reset	FieldName	Description
31:0	rw	'h0	alarm	set alarm value and enable alarm compare

11.9 _REG_CNT offset 0x18

Bits	Access	Reset	FieldName	Description
31:0	ro	'h0	timer_cnt	timer counter current value

.

11.10 _REG_CNT_VALID offset 0x1c

Bits	Access	Reset	FieldName	Description
0	ro	'h0	timer_cnt_vld	timer counter value is valid

11.11 _REG_CLKN offset 0x20

Bits	Access	Reset	FieldName	Description
31:0	ro	'h0	sync_timer_cnt	sync lock timer value

11.12 _REG_CLKN_VALID offset 0x24

Bits	Access	Reset	FieldName	Description
0	ro	'h0	sync_lock_vld	sync lock timer value is valid

11.13 _REG_ALARM1 offset 0x28

Bits	Access	Reset	FieldName	Description
31:0	rw	'h0	alarm1	set alarm1 value and enable alarm1 compare

11.14 _REG_ALARM2 offset 0x2c

Bits	Access	Reset	FieldName	Description
31:0	rw	'h0	alarm2	set alarm2 value and enable alarm2 compare for RX timer out, auto disable when sync

11.15 _REG_WKRF_CR offset 0x30

Bits	Access	Reset	FieldName	Description
0	rw	'h1	alarm1_wkrf_en	alarm1 function; 1-alarm1 for RF timing transmit-receive, 0-alarm1 is the same as alarm

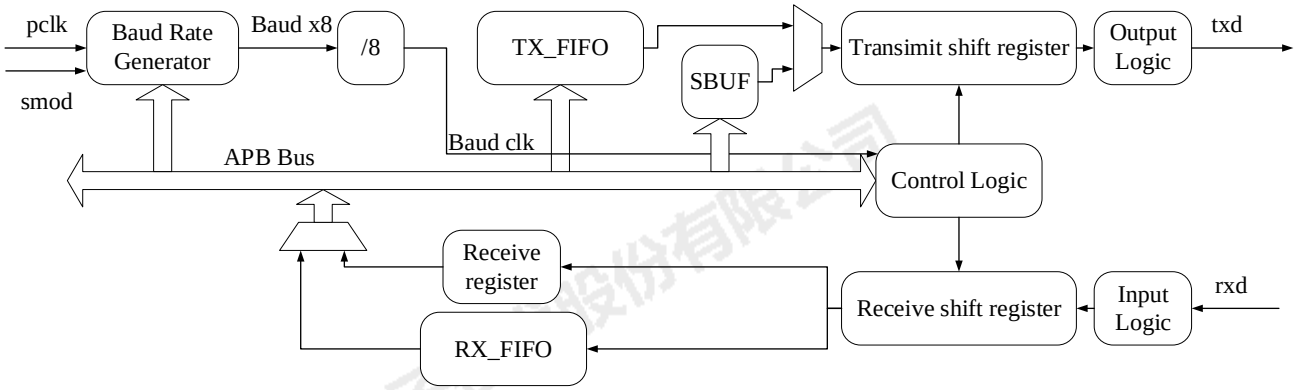
12 UART

12.1 overview

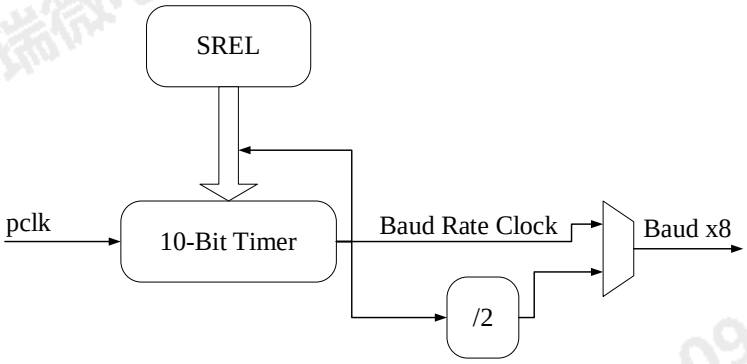
The UART is full-duplex, meaning it can transmit and receive simultaneously. It also has a tx fifo and a rx fifo for transmit and receive, respectively. Each fifo can be enabled/disabled by software and come with a status flag. The data counts saved in the fifo can be read through the fifo status register. It is possible to configure the tx fifo and tx fifo levels at which the Tx and RX interrupts are triggered.

12.2 module description

UART block diagram



The UART Baud Rate Generation



12.3 UART address map

Below is the UART address map:

Address Range	Description

.

0x4000 0300 to 0x4000 03FF	UART register space

12.4 UART register map

Offset address	Name	Access	Reset value	Description
Base address : 0x40000300				
0x00000	intr_status	RW	0x00000000	UART Interrupt Status Register
0X00004	intr_mask	RW	0x00000000	UART Interrupt Mask register
0X00008	SCON	RW	0x00000000	UART control register
0X0000C	SMOD	RW	0x00000001	UART SMOD register
0X00010	SREL	RW	0x000003d9	UART Baud Rate register
0X00014	SBUF	RW	0x00000000	UART buffer register
0X00018	FIFO_TH	RW	0x00000044	UART FIFO threshold
0X0001C	FIFO_ST	RW	0x00000000	UART FIFO status

12.5 UART register description

field	name	Access	Description
intr_status			
31:6	reseved	NA	
5	tx_fifo_crossing_th	RW	R: It will assert only when the tx fifo count change from threshold to threshold – 1 W: Write 1 to clear status
4	rx_fifo_above_th	RW	R: It will assert when the rx fifo count is larger than or equal to the RX_FIFO_TH. W: Write 1 to clear status
3	tx_busy	RO	0: No busy 1: Is busy
2	rx_busy	RO	0: No busy 1: Is busy
1	tx_int_status	RW	R: Transmit interrupt status W: write 1 to clear Transmit interrupt status
0	rx_int_status	RW	R: Receive interrupt status W: write 1 to clear Receive interrupt status

field	name	Access	Description
	spi_intr_mask		
31:6	reserved	NA	
5	tx_fifo_int_mask	RW	When 1, allows the interrupt to assert the interrupt.
4	rx_fifo_int_mask	RW	When 1, allows the interrupt to assert the interrupt.
3	reserved	NA	
2	reserved	NA	
1	tx_int_mask	RW	When 1, allows the interrupt to assert the interrupt.
0	rx_int_mask	RW	When 1, allows the interrupt to assert the interrupt.

field	name	Access	Description
	SCON		
31:6	Reserved	NA	
7	TX_FIFO_EN	RW	1: Enable tx fifo
6	RX_FIFO_EN	RW	1: Enable rx fifo
5:4	SM		01: mode0: 8-bit UART, variable baud rate 11: mode 1: 9-bit UART, variable baud rate
3	SM2	RW	In mode 1 and SM2 being set to 1, rx_int_status will not be activated if the received 9th data bit (RB8) is 0. In mode 0 and SM2 = 1, rx_int_status will not be activated if a valid stop bit has not been received.
2	REN	RW	Receive enable. Enables serial reception. Set by software to enable reception. Cleared by software to disable reception.
1:0	Reserved	NA	

field	name	Access	Description
	SMOD		
31:1	reserved	NA	
0	SMOD	RW	When set, the baud rate of UART is doubled.

field	name	Access	Description
	SREL		
31:10	reserved	NA	
9:0	SREL	RW	Baud Rate set

field	name	Access	Description
	SBUF		
31:8	reserved	NA	
7:0	SBUF	RW	Receive and transmit buffer of serial interface. Writing to SBUF loads the transmit register and initiates transmission. Reading out SBUF accesses a physically separate receive register

	FIFO_TH		
31:16	reserved	NA	
9:5	TX_FIFO_TH	RW	Threshold for Tx fifo if tx fifo count change from threshold to threshold – 1, a flag will be set by hardware. An interrupt will also be set if TX_FIFO_INT_MASK is set.
4:0	RX_FIFO_TH	RW	Threshold for Rx fifo when the rx fifo count is larger than or equal to the RX_FIFO_TH., a flag will be set by hardware. An interrupt will also be set if RX_FIFO_INT_MASK is set.

explanatory note：the thresholds of RX FIFO and TX FIFO can be set from 1 to 8, set the threshold as 0 will result in the interrupt not being triggered.

	FIFO_ST		
31:18	reserved	NA	
17:13	tx_fifo_cnt	RO	Current data count in tx fifo
12:8	rx_fifo_cnt	RO	Current data count in rx fifo
7:2	reserved	NA	
1	tx_fifo_full	RO	Indicate tx fifo is full
0	rx_fifo_empty	RO	Indicate rx fifo is empty

explanatory note: the depth of the RX FIFO and TX FIFO is 8, thus, the maximum value of tx_fifo_cnt and rx_fifo_cnt is 8.

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13 AES hardware accelerator (AES)

13.1 Futures

- Compliance with NIST “Advanced encryption standard (AES), FIPS publication 197” from November 2001
- 128-bit data block processing
- Support for cipher key lengths of 128-bit
- Encryption with multiple chaining modes
- Data-swapping and Key-swapping logic support

13.2 AES address map

Below is the AES module address map:

Address Range	Description
0x4000 0100 to 0x4000 01FF	AES register space

13.3 AES register map

Offset address	Name	Access	Reset value	Description
Base address : 0x40000100				
0x00000	AESCTRL	RW	0x00000000	
0X00004	AESDAT_1	RW	0x00000000	
0X00008	AESDAT_2	RW	0x00000000	
0X0000C	AESDAT_3	RW	0x00000000	
0X00010	AESDAT_4	RW	0x00000000	
0X00014	AESKEY_1	RW	0x00000000	
0X00018	AESKEY_2	RW	0x00000000	
0X0001C	AESKEY_3	RW	0x00000000	
0X00020	AESKEY_4	RW	0x00000000	
0X00024	AESINT	RW	0x00000000	
0X00028	AESINTMASK	RW	0x00000000	
0X0002C	SWAPCTRL	RW	0x00000001	

13.4 AES register description

field	name	Access	Description
	AESCTRL		

31:1	Reserved	NA	
0	aes_start	RW	Set by software, clear by hardware

field	name	Access	Description
AESDAT_1			
31:0	aes_dat[31:0]	RW	Write Plain byte3~byte0 Read Ciphered Byte3~byte0

field	name	Access	Description
AESDAT_2			
31:0	aes_dat[63:32]	RW	Write Plain byte7~byte4 Read Ciphered Byte7~byte4

field	name	Access	Description
AESDAT_3			
31:0	aes_dat[95:64]	RW	Write Plain byte11~byte8 Read Ciphered Byte11~byte8

field	name	Access	Description
AESDAT_4			
31:0	aes_dat[127:96]	RW	Write Plain byte15~byte12 Read Ciphered Byte15~byte12

field	name	Access	Description
AESKEY_1			
31:0	aeskey[31:0]	RW	AES key byte3~byte0

field	name	Access	Description
AESKEY_2			
31:0	aeskey[63:32]	RW	AES key byte7~byte4

field	name	Access	Description
AESKEY_3			
31:0	aeskey[95:64]	RW	AES key byte11~byte8

field	name	Access	Description
AESKEY_4			
31:0	aeskey[127:96]	RW	AES key byte15~byte12

field	name	Access	Description

.

	AESINT		
31:1	Reserved	RW	
0	aesint_status	RW	Set when AES is complete. Writing 1 to clear the interrupt Status.

field	name	Access	Description
	AESINTMASK		
31:1	Reserved	RW	
0	aesintmask	RW	AES done interrupt mask. When 1, allows the interrupt to assert the interrupt.

field	name	Access	Description
	SWAPCTRL		
31:2	Reserved	RW	
1	data_swap	RW	1: Plain and Ciphered data swap 0: No swap
0	Key_swap	RW	1: Key swap 0: No swap

14 Interrupt

14.1 Futures

- 13 (not including the sixteen MCU interrupt lines)
- 16 programmable priority levels (4 bits of interrupt priority are used)
- Low-latency exception and interrupt handling
- Implementation of System Control Registers

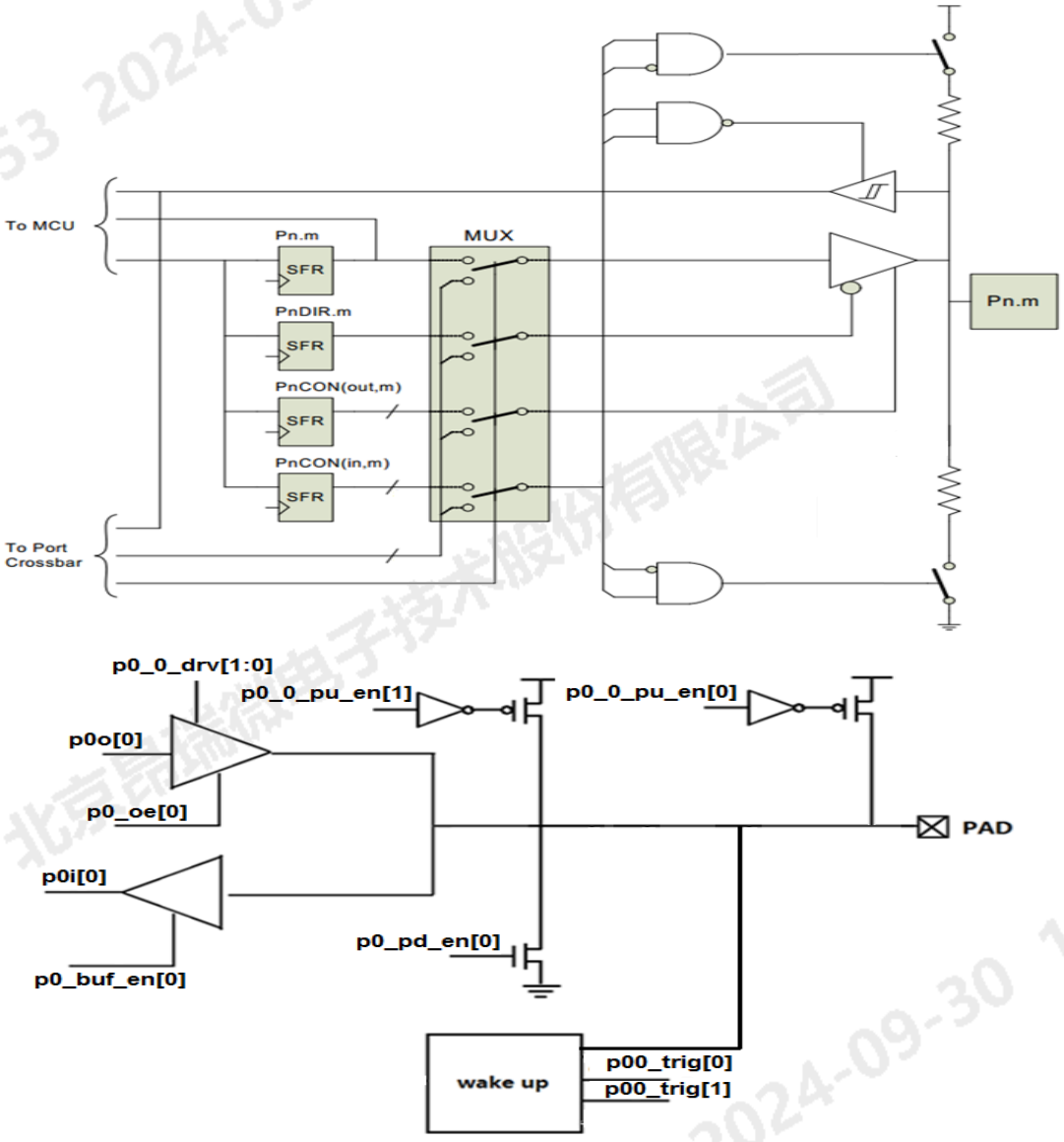
14.2 Interrupt vectors

position	Priority	Type of Priority	source	Address
	-	-	Reserved	0x0000_0000
	-3	fixed	Reset	0x0000_0004
	-2	fixed	NMI	0x0000_0008
	-1	fixed	HardFault	0x0000_000C
	0	settable	MemManage	0x0000_0010
	1	settable	BusFault	0x0000_0014
	2	settable	UsageFault	0x0000_0018
	-	-	Reserved	
	3	settable	SVCall	0x0000_002C
	4	settable	Debug Monitor	0x0000_0030
	-	-	Reserved	
	5	settable	PendSV	0x0000_0038
	6	settable	SysTick	0x0000_003C
0	7	RTC	RTC	0x0000_0040
1	8	USB_INT	USB	0x0000_0044
2	9	USB_WK_INT	USB	0x0000_0048
3	10	RF	RF	0x0000_004C
4	11	WAKEUP/GPIO	GPIO	0x0000_0050
5	12	TIMERT_INT	TIMER	0x0000_0054
6	13	1'b0	1'b0	0x0000_0058
7	14	WDT	Watchdog	0x0000_005C
8	15	FLASH	FLASH	0x0000_0060
9	16	ENCODER	ENCODER	0x0000_0064
10	17	RF_DMA_INT	RF	0x0000_0068
11	18	AES	AES	0x0000_006C
12	19	UART	UART	0x0000_0070
13	20	Sensor_scan		0x0000_0074
14	21	Ledc_int	LEDC	0x0000_0078

15 GPIO Purpose IO Port and Pin Assignment

The IO pins of the OM6239 are default set to general purpose IO for the MCU. The numbers of available IOs are 11 for the OM6239. The IO pins provide the input, output and in-out function to communicate with other components.

15.1 Block diagram



15.2 Features

- Configurable Direction
- Configurable Drive Strength
  - Output buffer on, normal drive strength

.

- Output buffer on, high drive strength

- Configurable Pull Up/Down
  - Input buffer on, no pull up/down resistor
  - Input buffer on, pull up resistor
  - Input buffer on, pull down resistor
  - Input buffer off

15.3 IO pin maps

The following conventions are used in all pin out maps:

pin	USB	JTAG	UART	PWM	LED RZ	2V SENSOR	1M pull up	Sensor Scan		Wheel Scan	LVD IO
P3.6											
P3.5											
P3.4											
P3.3					LED_RZ						LVD
P3.2				PWM							
P3.1											
P3.0											
P2.7											
P2.6				PWM							
P2.5		TMS	UART_TX				1M pull up			WHA	
P2.4										WHB	
P2.3							1M pull up				LVD
P2.2		TCK	UART_RX	PWM							
P2.1					LED_RZ					WHA	
P2.0										WHB	
P1.3						2V INPUT	1M pull up	MISO			
P1.2			UART_TX				1M pull up	MOSI	SDIO		
P1.1			UART_RX					SCK	SCK		
P1.0								CSN			
P0.7						2V INPUT					
P0.6			UART_RX	PWM			1M pull up			WHA	
P0.5				PWM	LED_RZ					WHB	
P0.4			UART_TX	PWM			1M pull up				
P0.3	D+										
P0.2	D-										
P0.1				PWM	LED_RZ					WHA	
P0.0				PWM						WHB	

15.4 AHB GPIO Register address map

Below is the AHB GPIO register address map:

Address Range	Description
0x4300 0000 to 0x4300 FFFF	AHB GPIO register space

15.5 AHB GPIO register map

Offset address	Name	Access	Reset value	Description
Base address: 0x43000000				
0x00000	DATA	RW	0X-----	Data value
0X00004	DATAOUT	RW	0X00000000	Data output Register value
0X00010	OUTENSET	RW	0X00000000	Output enable set
0X00008	DATAOUTSET	RW	0X00000000	Output set
0x00000c	DATAOUTCLR	RW	0x00000000	Ouptu clear
0X00014	OUTENCLR	RW	0X00000000	Output enable clear
0X00020	INTENSET	RW	0X00000000	Interrupt enable set
0X00024	INTENCLR	RW	0X00000000	Interrupt enable clear
0X00028	INTTYPE0SET	RW	0X00000000	Interrupt type0 set
0X0002C	INTTYPE0CLR	RW	0X00000000	Interrupt type0 clear
0X00030	INTPOLSET	RW	0X00000000	Polarity-level, edge IRQ cfg
0X00034	INTPOLCLR	RW	0X00000000	Polarity-level, edge IRQ cfg
0X00038	INTSTATUS	RW	0X00000000	IRQ status Register
0X00040	INTTYPE1SET	RW	0X00000000	Interrupt type1 set
0X00044	INTTYPE1CLR	RW	0X00000000	Interrupt type1 clear

15.6 AHB GPIO register description

field	name	Access	Description
DATA			
31:16	Reserved	NA	
15:0	data	RW	Write: To data output register Read: Sampled at pin

.

field	name	Access	Description
DATAOUT			
31:16	Reserved	NA	
15:0	dataout	RW	Write: To data output register Read: Current value of data output value

field	name	Access	Description
DATAOUTSET			
31:16	Reserved	NA	
15:0	dataoutset	RW	Write 1 to set the correspond gpio Write 0 have no effect

field	name	Access	Description
DATAOUTCLR			
31:16	Reserved	NA	
15:0	dataoutclr	RW	Write 1 to clear the correspond gpio Write 0 have no effect

field	name	Access	Description
OUTENSET			
31:16	Reserved	NA	
15:0	outenset	RW	Write: 1 Set the output enable bit Write: 0 No effect Readback 0 Indicates the signal direction as input Readback 1 Indicates the signal direction as output

field	name	Access	Description
OUTENCLR			
31:16	Reserved	NA	
15:0	outenclr	RW	Write: 1 Clear the output enable bit Write: 0 No effect Readback 0 Indicates the signal direction as input Readback 1 Indicates the signal direction as output

field	name	Access	Description
INTENSET			
31:16	Reserved	NA	
15:0	Intenset	RW	Write: 1 Set the enable bit Write: 0 No effect Readback 0 Interrupt disabled

			Readback 1 Interrupt enabled

INTENSET			
31:16	Reserved	NA	
15:0	Intenclr	RW	Write: 1 Clear the enable bit Write: 0 No effect Readback 0 Interrupt disabled Readback 1 Interrupt enabled

field	name	Access	Description
INTTYPE0SET			
31:16	Reserved	NA	
15:0	Inttype0set	RW	Write: 1 Set the interrupt type0 bit Write: 0 No effect Readback 0 For Low or High level Readback 1 For falling or rising edge

field	name	Access	Description
INTTYPE0CLR			
31:16	Reserved	NA	
15:0	Inttype0clr	RW	Write: 1 Clear the interrupt type0 bit Write: 0 No effect Readback 0 For Low or High level Readback 1 For falling or rising edge

field	name	Access	Description
INTPOLSET			
31:16	Reserved	NA	
15:0	Intpolset	RW	Write: 1 Set the interrupt polarity bit Write: 0 No effect Readback 0 For Low level or falling edge Readback 1 For High level or rising edge

field	name	Access	Description
INTPOLCLR			
31:16	Reserved	NA	
15:0	Intpolclr	RW	Write: 1 Clear the interrupt polarity bit Write: 0 No effect Readback 0 For Low level or falling edge

			Readback 1 For High level or rising edge

field	name	Access	Description
INTSTATUS			
31:16	Reserved	NA	
15:0	Intstatus	RW	Write: 1 clear the interrupt request Write: 0 No effect Readback IRQ status Register

field	name	Access	Description
INTTYPE1SET			
31:16	Reserved	NA	
15:0	Inttype1set	RW	Write: 1 Set the interrupt type1 bit Write: 0 No effect Readback 0 For Low or High level or falling edge or rising edge Readback 1 For falling and rising edge

field	Name	Access	Description
INTTYPE1CLR			
31:16	Reserved	NA	
15:0	Inttype1clr	RW	Write: 1 Clear the interrupt type1 bit Write: 0 No effect Readback 0 For Low or High level or falling edge or rising edge Readback 1 For falling and rising edge

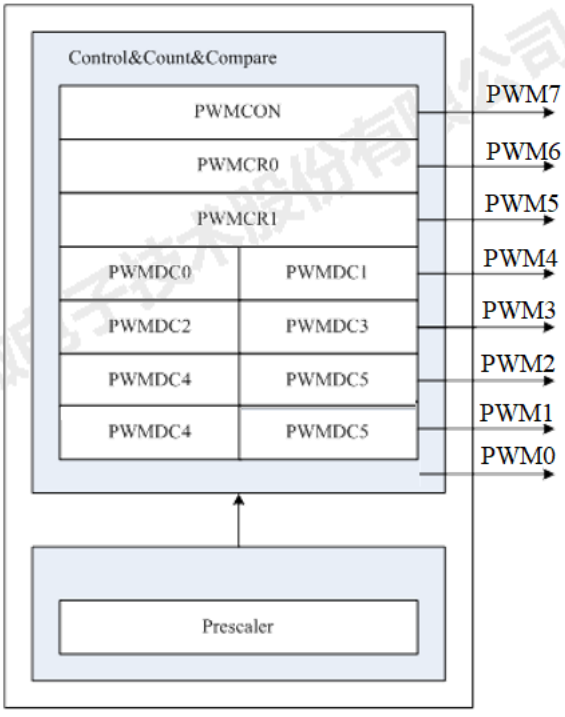
16 PWM

The OM6239 includes a six channel Pulse-Width Modulation (PWM) module. The eight channels (PWM0, PWM1, PWM2,PWM3, PWM4 PWM5 PWM6and PWM7) share a common enable/disable PWM and frequency pre-scale factor register, a period setting register, a controlled Initial level register and a controlled active level register, and have an individually controlled active level width register. The six PWM waveforms are the same period, and can have different Initial level, active level and inactive level width.

16.1 Features

- Six-channel output
- Frequency-range from 2KHz to 8 MHz
- Compact control using few registers for enabling, frequency pre-scale, period, Initial level, active level and inactive level width

16.2 Block diagram



16.3 Functional description

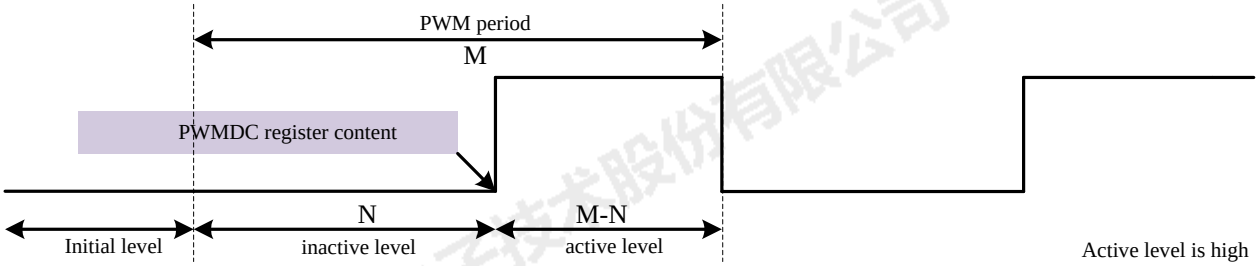
The OM6239 PWM is a eight-channel PWM with a ten registers interface. The first register PWMCON enables or disables the PWM function and sets the PWM frequency pre-scale factor. The register PWMPRD sets the PWM period length, which is the number of counter clock cycles for one PWM period. The register PWMCR0 controls the PWM initial level. The register PWMCR1 controls the PWM active level. The registers, PWMDC0, PWMDC1, PWMDC2, PWMDC3, PWMDC4 PWMDC5 PWMDC6and PWMDC7 control the inactive level for each PWM channel.

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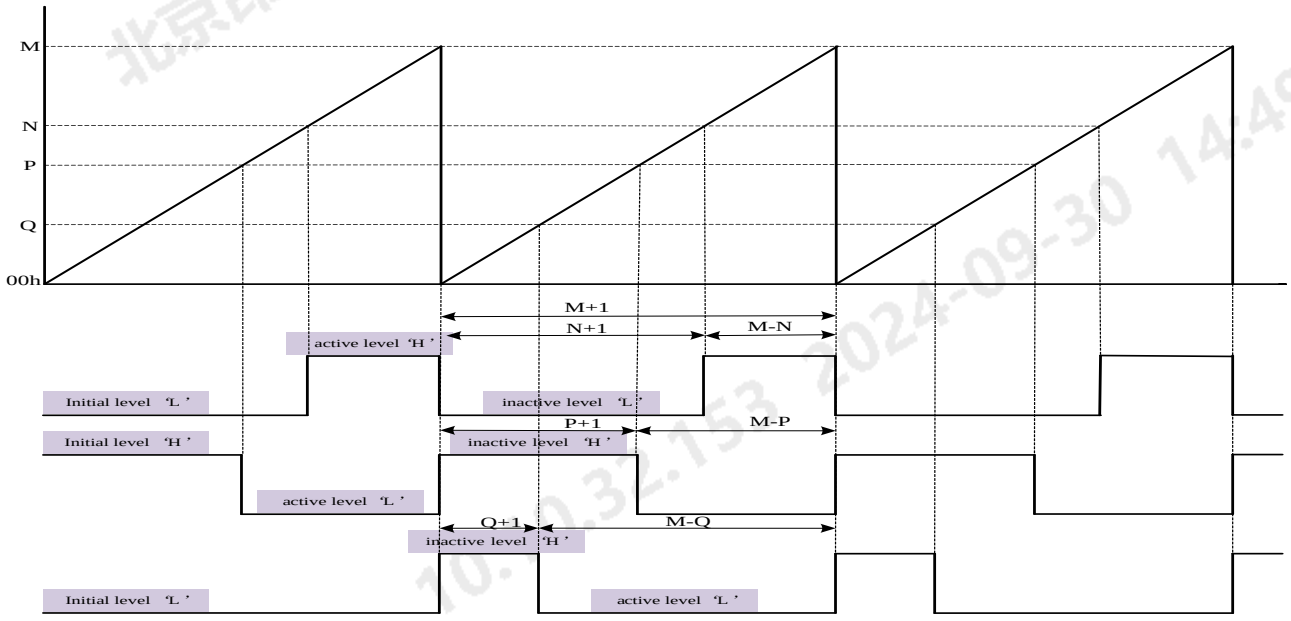
If you disable one of the six PWMs, the PWM output pin retains the inactive level after the output is changed by matching the period. The following table shows how the PWM is controlled by the PWM registers. Frequency-range is approximately 2KHz-8MHz.

Item	Specification
Count source	1/1, 1/2, 1/8, 1/32 of PWM clock (PWMDIV)
Count operation	Increment
PWM waveform	PWM period: 1/frq × (M + 1) Active level width: 1/frq × (M- N) Inactive width: 1/frq × (N + 1) frq: Frequency of count source M: Value set in PWMPRD register N:Value set in PWMDCj register (j = 0, 1, 2, 3, 4, 5, 6, 7) refer to the following figure
Selectable functions	Initial level selectable for each individual pin (PWMCR0) Active level selectable for each individual pin (PWMCR1)

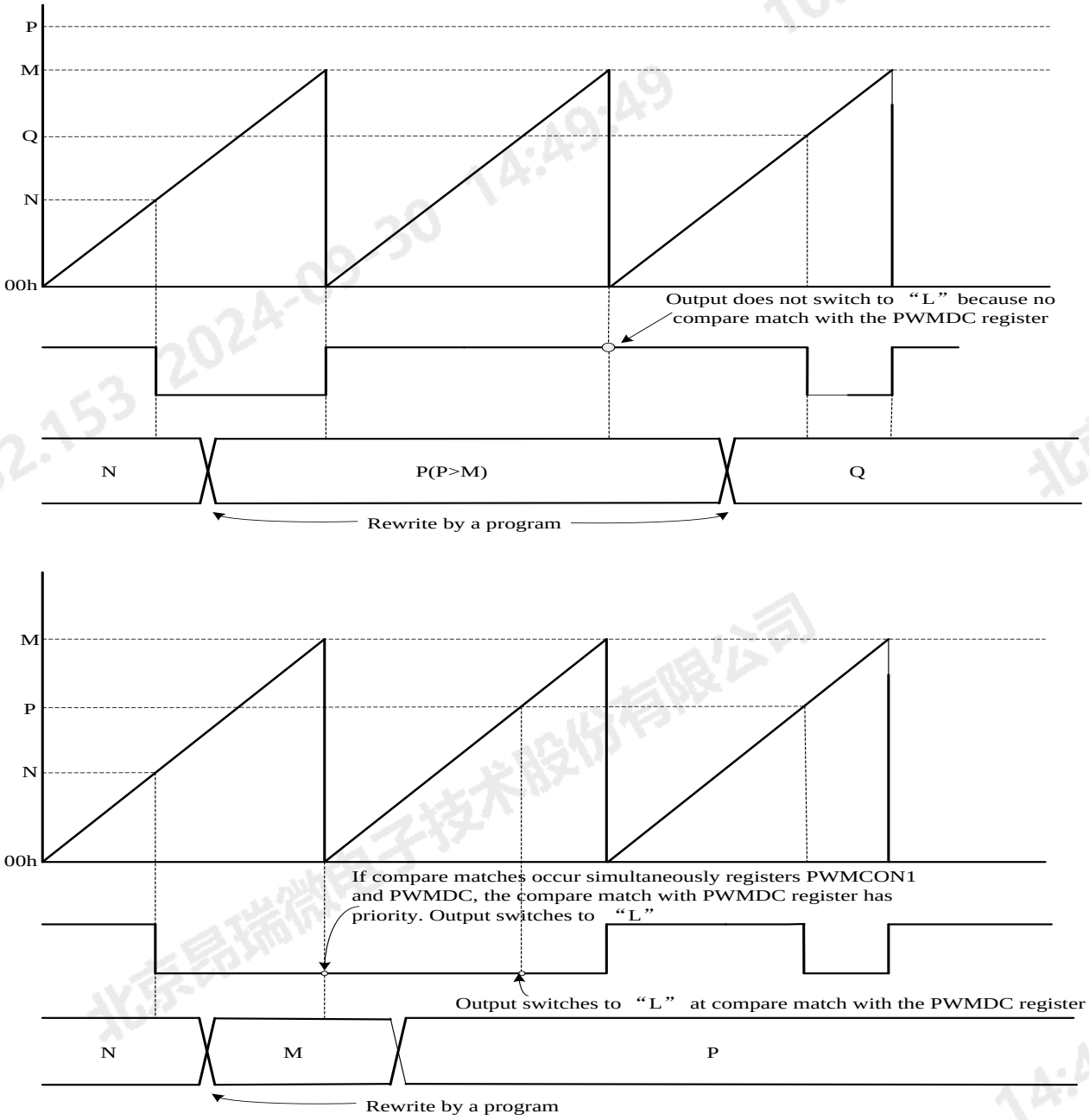
Upon matching PWMDC register contents, the output signal changes from inactive level to active level. It goes back to inactive level by matching the period (refer to the PWMPRD register content).



Operating example in PWM mode is shown in figure below.



Operating example in PWM mode is shown in figure below.



16.4 PWM Register address map

Below is the PWM register address map:

Address Range	Description
0x4000 0800 to 0x4000 08FF	PWM register space

.

16.5 PWM register map

Offset address	Name	Access	Reset value	Description
Base address : 0x40000800				
0x00000	PWMCON	RW	0x00000000	
0X00004	PWMCR0	RW	0x00000000	
0X00008	PWMCR1	RW	0x00000000	
0X0000C	PWMDIV	RW	0x00000000	
0X00010	PWMPRD	RW	0x00000000	
0X00014	PWMDC0	RW	0x00000000	
0X00018	PWMDC1	RW	0x00000000	
0X0001C	PWMDC2	RW	0x00000000	
0X00020	PWMDC3	RW	0x00000000	
0X00024	PWMDC4	RW	0x00000000	
0X00028	PWMDC5	RW	0x00000000	
0X00029	PWMDC6	RW	0x00000000	
0X0002a	PWMDC7	RW	0x00000000	

16.6 PWM register description

field	name	Access	Description
PWMCON			
31:6	Reserved	NA	
7	en_7	RW	Enable bit for channel 7(pwm7) 1:enable 0:disable
6	en_6	RW	Enable bit for channel 6(pwm6) 1:enable 0:disable
5	en_5	RW	Enable bit for channel 5(pwm5) 1:enable 0:disable
4	en_4	RW	Enable bit for channel 4(pwm4) 1:enable 0:disable
3	en_3	RW	Enable bit for channel 3(pwm3) 1:enable 0:disable
2	en_2	RW	Enable bit for channel 2(pwm2) 1:enable

			0:disable
1	en_1	RW	Enable bit for channel 1(pwm1) 1:enable 0:disable
0	en_0	RW	Enable bit for channel 0(pwm0) 1:enable 0:disable

field	name	Access	Description
	PWMCR0		
31:6	Reserved	NA	
7	Initial_ctl7	RW	Initial output selected bit for channel 7(pwm7) 1: Initial output selected as active level 0: Initial output selected as non-active level
6	Initial_ctl6	RW	Initial output selected bit for channel 6(pwm6) 1: Initial output selected as active level 0: Initial output selected as non-active level
5	Initial_ctl5	RW	Initial output selected bit for channel 5(pwm5) 1: Initial output selected as active level 0: Initial output selected as non-active level
4	Initial_ctl4	RW	Initial output selected bit for channel 4(pwm4) 1: Initial output selected as active level 0: Initial output selected as non-active level
3	Initial_ctl3	RW	Initial output selected bit for channel 3(pwm3) 1: Initial output selected as active level 0: Initial output selected as non-active level
2	Initial_ctl2	RW	Initial output selected bit for channel 2(pwm2) 1: Initial output selected as active level 0: Initial output selected as non-active level
1	Initial_ctl1	RW	Initial output selected bit for channel 1(pwm1) 1: Initial output selected as active level 0: Initial output selected as non-active level
0	Initial_ctl0	RW	Initial output selected bit for channel 0(pwm0) 1: Initial output selected as active level 0: Initial output selected as non-active level

field	name	Access	Description
	PWMCR1		
31:6	Reserved	NA	
7	active_ctl7	RW	Active output selected bit for channel 7(pwm7) 1:output level selected as high active 0:output level selected as low active

6	active_ctl6	RW	Active output selected bit for channel 6(pwm6) 1:output level selected as high active 0:output level selected as low active
5	active_ctl5	RW	Active output selected bit for channel 5(pwm5) 1:output level selected as high active 0:output level selected as low active
4	active_ctl4	RW	Active output selected bit for channel 4(pwm4) 1:output level selected as high active 0:output level selected as low active
3	active_ctl3	RW	Active output selected bit for channel 3(pwm3) 1:output level selected as high active 0:output level selected as low active
2	active_ctl2	RW	Active output selected bit for channel 2(pwm2) 1:output level selected as high active 0:output level selected as low active
1	active_ctl1	RW	Active output selected bit for channel 1(pwm1) 1:output level selected as high active 0:output level selected as low active
0	active_ctl0	RW	Active output selected bit for channel 0(pwm0) 1:output level selected as high active 0:output level selected as low active

field	name	Access	Description
PWMDIV			
31:2	Reserved	NA	
1:0	pre-scale	RW	Frequency pre-scale factor 00: 1/1 clock 01: 1/2 clock 10: 1/8 clock 11: 1/32 clock

field	name	Access	Description
PWMPRD			
31:12	Reserved	NA	
11:0	pwm_prd	RW	PWM period length

field	name	Access	Description
PWMDC0			
31:12	Reserved	NA	
11:0	pwmdc0	RW	PWM inactive level width control for channel 0

field	name	Access	Description
	PWMDC1		
31:12	Reserved	NA	
11:0	pwmdc1	RW	PWM inactive level width control for channel 1

field	name	Access	Description
	PWMDC2		
31:12	Reserved	NA	
11:0	pwmdc2	RW	PWM inactive level width control for channel 2

field	name	Access	Description
	PWMDC3		
31:12	Reserved	NA	
11:0	pwmdc3	RW	PWM inactive level width control for channel 3

field	name	Access	Description
	PWMDC4		
31:12	Reserved	NA	
11:0	pwmdc4	RW	PWM inactive level width control for channel 4

field	name	Access	Description
	PWMDC5		
31:12	Reserved	NA	
11:0	pwmdc5	RW	PWM inactive level width control for channel 5

field	name	Access	Description
	PWMDC6		
31:12	Reserved	NA	
11:0	Pwmdc6	RW	PWM inactive level width control for channel 6

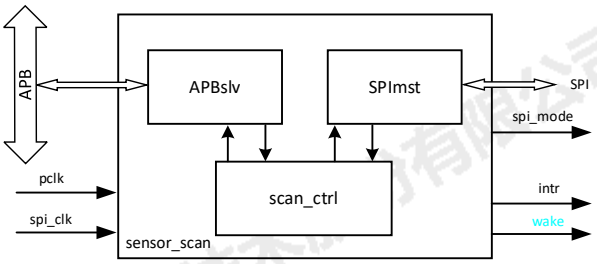
field	name	Access	Description
	PWMDC7		
31:12	Reserved	NA	
11:0	Pwmdc7	RW	PWM inactive level width control for channel 7

17 Sensor scan (SENSCAN)

17.1 Features

- APB interface
- 4-/2-wire SPI
- single mode and continuous mode
- Scanning time can be configured
- Re-synchronization time can be adjusted
- data format: 8-bit, 12-bit and 16-bit
- Interrupt
- sensor configure/scan mode

17.2 Block diagram



The spi_clk is 2MHz and SCLK is 1MHz.

***Interface*:**

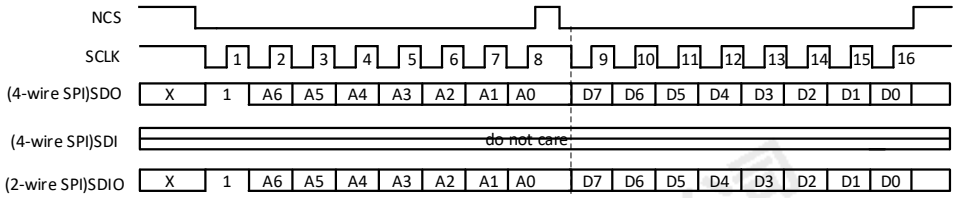
Port Name	I/O	Description
pclk	I	APB clock signal
presentn	I	APB async reset signal(active low)
paddr[7:0]	I	APB address bus
penable	I	APB strobe signal
pwrite	I	APB write signal
psel	I	APB peripheral select signal
pwdata[31:0]	I	APB write data bus
prdata[31:0]	O	APB read data bus
spi_clk	I	SPI clock signal
spi_rstn	I	SPI async reset signal(active low)
sclk	O	serial clock signal
ncs	O	chip select(active low)
txd	O	transmit data signal

rxd	I	receive data signal
ss_oe	O	output enable(The signal is valid when SPIMODE=1)
sensor_intr	O	senscan interrupt
spi_mode	O	SPI mode
wake	O	唤醒信号

17.3 Functional Description

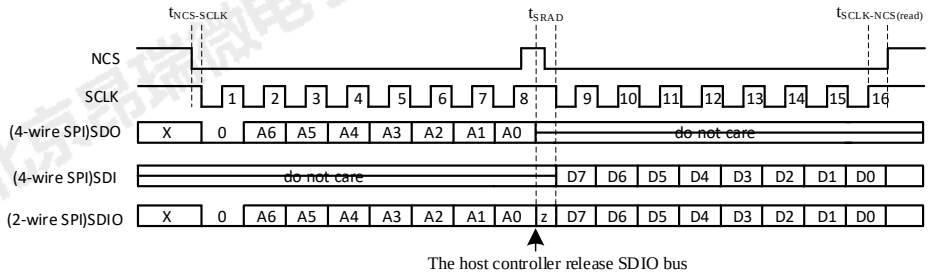
17.3.1 SPI Operation

Write Operation



Read Operation

The read operation contains a total of two Byte of data, the first byte is sent by the host (MOSI) and is the register address of the sensor (A6~A0), where the 7th bit is 0, representing the read operation. The second byte is sent by the sensor (MISO) and is the data in the sensor register. The timing diagram is as follows:



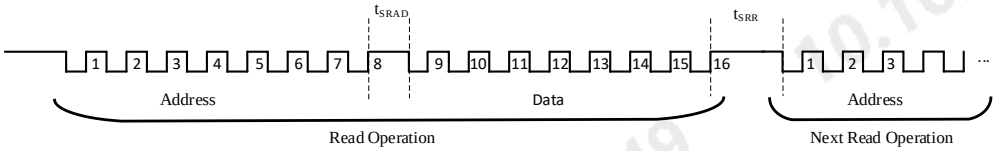
Note：

1. $t_{NCS-SCLK}$: From last NCS falling edge to first SCLK rising edge.

2. $t_{SCLK-NCS\ (read)}$: From last SCLK rising edge to NSC rising edge, for vaild MISO data transfer.

3. t_{SRAD} : From rising SCLK for last bit of the address byte, to falling SCLK for first bit of data being read.

A delay is required between two consecutive read operations to ensure that the sensor has time to prepare the data. As shown in the figure below:

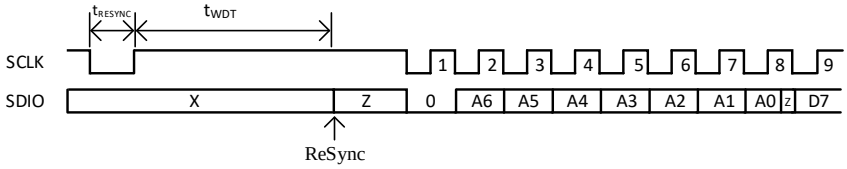


t_{SRR}: From rising **NCS** for last bit of the first data byte, to falling NCS for first bit of the address byte if the next commad.

Re-synchronization of the SPI

resync : For 2-wired SPI when the read sensor ID and the configured ID don't match, it will do re-sync. the SCLK will be pulled down for the first period of time and then pulled up again, the time of SCLK pulling down and pulling up can be matched, as follows.

In two-wire configuration mode, resync can be initiated proactively by writing RESYNC_STRT.



17.3.2 Data Format

The sensor coordinate data format supports 8-bit, 12-bit and 16-bit and is configurable via delta_format in the SENSCAN_CSR register.

When the coordinate data format is 8-bit, you need to configure SENSCAN_XID (Sensor Delta_X low address) and SENSCAN_YID (Sensor Delta_Y low address), and you also need to configure the SENSCAN_XYID1 and SENSCAN_XYID2 registers to 0xFF (invalid address). When scanning, the value of Sensor Delta_X low and Delta_Y low will be read in turn, where bit7 is the sign bit.

When the coordinate data format is 8-bit, if the configuration in the SENSCAN_XYID1 or SENSCAN_XYID2 register is not 0x7F, the register with the corresponding address in the sensor will be read when scanning, but the read data is invalid.

When the coordinate data format is 12-bit, you need to configure SENSCAN_XID (Sensor Delta_X low-bit address) and SENSCAN_YID (Sensor Delta_Y low-bit address), and if the high bits of Delta_X and Delta_Y share a common register, you need to configure SENSCAN_XYID1 as the address of the high-bit shared register, and SENSCAN_XYID2 is configured as an invalid address. When scanning, the Delta_X low register, Delta_Y low register and the shared high register of the sensor will be read sequentially. If the deltaxy_rev field in SENSCAN_CSR is 0x0, bit7~bit4 in the read out high bit data indicates the high bit of Delta X, and bit3~bit0 indicates the high bit of Delta Y. If the deltaxy_rev field in SENSCAN_CSR is 0x1, bit7 ~bit4 denotes the high bit of Delta Y, and bit3~bit0 denotes the high bit of Delta X.

When the coordinate data format is 12-bit, if the high level data of the sensor Delta_X and Delta_Y are stored in two registers, you need to configure SENSCAN_XYID1 as the address of the Delta_X high register and

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SENSCAN_XYID2 as the address of the Delta_Y high register. When scanning, the Delta_X low register, Delta_Y low register, Delta_X high register and Delta_Y high register of the sensor will be read sequentially.

When the coordinate data format is 16-bit, you need to configure SENSCAN_XID (Sensor Delta_X low address), SENSCAN_YID (Sensor Delta_Y high address), SENSCAN_XYID1 (Delta_X high address), and SENSCAN_XYID2 (Delta_Y high address). When scanning, the Delta_X low register, Delta_Y low register, Delta_X high register and Delta_Y high register of the sensor will be read sequentially.

SENSCAN will splice the read data, accumulate it, and store it in the scan data registers (SENSCAN_X and SENSCAN_Y).

When there is valid data in the Scan Data Register, DATAVLD=1. This bit is set by hardware to 1. This bit clears the zero condition:

1) When all deltas are read. Reading only one Delta (e.g., reading only Delta X) or reading the same Delta repeatedly (e.g., reading Delta X repeatedly) does not clear this bit. The bit is cleared only when Delta X and Delta Y are read (in no particular order).

2) Restart scanning. Aborting a scan or stopping a scan without reading data clears the bit when the scan is started again.

3) This bit is cleared when configuration senscan_csr[10]=1.

Scan data register clear condition as above.

The scan data is a 24bit signed number. The register data is cleared after the data is read, when the scan is restarted, or when DELTA_CLR=1 is configured.

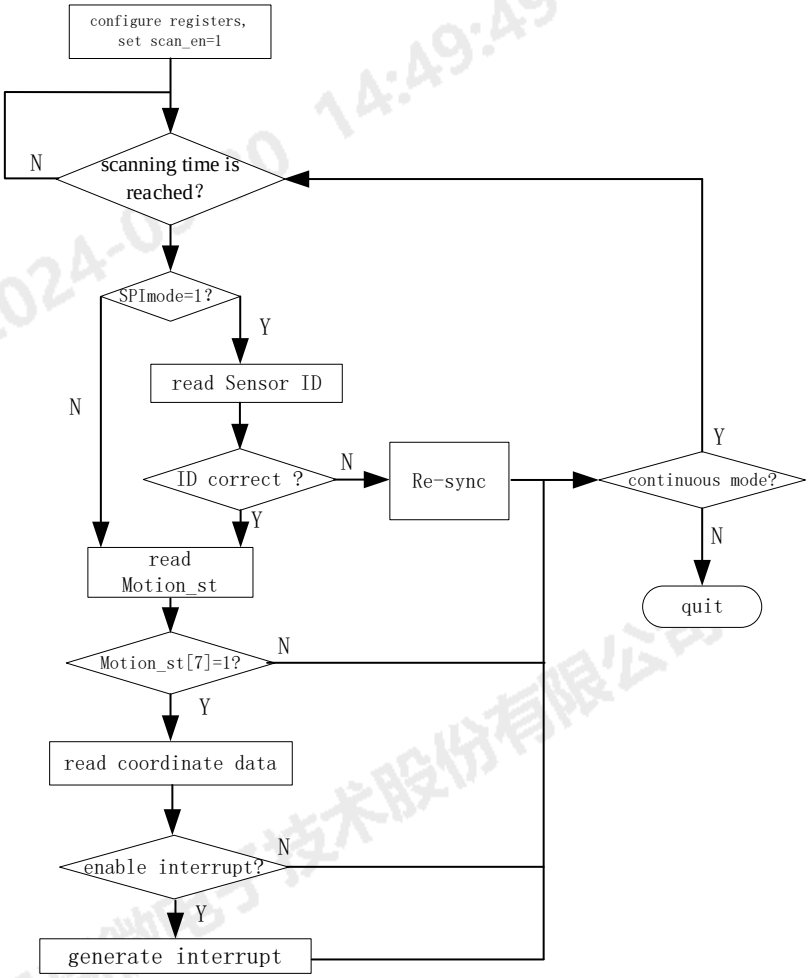
17.3.3 Scanning Mode

Hardware Scanning Process

1. Start the scan.
2. For 2-wire SPI: read the sensor ID, if the sensor ID read does not match the ID configured in SENSCAN, then Re-Sync will be performed;
3. Reads the sensor's Motion_st register:
    a. If the sensor's Motion_st[7]=0, the single scan will end directly, and the continuous scan will wait until the next scan count reaches the set value, then execute 3.
    b. If sensor's Motion_st[7]=1, it means that sensor has data to be read, execute 4. (When sensor's Motion_st register is read, it locks the coordinate data register, and the coordinate data is cleared to zero after the coordinate data is read).
4. According to the sensor data register address, read the sensor coordinate data in order: DetlaX_L, DetlaY_L, DetlaX_H, DetlaY_H. (The data register address is invalid when it is configured to be 0xFF).
5. If the scan_done interrupt is turned on, an interrupt will be generated each time a read is completed.

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6. Repeat 2~5 if in continuous scan mode.



17.3.4 Single Mode

The scan_mode field of 0 in SENSCAN_CSR is a single scan. Configure the sensor related registers and scan mode, writing scan_en=1 will clear the scan count and coordinate data registers and start the scan timer. When the scanning timer reaches the time, it will read the IP (two-wire SPI), Motion status and coordinate data of the sensor in turn, after the scanning is finished, scan_en will be pulled down automatically by the hardware, and the next scanning will need to reset scan_en. if scan_done interrupt is enabled, the interrupt will be generated when the scanning is finished, so it is necessary to read the coordinate data and clear the interrupt in time. If scan_done interrupt is enabled, the interrupt will be generated when the scan is completed, and the coordinate data should be read and cleared in time. If the data is not read in time, the coordinate data will be cleared next time when scan_en=1 is written or the corresponding registers are read. During a single scan, if bit7 of Motion_st is 0, it will clear scan_en to zero. Or if the clock data is not synchronised, it will clear scan_en at the end of Resync (2-wire SPI).

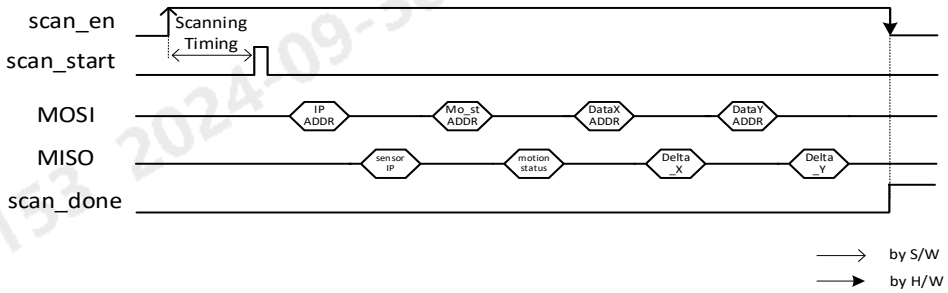
Writing scan_en=1 locks the scan configuration, including: scan mode, data format, sensor ID and sensor register

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address, scan interval and resync time.

Writing scan_stp=1 stops the current scan, clears the scan timer count, and the data obtained from the scan is not stored in the data register. The hardware automatically clears scan_stp and scan_en when the scan stops.

If the sensor's registers are being read when scan_stp=1 is written, scan_stp and scan_en are cleared when the register read is complete. If a Resync is in progress when scan_stp=1 is written, scan_stp and scan_en are cleared when the Resync is completed.



17.3.5 Continuous Mode

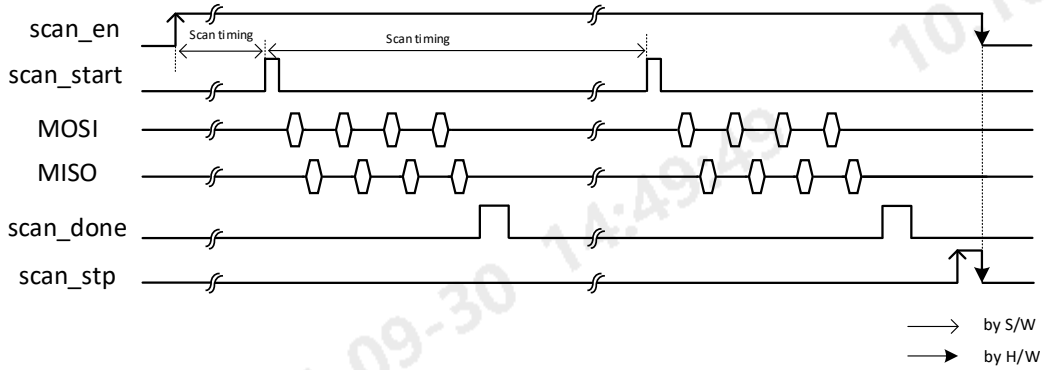
The scan_mode field in SENSCAN_CSR is 1 for continuous scanning. Configure the sensor related registers and scan mode, writing scan_en=1 will clear the scan count and coordinate data registers and start the scan timing. After reaching the scanning timer, it will read the IP (two-wire SPI), Motion status and coordinate data of the sensor in turn, and scan_en will not be pulled down by hardware after the scanning is completed. The scan_en will be automatically cleared and restarted when the scan_timer reaches the scan_done time. If scan_done interrupt is enabled, it will be generated at the completion of each scan, coordinate data need to be read in time and clear the interrupt, if it fails to read the data in time, the coordinate data will be accumulated at the completion of the next scan.

Writing scan_en=1 locks the scan configuration, including: scan mode, data format, sensor ID and sensor register address, scan interval and resync time.

Writing scan_stp=1 stops the current scan and clears the scan timer count, but the previous coordinate data will not be cleared, and the coordinate data will be cleared the next time scan_en=1 is written or automatically after a read. When scanning stops, the hardware will automatically clear scan_stp and scan_en.

If the sensor's registers are being read when scan_stp=1 is written, scan_stp and scan_en are cleared when the register read is complete. If a Resync is in progress when scan_stp=1 is written, scan_stp and scan_en are cleared when the Resync completes.

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17.3.6 Frequency of Scanning

The scan frequency can be configured via the SENSCAN_FREQ register and the t_{scan} is: (SENSCAN_FREQ +1)*0.25ms.

17.3.7 Procedure to Use SENSCAN

1. Configure the sensor ID register address (SENSCAN_IDADDR), the sensor ID (SENSCAN_ID), the sensor motion status register address (SENSCAN_MOADDR) and the sensor coordinate data register addresses (SENSCAN_XADDR, SENSCAN_ YADDR, SENSCAN_HADDR1 and SENSCAN_HADDR2);
2. Configure the scan interval and resync time;
3. Enable scan_done interrupt;
4. Configure the scan mode (SCAN_MODE), data format (SCAN_FORMAT, etc.);
5. Ensure that CFG_CSR.SENCFG_EN=0 and write SCAN_EN=1;
6. Wait for an interrupt to read the coordinate data;

If you need to stop scanning, you need to write SCAN_STP=1 and wait for SCAN_EN and SCAN_STP to be pulled low.

17.3.8 Scan mode switching

Continuous mode switching to single mode: you need to stop continuous scanning first and wait for both SENSCAN_STP and SENSCAN_EN to be pulled down before turning on word scanning.

Switching from single mode to continuous mode: you need to start continuous scanning when both SENSCAN_STP and SENSCAN_EN are 0.

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17.3.9 sensor configuration mode

When CFG_CSR.SENCFG_EN=1, the sensor register can be configured and read by SENSCAN. It is necessary to configure the sensor register address (CFG_ADDR) and direction (CFG_CSR.SENCFG_DIR), when CFG_CSR.SENCFG_DIR=1 for write operation, when CFG_CSR.SENCFG_DIR=0 for read operation. For write operation, data needs to be written to CSR_DATA For read operation, read CSR_DATA register to get data. Read/write completion: If interrupt is enabled (CFG_CSR.SENCFG_INTEN=1),interrupt will be generated when read/write is completed, or by polling CFG_CSR.SENCFG_STRT, CFG_CSR.SENCFG_STRT will be pulled down automatically when the operation is completed.

17.3.10 Configuration mode and scan mode switching

If it is in scan mode, you need to stop the current scan first, configure SENSCAN_STP=1, wait for SENSCAN_STP=0, then CFG_CSR.SENCFG_EN=1, and do the sensor configuration.

If you are in config mode, you need to change to scan mode (CFG_CSR.SENCFG_EN=0) first after the configuration is finished, and then do the sensor scan.

17.4 Interrupts

Event flag	Enable control bit	Interrupt event
CFG_DONE	CFG_DONE_IE	Configuration mode, read or write data completion.
SCAN_RESYNC	SCAN_RESYNC_IE	scanning timer is reached during the Re-synchronization period.
RESYNC_DONE	RESYNC_DONE_IE	The resync completes in two-wire mode.
RESYNC_STRT	RESYNC_STRT_IE	the 2-wired SPI is out of synchronization.
SCAN_DONE	SCAN_DONE_IE	the scanning is completed.

17.5 SENSCAN Register address map

Below is the SENSCAN register address map:

Address Range	Description
0x40019000 to 0x40019FFF	SENSCAN register space

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17.6 SENSCAN Register map

Offset	Name	Access	Reset	Description
0x00	SENSCAN_CSR	RW/RS	0x0	SENSCAN control and status register
0x04	SENSCAN_ISR	RC_W1	0x0	SENSCAN interrupt status register
0x08	SENSCAN_IER	RW	0x0	SENSCAN interrupt enable register
0x0C	SENSCAN_IDADDR	RW	0x0	sensor ID address register
0x10	SENSCAN_ID	RW	0x0	sensor ID
0x14	SENSCAN_MOADDR	RW	0x0	sensor's motion status register address
0x18	SENSCAN_XADDR	RW	0x7F	DeltaX lower-bit data register address
0x1C	SENSCAN_YADDR	RW	0x7F	DeltaY lower-bit data register address
0x20	SENSCAN_HADDR1	RW	0x7F	upper-bit data register address 1
0x24	SENSCAN_HADDR2	RW	0x7F	upper-bit data register address 2
0x28	SENSCAN_X	RO	0x0	DeltaX
0x2C	SENSCAN_Y	RO	0x0	DeltaY
0x30	SENSCAN_TR	RW	0x08	t_{SRR} & t_{SRAD} Configuration Register
0x34	SENSCAN_TNS	RW	0x0	$t_{NCS-SCLK}$ & $t_{SCLK-NCS}$ Configuration Register
0x38	SENSCAN_FREQ	RW	0x0	frequency of scanning
0x3C	SENSCAN_RESYNC	RW	0x090E	control of re-synchronization
0x40	SENCFG_CSR	RW	0x0	Configuration mode: control register
0x44	SENCFG_ADDR	RW	0x0	Configuration mode: register address of sensor
0x48	SENCFG_DATA	RW	0x0	Configuration mode: write/ read data register

Note: RC_W1: Software can read as well as clear this bit by writing 1. Writing 0 has no effect on the bit value.

Note: RS: Software can read as well as set this bit. Writing 0 has no effect on the bit value.

17.7 SENSCAN register description

SENSCAN_CSR

Bit	Field	Access	Description

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31:17	Reserved	NA	-
16	WAKE_EN	RW	Enable sensor_wake. sensor_wake signal is generated when SENSCAN_ISR has state pulled high. When the corresponding state in SENSCAN_ISR is cleared, the sensor_wake signal is cleared.
15:12	Reserved	NA	-
11	SCAN_BUSY	RO	If this bit is 1, it indicates that scanning is in progress.
10	DELTA_CLR	RS	Clear the Delta register. Hardware auto-clear.
9	Reserved	NA	-
8	SPI_MODE	RW	SPI mode 0x0: 4-wired SPI 0x1: 2-wired SPI
7	RESYNC_FLAG	RO	If this bit is 1, it indicates that re-synchronisation is in progress.
6:5	DATA_FORMAT	RW	sensor coordinate data format 0x2: 16-bit 0x1: 12-bit 0x0: 8-bit
4	DELTAXY_REV	RW	Data format of upper-bit coordinate data 1: bit3-bit0 is DeltaX upper 4 bits data 0: bit7-bit4 is DeltaX upper 4 bits data
3	SCAN_MODE	RW	SENCAN mode 1: continuous mode 0: single mode
2	DATA_VLD	RO	data valid flag
1	SCAN_STP	RS	scan stop 1: stop the current scanning
0	SCAN_EN	RS	scan enable 1: enable scanning

SENSCAN_ISR

Bit	Field	Access	Description
31:5	Reserved	NA	-
4	CFG_DONE	RC_W1	The sensor register read/write operation is complete and write 1 is cleared.

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3	SCAN_RESYNC	RC_W1	Scanning timer is reached during the Re-synchronization period. Clear this bit by writing 1.
2	RESYNC_DONE	RC_W1	Re-synchronisation is complete. Clear this bit by writing 1.
1	RESYNC_STRT	RC_W1	The 2-wired SPI is out of synchronization. Clear this bit by writing 1.
0	SCAN_DONE	RC_W1	Scanning is completed. Clear this bit by writing 1.

SENSCAN_IER

Bit	Field	Access	Description
31:5	Reserved	NA	-
4	CFG_DONE_IE	RW	Enable/Disable CFG_DONE interrupt.
3	SCAN_RESYNC_IE	RW	Enable/Disable SCAN_RESYNC interrupt. 0x0: disable 0x1: enable
2	RESYNC_DONE_IE	RW	Enable/Disable RESYNC_DONE interrupt. 0x0: disable 0x1: enable
1	RESYNC_STRT_IE	RW	Enable/Disable RESYNC_STRT interrupt. 0x0: disable 0x1: enable
0	SCAN_DONE_IE	RW	Enable/Disable SCAN_DONE interrupt. 0x0: disable 0x1: enable

SENSCAN_IDADDR

Bit	Field	Access	Description
31:7	Reserved	NA	-
6:0	SENSOR_ID_ADDR	RW	sensor ID address register

SENSCAN_ID

Bit	Field	Access	Description
31:8	Reserved	NA	-

.

7:0	SENSOR_ID	RW	sensor ID

SENSCAN_MOADDR

Bit	Field	Access	Description
31:7	Reserved	NA	-
6:0	SENSCAN_MOADDR	RW	sensor's motion status register address

SENSCAN_XADDR

Bit	Field	Access	Description
31:7	Reserved	NA	-
6:0	SENSCAN_XADDR	RW	sensor's DeltaX lower-bit data register address

SENSCAN_YADDR

Bit	Field	Access	Description
31:7	Reserved	NA	-
6:0	SENSCAN_YADDR	RW	sensor's DeltaY lower-bit data register address

SENSCAN_HADDR1

Bit	Field	Access	Description
31:7	Reserved	NA	-
6:0	SENSCAN_HADDR1	RW	sensor's upper-bit coordinate data register address 1

SENSCAN_HADDR2

Bit	Field	Access	Description
31:7	Reserved	NA	-
6:0	SENSCAN_HADDR2	RW	sensor's upper-bit coordinate data register address 2

SENSCAN_X

Bit	Field	Access	Description
31:24	Reserved	NA	-
23:0	SENSCAN_XL	RO	DeltaX lower-bit data

.

SENSCAN_Y

Bit	Field	Access	Description
31:24	Reserved	NA	
23:0	SENSCAN_YL	RO	DeltaY lower-bit data

SENSCAN_TR

Bit	Field	Access	Description
31:8	Reserved	NA	-
7:4	TSRR	RW	From rising SCLK for last bit of the first data byte, to falling SCLK for first bit of the address byte if the next commad. This configuration is invalid when SENCFG_EN=1. Tsrr = (t_srr+5)* T_spiclk
3:0	TSRAD	RW	From rising SCLK for last bit of the address byte, to falling SCLK for first bit of data being read. Tsrad = (t_srad+1)*T_spiclk

SENSCAN_TNS

Bit	Field	Access	Description
31:8	Reserved	NA	-
7:4	TNS	RW	From last SCLK rising edge to first SCLK rising edge. Tncs-sclk=(t_ncs_sclk+1)* T_spiclk
3:0	TSN	RW	From last SCLK rising edge to NSC rising edge, for vaild MISO data transfer. Tsclk-ncs=(t_sclk_ncs+1)* T_spiclk

SENSCAN_FREQ

Bit	Field	Access	Description
31:11	Reserved	NA	-
10:0	SENSCAN_FREQ	RW	Frequency of scanning. Scanning time t_{scan} (ms) is: (SENSCAN_FREQ +1) *0.25ms

SENSCAN_RESYNC

.

Bit	Field	Access	Description
31:13	Reserved	NA	-
12:8	T_WDT	RW	Valid only when SPIMODE=1, t_{WDT}: 0x0: 1ms 0x1: 2ms 0x2: 4ms 0x3: 6ms 0x4: 8ms 0x5: 12ms 0x6: 16ms 0x7: 20ms 0x8: 24ms 0x9: 32ms (default) 0xA: 40ms 0xB: 48ms 0xC: 64ms 0xD: 128ms 0xE: 256ms 0xF: 512ms
7:5	Reserved	NA	-
4:0	T_RESYNC	RW	Valid only when SPIMODE=1, t_{RESYNC}: 0x0: 2us 0x1: 4us 0x2: 8us 0x3: 16us 0x4: 32us 0x5: 64us 0x6: 128us 0x7: 256us 0x8: 512us 0x9: 1ms 0xA: 2ms 0xB: 4ms 0xC: 6ms 0xD: 8ms 0xE: 10ms (default) 0xF: 12ms 0x10: 16ms 0x11: 24ms 0x12: 32ms

CFG_CSR

Bit	Field	Access	Description
31:8	Reserved	NA	-
7	RESYNC_STRT	RS	Resync is initiated in cfg mode, valid for two-wire SPI only. The hardware is automatically cleared. When initiating resync, ensure that senscan is in cfg mode (SENCFG_EN=1) and two-wire SPI mode (SPIMODE=1), and configure the resync time (SENSCAN_RESYNC register).
6	SENCFG_DIR	RW	Configuration mode data direction. 0x0: Read operation 0x1: Write operation
5	Reserved	NA	-
4	SENCFG_STRT	RS	Read/write start control bit in configuration mode. Hardware is automatically cleared. When performing a CFG, ensure that senscan is in configuration mode (SENCFG_EN=1) and that TSRAD, TNS and TSN are configured.
3:1	Reserved	NA	-
0	SENCFG_EN	RW	0：sensor scan 1: sensor config

CFG_ADDR

Bit	Field	Access	Description
31:7	Reserved	NA	-
6:0	SENCFG_ADDR	RW	Configuration mode, sensor register address

CFG_DATA

Bit	Field	Access	Description
31:8	Reserved	NA	-
7:0	SENCFG_DATA	RW/RO	Read/write data in cfg mode. rdata register (RO) for SENCFG_DIR=0; Wdata register (RW) for SENCFG_DIR=1.

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18 LEDC (Return Zero)

18.1 Introduction

The RGB LED Controller uses a single-wire communication method and sends signals through a return to zero code. Data is output through the DO pin, and after sending 24 bits of data, a low-level RESET code that lasting over 80 microseconds is output. Once the LED receives the RESET code, it outputs the received RGB data to the R, G, and B pins.

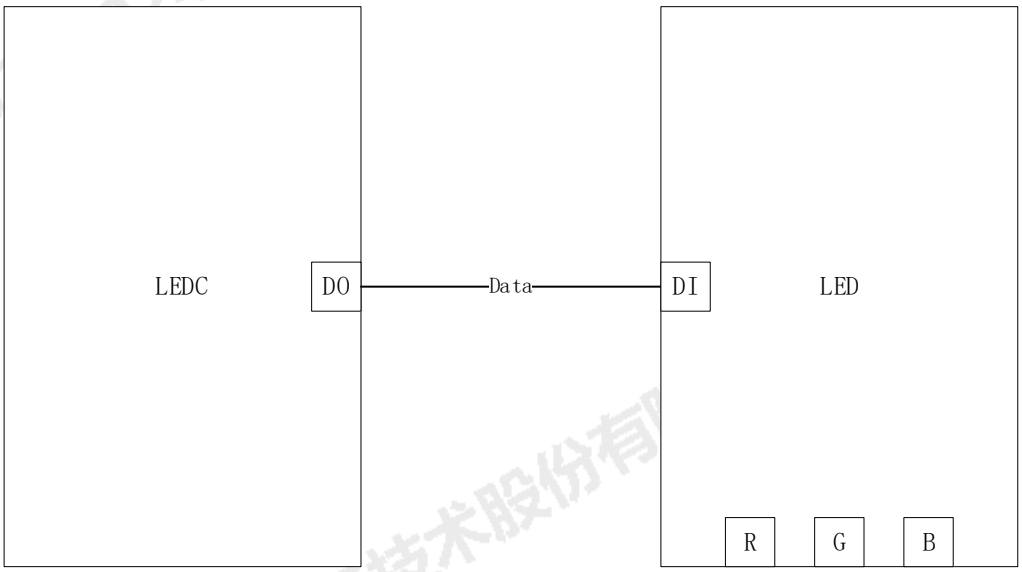


Figure18.1 System Diagram

18.2 Features

Features of LEDC include:

- RZ code
  - The length of reset period is configurable
- DMA interface
  - 3 level 24bit buffer。

18.3 Block diagram

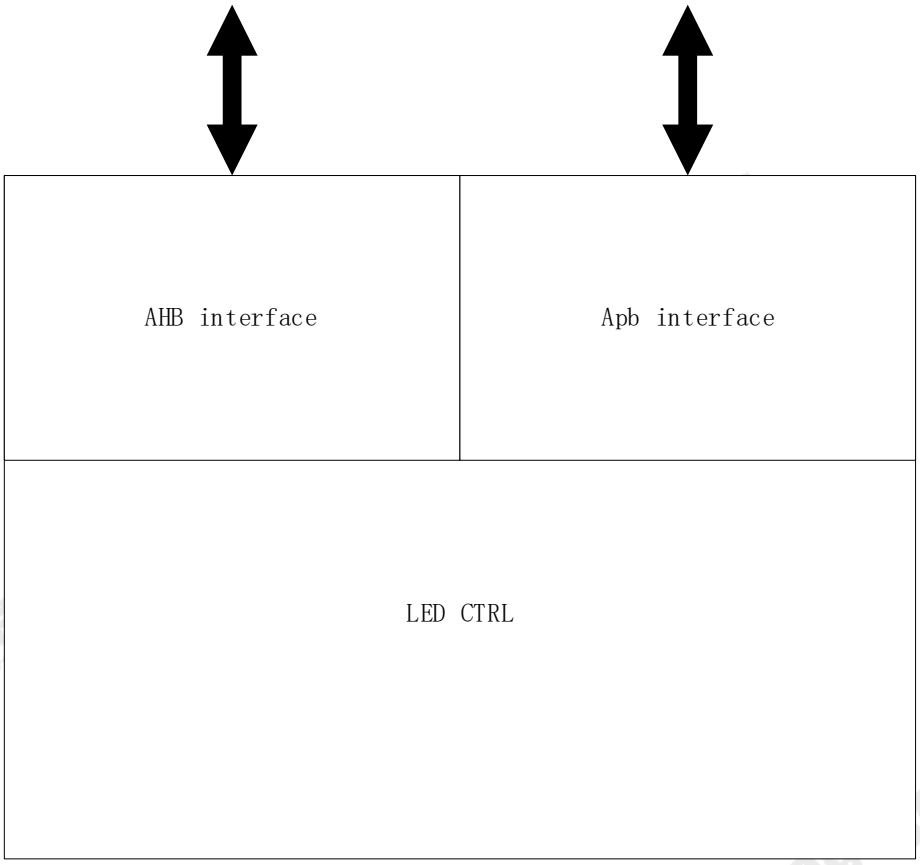


Figure18.2 block diagram

18.4 Reference document

3010 RGB-1815S spec

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18.5 Interface Signals

Table 18.1 Interface signals list

name	I/O	Width	Description
Clock and Reset			
hclk	I	1	AHB/APB bus clock
hreset_n	I	1	APB bus reset low active
APB Slave Bus signal			
psel	I	1	APB Bus PSEL
paddr	I	32	APB Bus address
penable	I	1	APB Bus PENABLE, High Active
pwrite	I	1	APB Bus WRITE High active
pwdata	1	32	APB Bus PWDATA 32bit write data
prdata	O	32	APB Bus PRDATA 32bit read data
AHB master signal			
haddr_m	O	32	AHB Bus address
htrans_m	O	2	AHB Bus htrans
hwrite_m	O	1	AHB Bus HWRITE (tie 0, for there is only read transition)
hsize_m	O	3	AHB Bus HSIZE
hburst_m	O	3	AHB Bus HBURST
hprot_m	O	4	AHB Bus HPROT
hlock_m	O	1	AHB Bus HLOCK
hwdata_m	O	32	AHB Bus HWDATA (tie 0， for there is only read transition)
hready_m	I	1	AHB Bus hready
hrdata_m	I	32	AHB Bus hrdata
LED 接口			
DO	0	1	
	O	8	
	O	8	

18.6 Input code

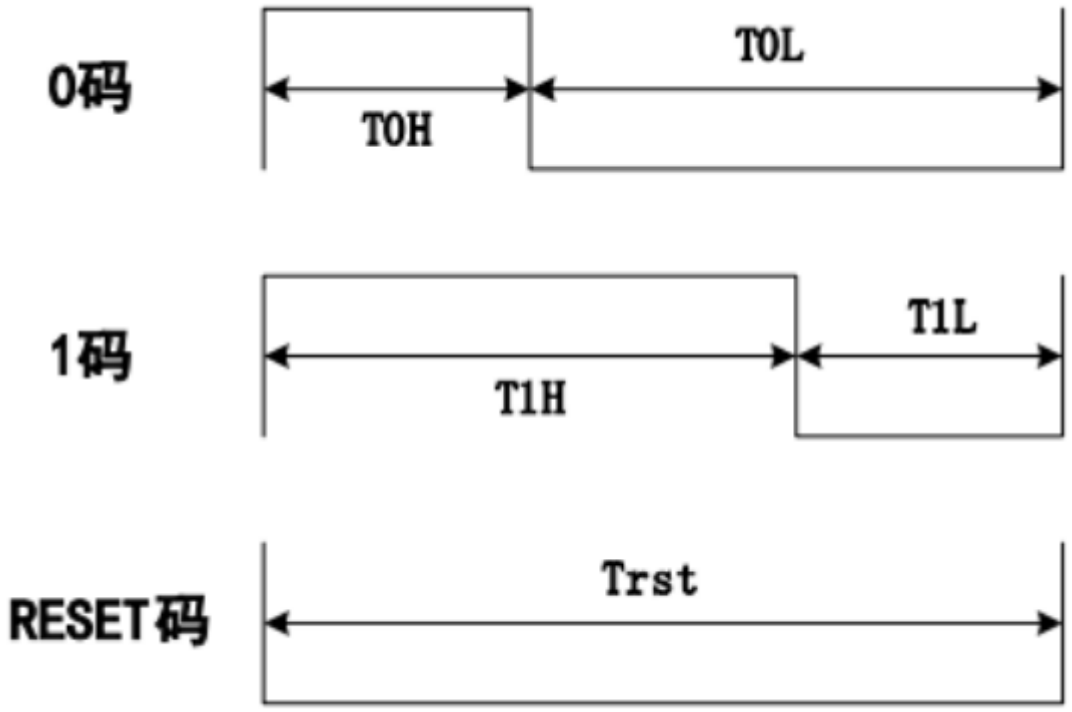


Figure18.3 input code

18.7 Code timing

Name	Description	Typical value	Tolerance
T0H	Code 0 high level timing	0.295us	+- 0.05us
T1H	Code 1 high level timing	0.595us	+- 0.05us
T0L	Code 0 low level timing	0.595us	+- 0.05us
T1L	Code 1 low level timing	0.295us	+- 0.05us
Trst	Code RESET low level timing	>=80us	

18.8 Architecture

The APB_REG is used for configuring the registers. The AHB_Master retrieves the data to be sent from RAM based on the software DMA configuration and places it into an 8-byte FIFO. When the FIFO has data, the CTRL will convert the corresponding data into RZ code and send it to the DO pin. Once all data has been transmitted, the hardware sends a reset code.

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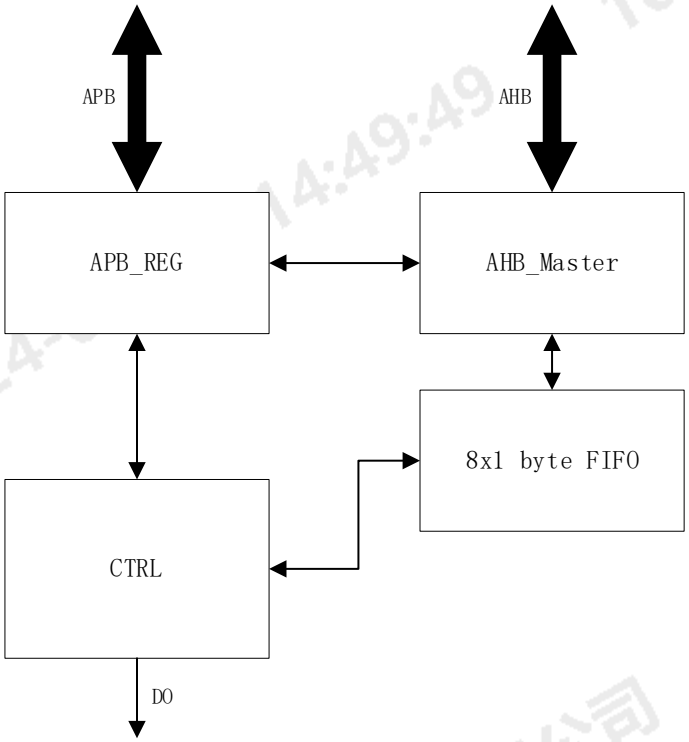


Figure 18.4 System Diagram

18.9 Register map table

All undefined bits in the table below are redundant. They are read out as '0'.

Offset	Name	Description
0x000	LEDC_CR	LEDC control register
0x004	SIG_TIM	Signal Timing control
0x008	RST_TIM	Reset Timing control
0x00C	DMA_CTL	DMA control

18.10 LEDC_CR Address: 00h

Bit	Symbol	Reset	R/W	Description

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31:17	Reserved	10'h0	RW	For future used
16	INT_RAW	1'b0	RW	Interrupt raw status register write 1 to clear the interrupt 1'b1: LEDC interrupt is happened 1'b0: No interrupt is happened
15:9	RSV	-	-	Reserved
8	INT_MASK	1'b0	RW	Interrupt Mask reg 1: Enable the interrupt 0: Mask the interrupt
7:1	RSV	-	-	Reserved
0	LEDC_EN	0	RW	Enable LEDC. Software set, hardware clear 1'b1: Write 1 to enable the LEDC 1'b0: When hardware finish it will clear the bit

18.11 SIG_TIM Address: 04h

This register only works when PKTCRTL0.STRUCT_SEL==0.

Bit	Symbol	Reset	R/W	Description
31:15	TL	2'b01	RW	Long pulse time control. It's the number of cycles of 48MHz clk 0: 1 cycles … N: N+1 cycles
15:0	TS	2'h0	RW	Short pulse time control. Itis the number of cycles of 48MHz clk 0: 1 cycles … N: N+1 cycles

18.12 RST_TIM Address: 08h

Bit	Symbol	Reset	R/W	Description
32:0	RST_T	27'h0	RW	Reset low time control. It is the number of cycles of 48MHz clk 0: 1 cycles … N: N+1 cycles

18.13 DMA_CTL Address: 0Ch

This register only fit for frame A

Bit	Symbol	Reset	R/W	Description
29:16	DMA_ADDR	0	RW	DMA start addr
8:0	DMA_LEN	0	RW	DMA length

				0: 0 byte 1: 1 byte … N：N byte

18.14 Programming Guide

1. Software prepare data in the sram.

   Uint8 data[10] = {0x00,0x01,0x02,0x03,0x04,0x05,0x06,0x07,0x08,0x09};

2. Program the DMA_addr and DMA_LEN register.

   Uint32* dma_addr = (uint32 *) data;

   WriteREG32(DMA_ADDR, *dma_addr);

   WriteREG32(DMA_LEN, 10);

3. Program the correct TS, TL, RST_T value

   WriteREG32(SIG_TIM, 28<<16|13);

   WriteREG32(RST_TIM, 3845);

4. Program the LEDC bit to Enable the LEDC

   WriteREG32(LEDC_CR, 1)

5. Polling the INT_RAW reg or enable interrupt mask and wait interrupt

   Do {

   Rdata = ReadREG32(LEDC_CR) & 0x100;

   } while(rdata==0x0)

6. Repeat 1 to 5

.

19 ENCODER

The ENCODER supports hardware scanning of the wheel's status. When the wheel scrolls up, Wheel_data increments by one, and when the wheel scrolls down, Wheel_data decrements by one. When the wheel state changes, an interrupt is sent to the software.

The following is the ENCODER process:

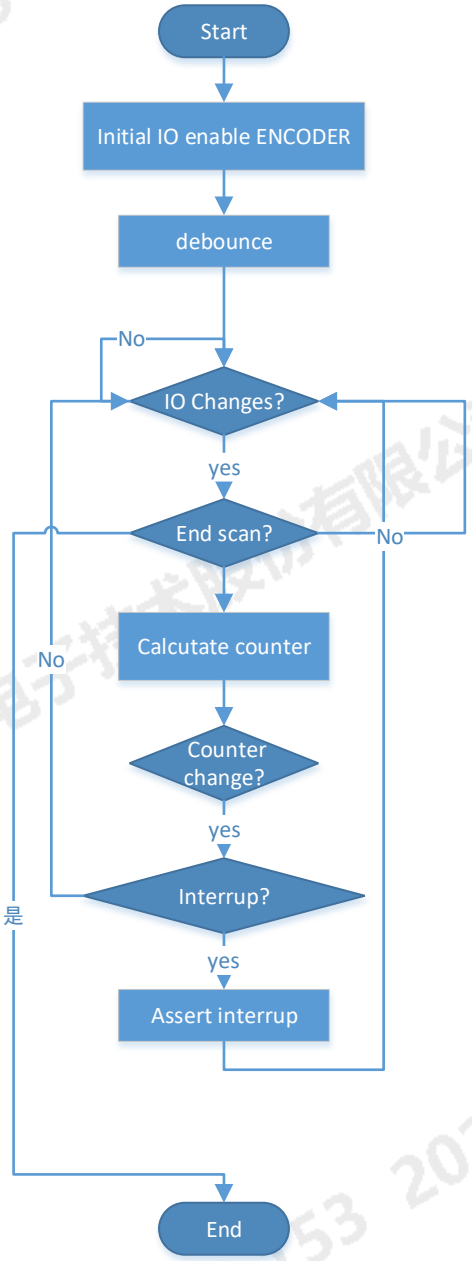


Figure 19.1 ENCODER process

19.1 Encoder register map

Offset address	Name	Access	Reset value	Description
Base address: 0x40000B00				
0x00000	ENCODER_EN	RW	0X00000000	Encoder enable
0x00004	ENCODER_CNT	RW	0X00000000	Encoder count
0x00008	ENCODER_MODE	RW	0X00000000	Encoder mode control
0X0000c	ENCODER_SAM	RW	0X00000000	Encoder sample control
0X00010	ENCODER_DBG	RW	0X00000000	Output enable clear
0x00014	ENCODER_VLD	R	0X00000000	Encoder_cnt valid

ENCODER_EN address offset: 0x0000

Bit	R/W	Reset	Name	Description
31:18	N/A	0x0	N/A	reserved
17:16	RW	0x0	Encoder_sel	Pinmux encoder select 0: p0.0,p0.1 select as encoder input pins 1: p0.5,p0.6 select as encoder input pins 2: p2.0 p2.1 select as encoder input pins 3: p2.4 p2.5 select as encoder input pins
15:0	N/A	0x0	N/A	Reserved
2	RW	0x0	wken	Encoder wakeup enable 1: enable 0: disable
1	RW	0x0	ien	interrupt enable: 1: enable 0: disable
0	RW	0x0	cen	encoder enable: 1: enable 0: disable

ENCODER_CNT address offset: 0x0004

Bit	R/W	Reset	Name	Description
31:4	N/A	0x0	N/A	reserved

3:0	R	0x0	Wheel_data	Wheel data value. read only. Read will clear the wheel_data

ENCODER_MODE address offset: 0x0008

Bit	R/W	Reset	Name	Description
31:4	N/A	0x0	N/A	reserved
3	RW	0x0	cc2p	ti2 polarity. 0: ti2 not inverted; 1: ti2 inverted.
2	RW	0x0	cc1p	ti1 polarity. 0: ti1 not inverted; 1: ti1 inverted.
1:0	N/A	0x0	N/A	reserved

ENCODER_SAM address offset: 0x00C

Bit	R/W	Reset	Name	Description
31:4	N/A	0x0	N/A	reserved

3:0	RW	0x0	icf	Debounce time
				icf=0: no debounce
				icf=1: 62.5us,
				icf=2: 125us
				icf=3: 187.5us
				icf=4: 250us
				icf=5: 312.5us
				icf=6: 375us;
				icf=7: 437.5us
				icf=8: 500us
				icf=9: 562.5us
				icf=10: 625us
				icf=11: 687.5us
				icf=12: 750us;
				icf=13: 812.5us;
				icf=14: 875us
				icf=15: 1ms;

ENCODER_DBG address offset: 0x0010

Bit	R/W	Reset	Name	Description
31:**11**	N/A	0x0	N/A	reserved
10:10	R	0x0	Cnd_vld	Cnt_vld
9:9	R	0x0	Cnt_vld_src	Cnt_vld_src
8:8	R	0x0	Bcorr_a_status	Bcurr_a_status

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7:7	R	0x0	Bcurr_b_status	Bcurr_b_status
6:6	R	0x0	Bno_status_valid	Bno_status_valid
5:5	R	0x0	Bpre_ab_no_status	Bpre_ab_no_status
4:4	R	0x0	Bpre_ab_status	Bpre_ab_status
3:0	R	0x0	Wheel_data_source	original wheel_data value. read only.(for debug)

ENCODER_VLD address offset: 0x0014

Bit	**R/W**	**Reset**	**Name**	**Description**
31:**1**	N/A	0x0	N/A	reserved
0:0	R	0x0	Wheel_data_vld	Wheel_data valid Read the ENCODER_CNT will clear this bit 1: The wheel data is ready for read 0: wheel data is not ready